CompTIA Server+

Troubleshooting Network Connectivity Issues

Introduction

Server+
Network Troubleshooter
IPv4 and IPv6
Ipconfig
Ping
Nbtstat

Welcome to the **Troubleshooting Network Connectivity Issues** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Network connectivity can sometimes be an issue. This module will explore some common tools in Windows as well as some command-line tools for Windows and Linux.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Common Problems
- Exercise 2 - Command Line Tools and Techniques

After completing this module, you should be able to:

- Explore Host File Misconfiguration
- Work with Network Troubleshooter
- Explore Internet Options

- Configure Network Adapters
- Configure Network Adapters IPv4 and IPv6 Properties
- Install and Configure DHCP Server
- Explore Windows Commands Commonly used in Troubleshooting
- Explore Linux Commands

Exam Objectives

The following exam objectives are covered in this module:

4.5 Given a scenario, troubleshoot network connectivity issues

- Common problems
- Causes of common problems
- Tools and techniques

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PLABDC01 -** (Windows Server 2019 - Domain Controller)
- **PLABDM01 -** (Windows Server 2022 - Domain Member Server)
- **PLABWIN10 -** (Windows 10 - Domain Member Workstation)
- **PLABALMA -** (Alma Linux 8.7 - Stand-alone Linux Workstation)

- **PLABUBUNTU -** (Linux Ubuntu - Stand-alone Linux Server)

Click **Next** to proceed to the first exercise.

Exercise 1 - Common Problems

There are common problems that happen with networks that can result in limited to no connectivity. These happen due to hardware failures, configuration errors, and other reasons. There are tools built into the operating systems that can help troubleshoot and diagnose these network issues.

In this exercise, different tools in the Windows Server 2019 operating system that can be used to troubleshoot network connectivity issues will be discussed.

Learning Outcomes

After completing this exercise, you should be able to:

- Explore Host File Misconfiguration
- Work with Network Troubleshooter
- Explore Internet Options
- Configure Network Adapters
- Configure Network Adapters IPv4 and IPv6 Properties
- Install and Configure DHCP Server

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABDC01 -** (Windows Server 2019 - Domain Controller)

Task 1 - Host File Misconfiguration

The **hosts** file is present on every Windows, MacOS, and Linux machine. This document doesn't go into effect automatically. It exists, but all it does is explain what it is and how to use it. When entries are added to the hosts file in the proper format and without # for comments, it becomes the computer's primary DNS mappings and will be accessed before any DNS service. It gives users the ability to manually link a domain name like practicelabs.com to an IP address.

Before the Domain Name System (DNS) was invented, the host file was the main method for linking a hostname to an IP address.

In this task, you will access the hosts file and manipulate the rule to prevent some diagnostic tools from working.

Step 1

Connect to **PLABDC01.**

Minimize the **Server Manager** window.

Click the **File Explorer** icon on the **Taskbar.**

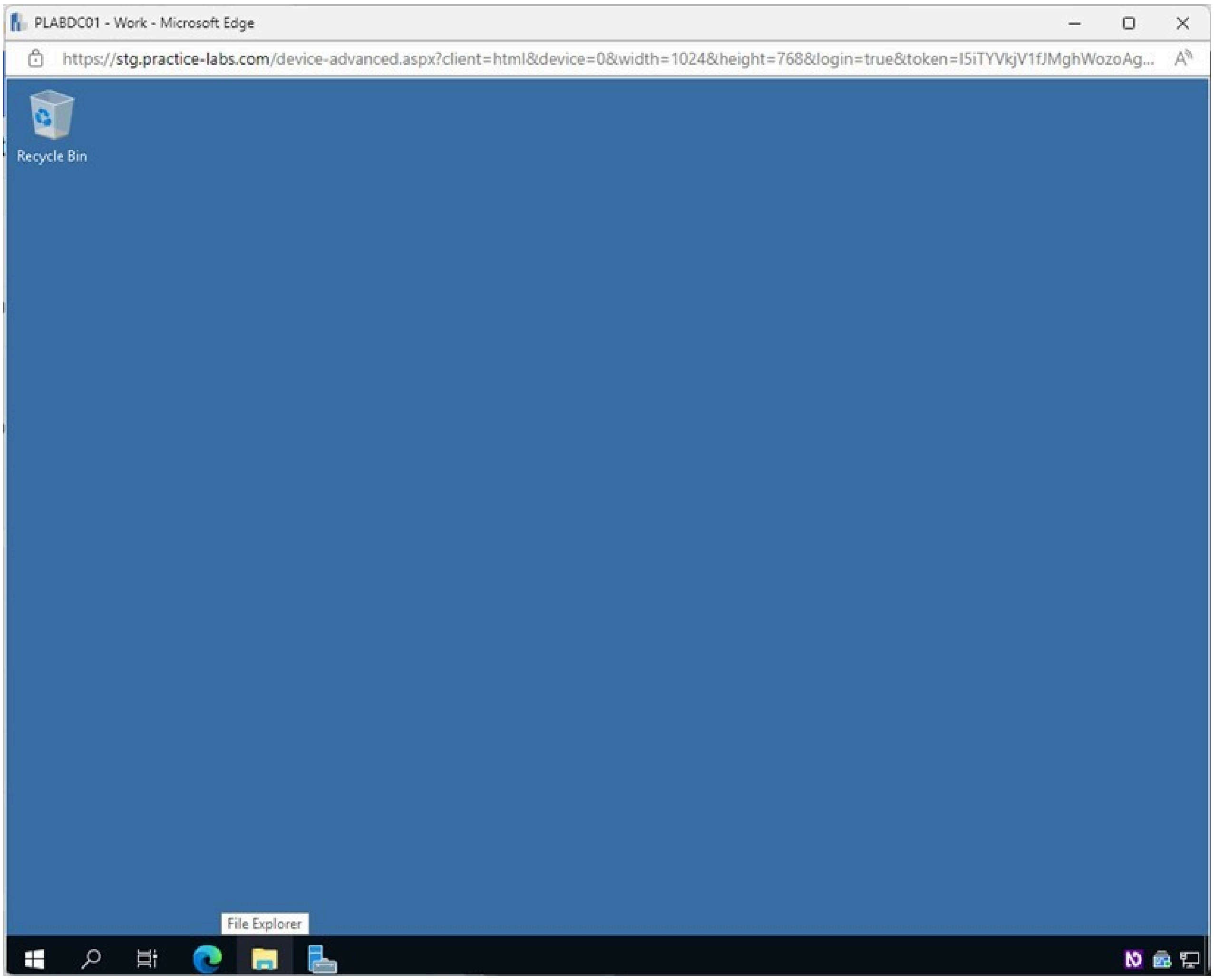

Figure 1.1 Screenshot of PLABDC01: Displaying opening File Explorer from the Taskbar.

Step 2

In the **File Explorer** window, select **System32** in the left pane.

Scroll down and double-click on **drivers** in the right details pane.

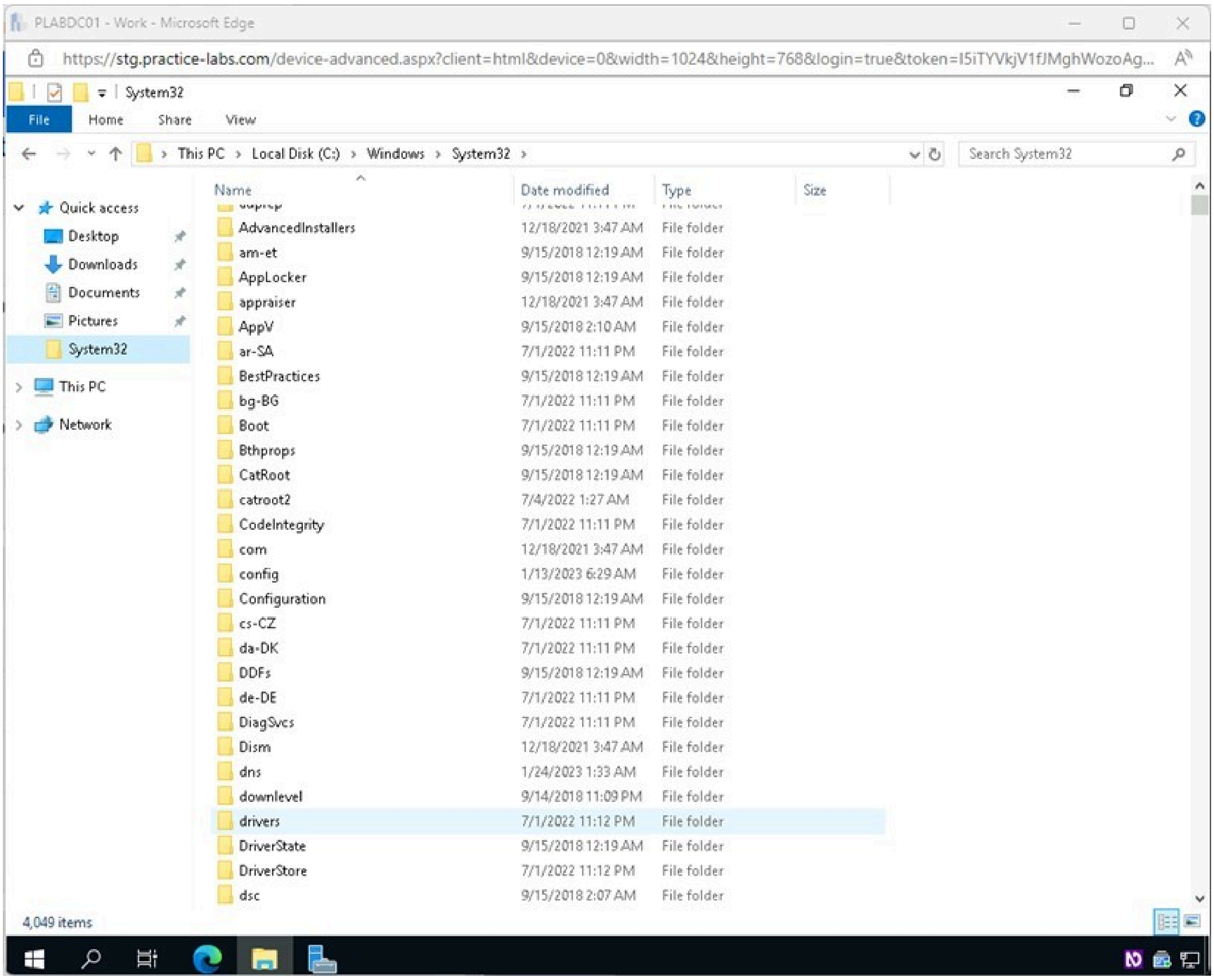

Figure 1.2 Screenshot of PLABDC01: Displaying the File Explorer window, showing drivers selected on the right details pane.

Step 3

Double-click on **etc** in the right details pane.

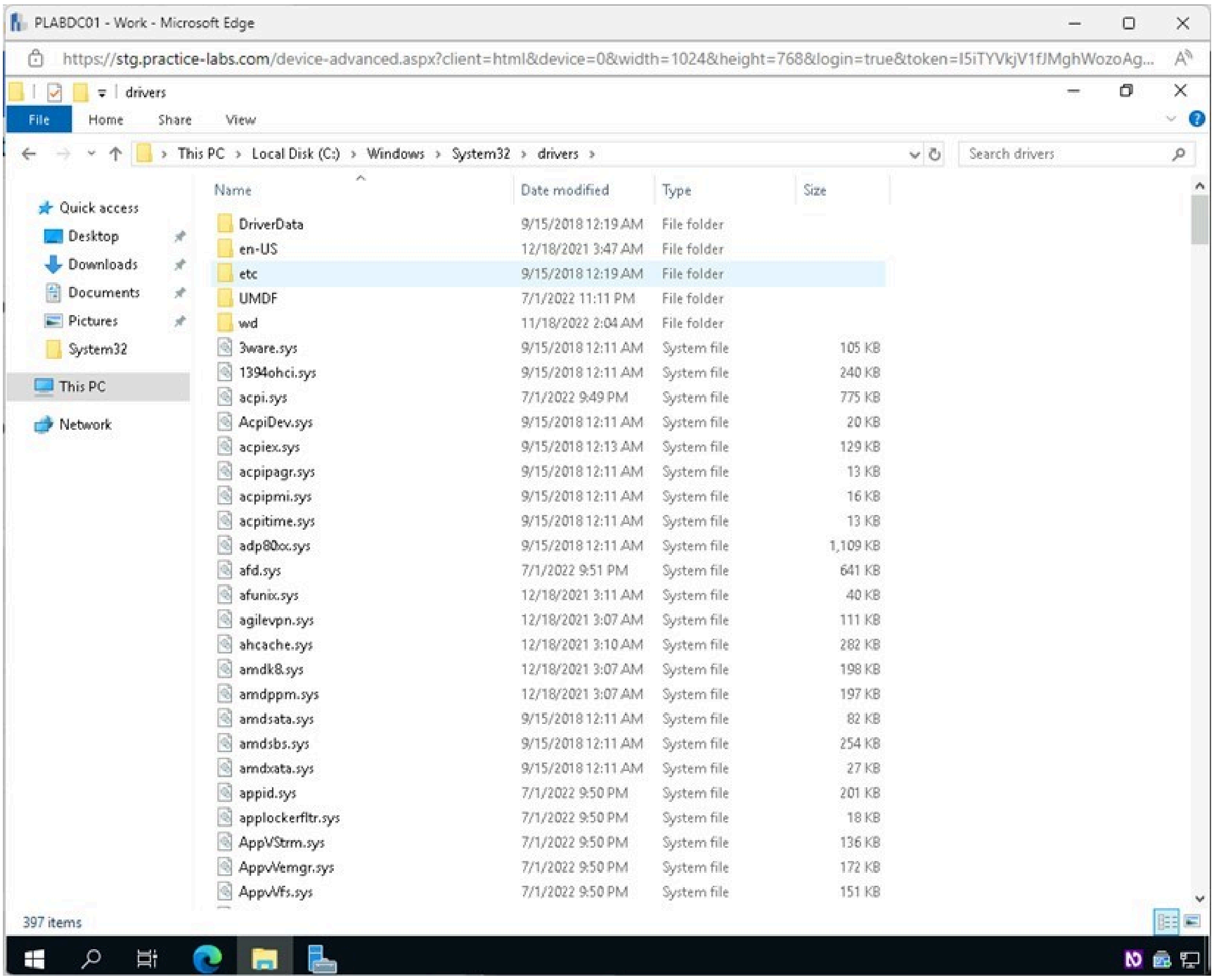

Figure 1.3 Screenshot of PLABDC01: Displaying selecting the etc folder in the File Explorer window.

Step 4

Double-click on **hosts** in the right details pane.

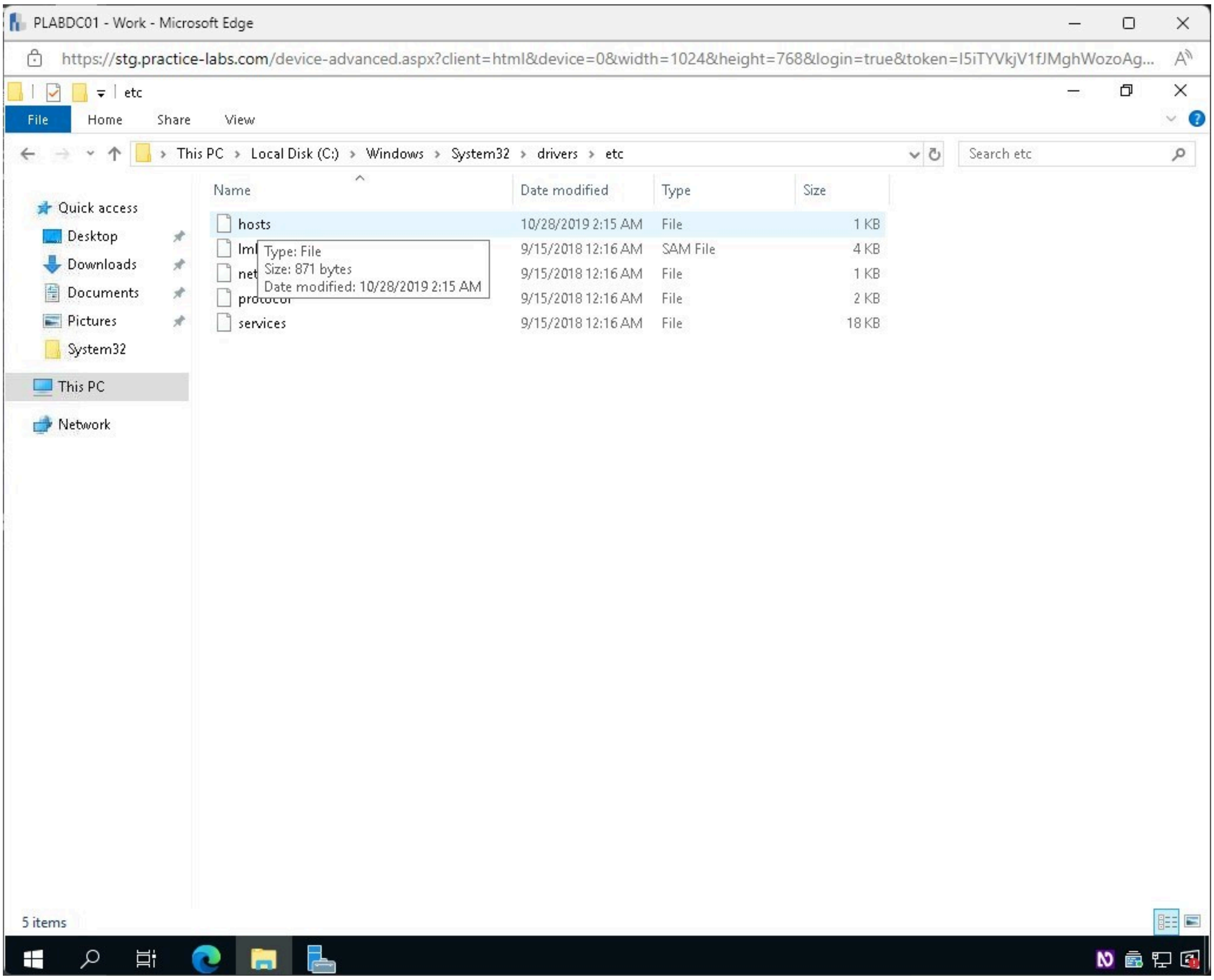

Figure 1.4 Screenshot of PLABDC01: Displaying selecting hosts in the File Explorer window.

Step 5

In the **How do you want to open this file?** pop-up window, select **Notepad** and click **OK.**

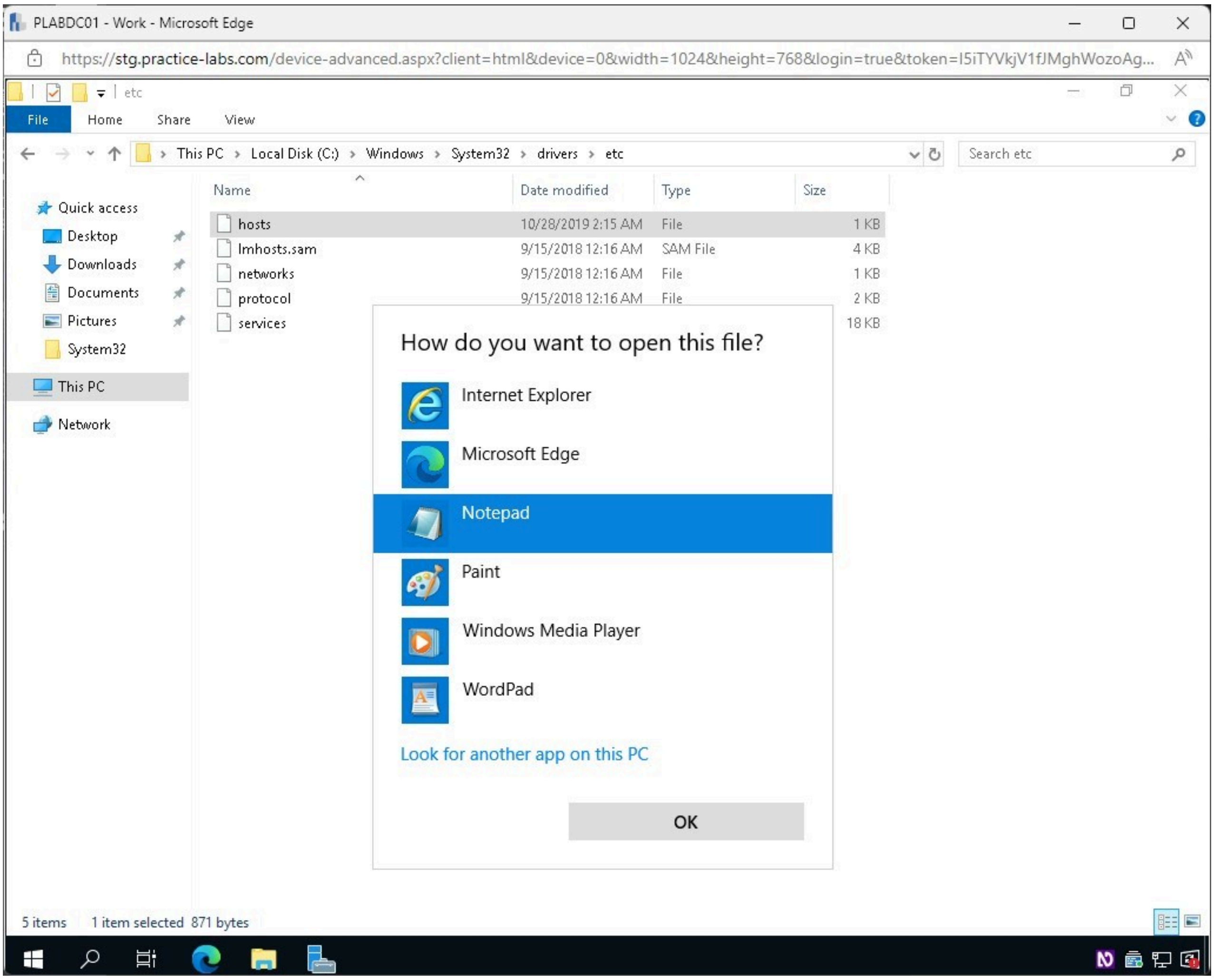

Figure 1.5 Screenshot of PLABDC01: Displaying the How do you want to open this file? pop-up window, showing Notepad selected.

Step 6

In the **hosts - Notepad** window, the first rule created **192.168.255.13 Proxy** is adding additional restrictions that will affect the diagnostics later in the lab.

Erase **192.168.255.13 Proxy** from the list.

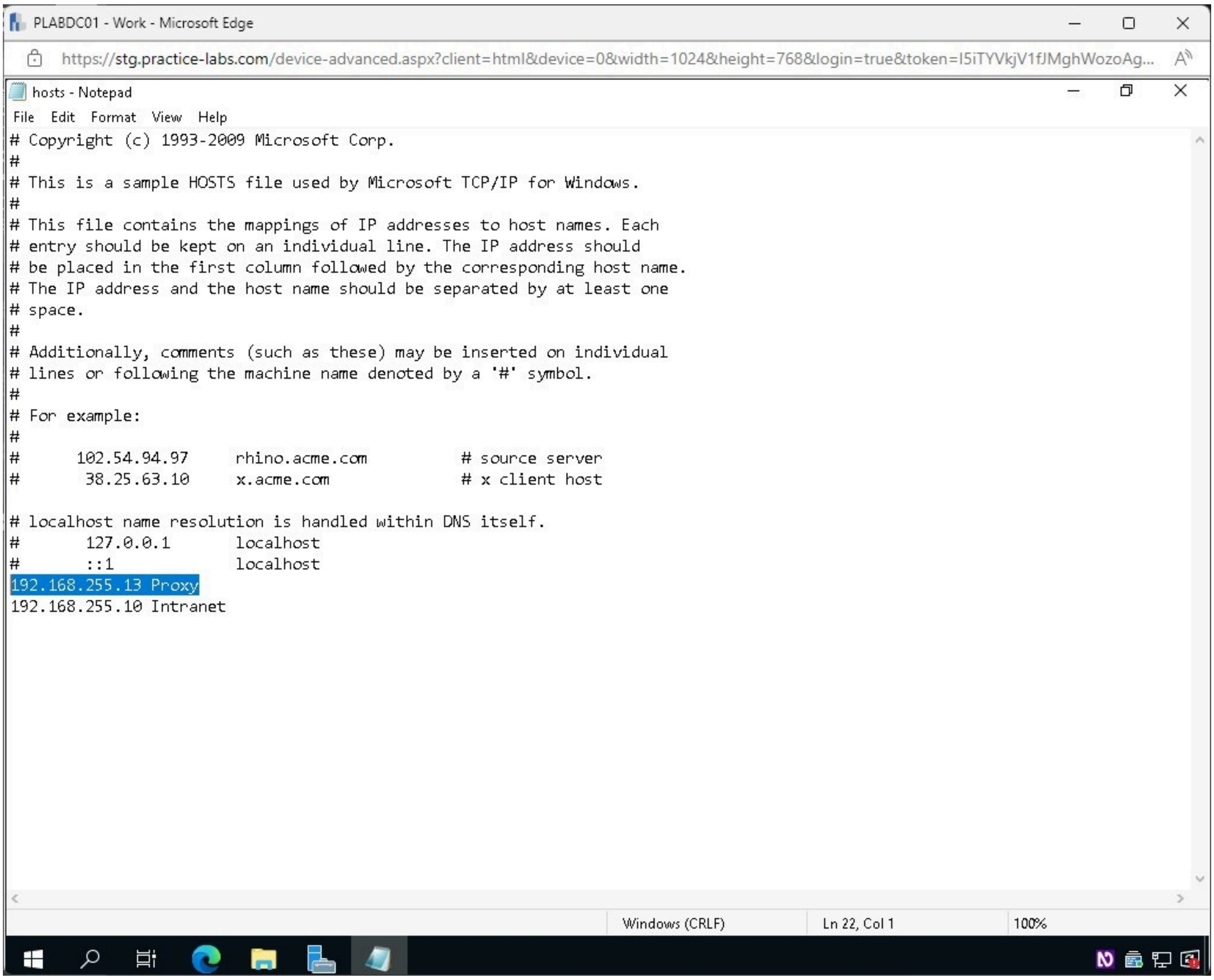

Figure 1.6 Screenshot of PLABDC01: Displaying the hosts - Notepad window, showing the required settings performed.

Step 7

Click the **File** menu and select **Save**.

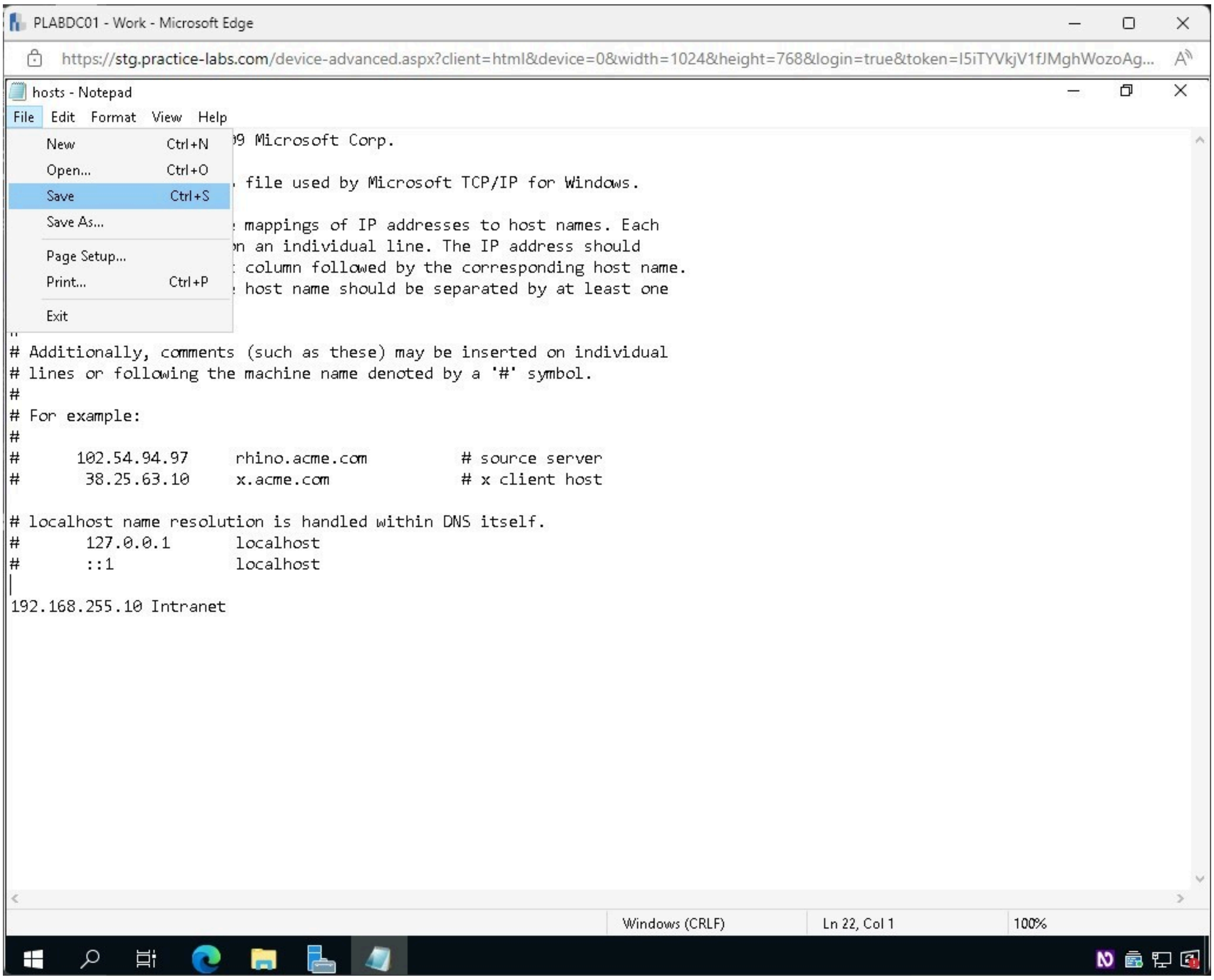

Figure 1.7 Screenshot of PLABDC01: Displaying accessing the File menu and selecting Save in the hosts - Notepad window.

Step 8

Close the **hosts - Notepad** file.

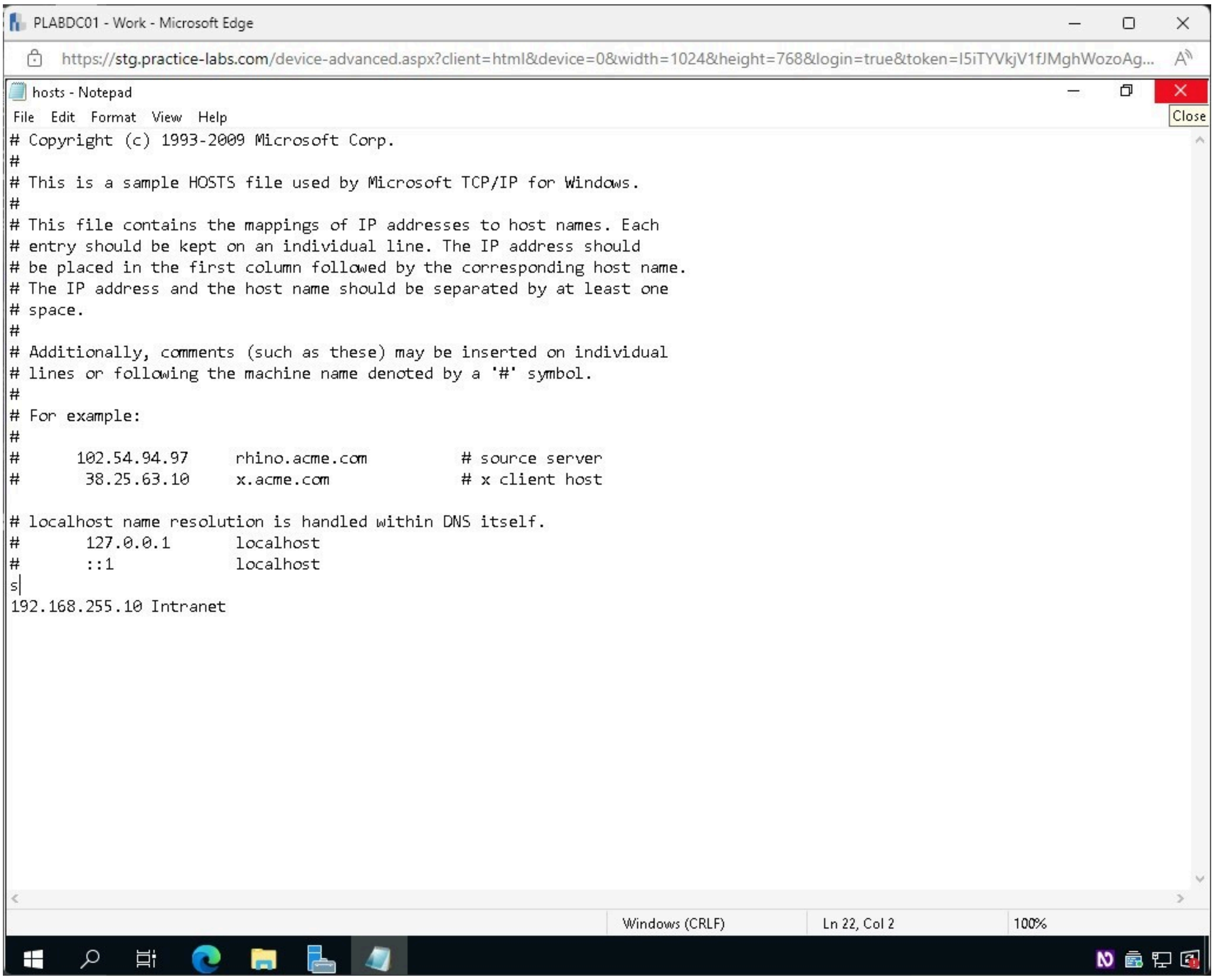

Figure 1.8 Screenshot of PLABDC01: Displaying closing the hosts - Notepad file.

Step 9

Close the **File Explorer** window.

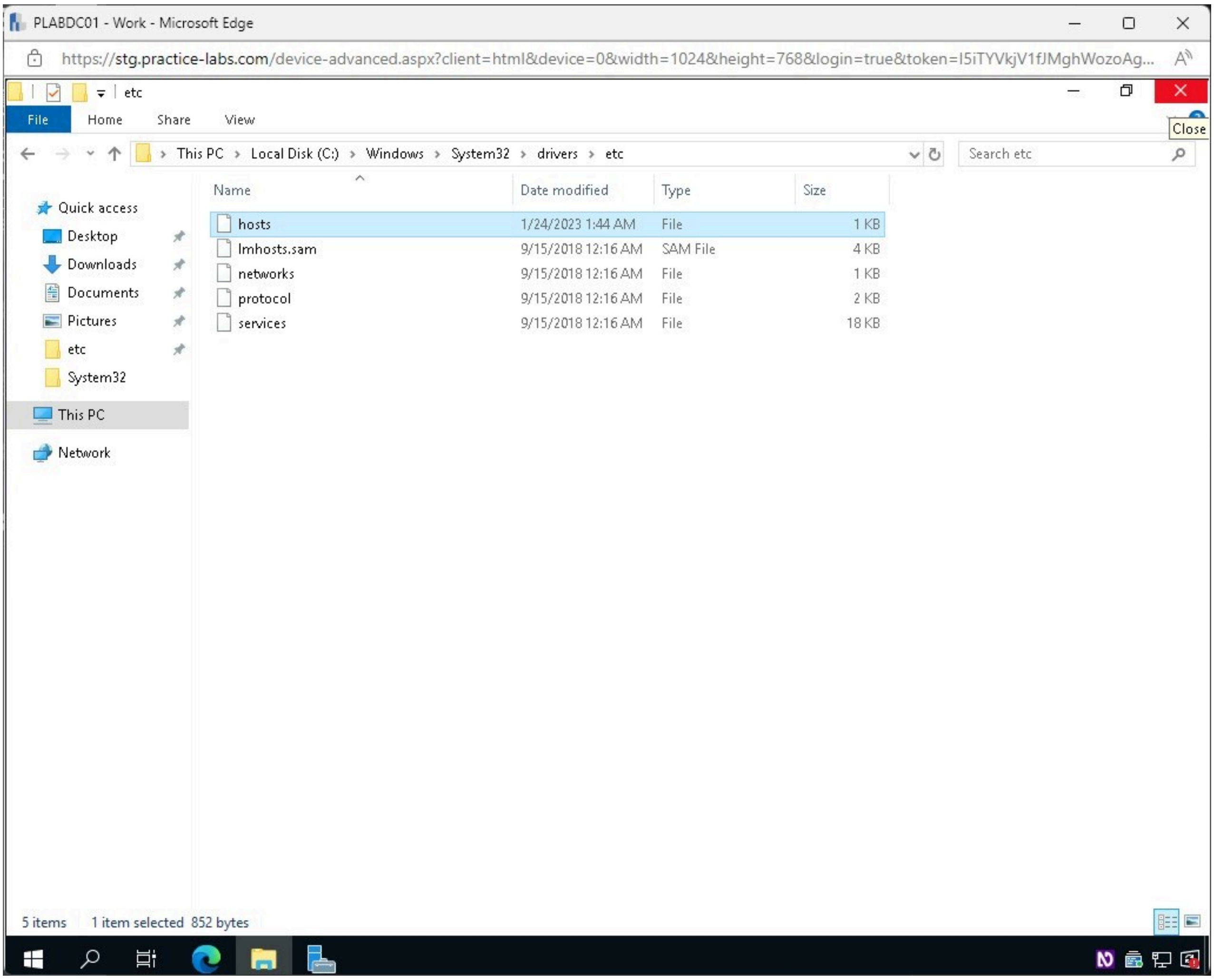

Figure 1.9 Screenshot of PLABDC01: Displaying closing the File Explorer window.

Task 2 - Network Troubleshooter

When users use Windows Troubleshooting tools, Windows will automatically fix problems such as corrupt settings that prevent essential services from running. It can make changes to work with your hardware and make any other specialized adjustments required for Windows to function with the hardware, apps, and settings desired. This tool is most typically used when users are unable to connect to the internet, but if users have connectivity difficulties after upgrading to Windows 11, the network troubleshooter can assist in resolving access to local network resources. If the tool executes a network reset, it will reset all your network adapters and their configurations.

In this task, you will learn about the Network Troubleshooter tool.

Step 1

Ensure you are connected to **PLABDC01.**

Right-click the **Network Internet access** icon on the right of the toolbar and select **Troubleshoot problems**.

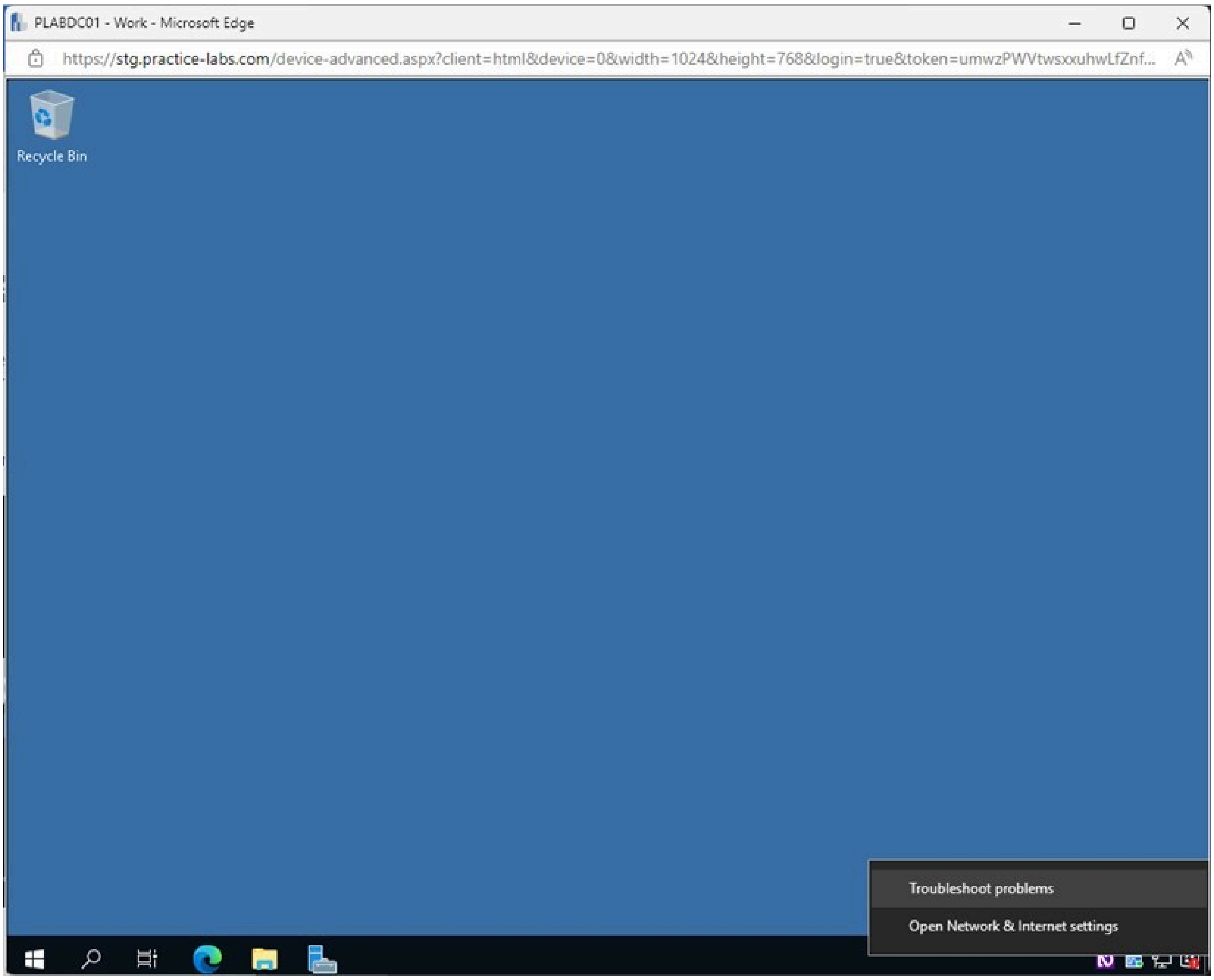

Figure 1.10 Screenshot of PLABDC01: Displaying right-clicking on the internet access icon and selecting Troubleshoot problems.

Step 2

In the **Windows Network Diagnostics** window, the tool works to detect problems with connectivity.

Wait for the process to complete.

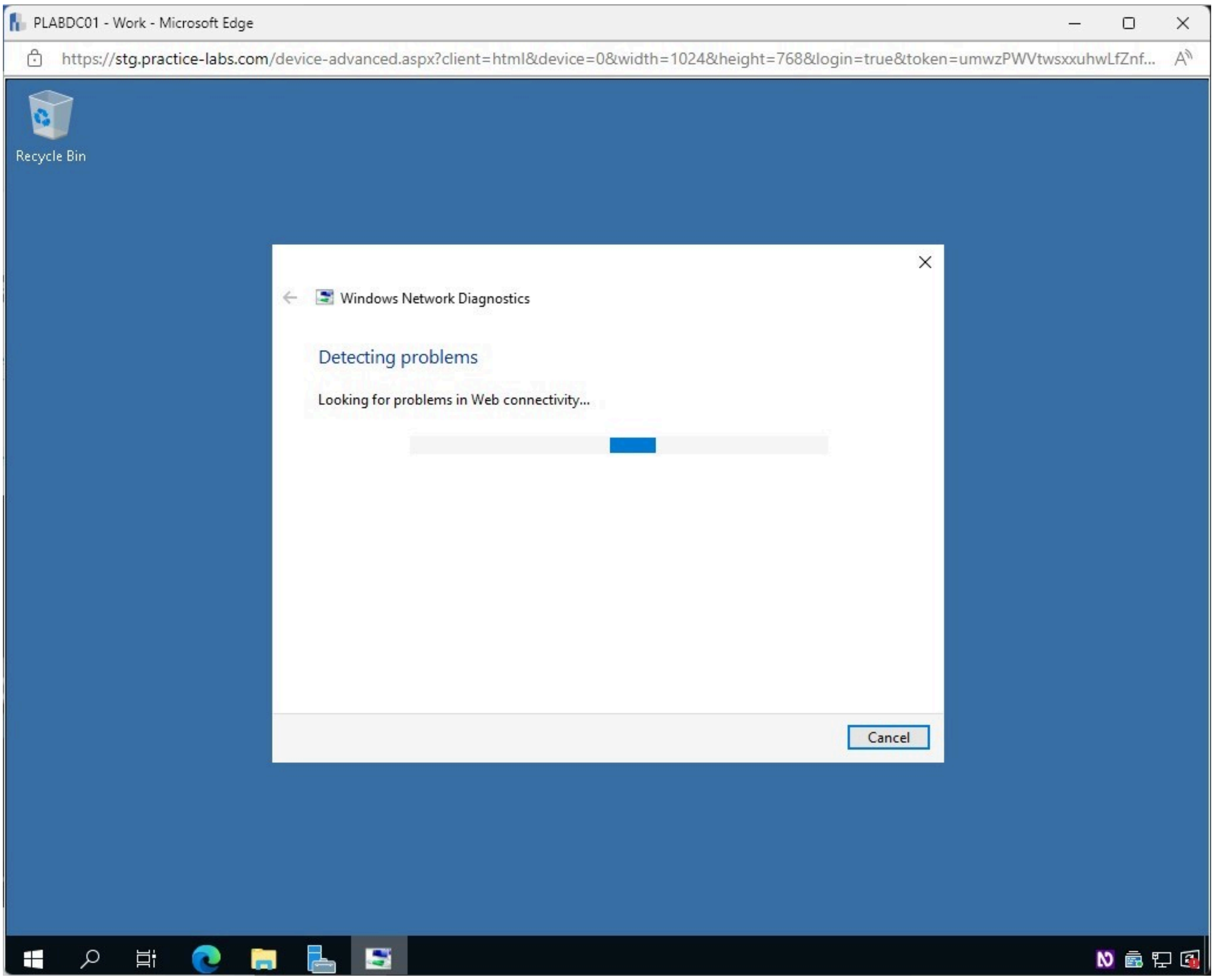

Figure 1.11 Screenshot of PLABDC01: Displaying the Windows Network Diagnostics window.

Step 3

In the **Windows Network Diagnostics** window, notice the troubleshooting is complete, and there appears to be an error with **DNS**.

This is due to the line that was removed in the **hosts - Notepad** file in the earlier task.

Click **Close**.

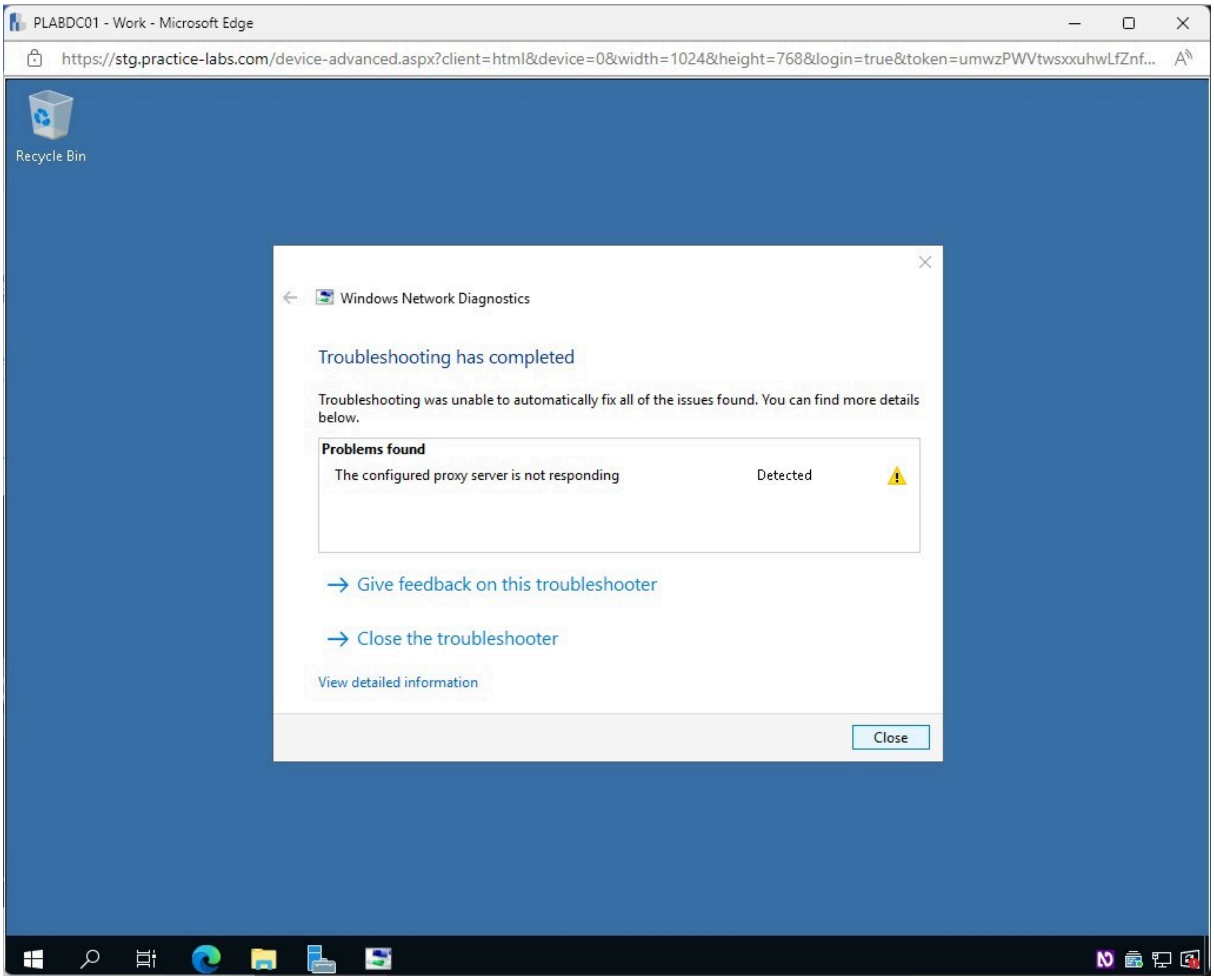

Figure 1.12 Screenshot of PLABDC01: Displaying closing the Windows Network Diagnostics window.

Task 3 - Internet Options

The Internet Options applet in the Control Panel allows users to customize the level of protection for their browser, modify personal settings, restrict or allow access to websites, and more.

In this task, you will explore the Internet Options applet.

Step 1

In **PLABDC01**, click the **Start** charm and type the following:

```
control panel
```

Select **Control Panel** from the **Best match** pop-up menu.

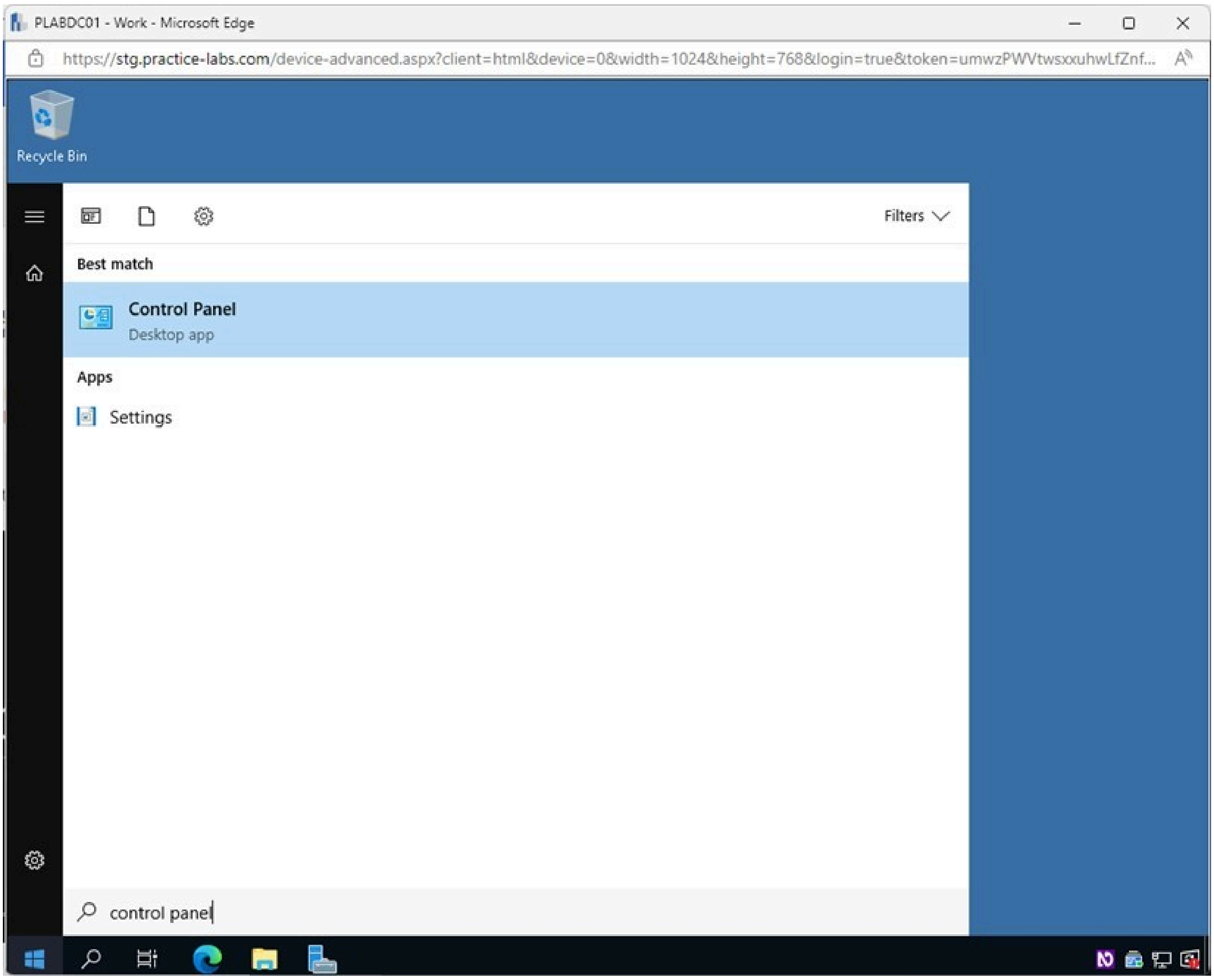

Figure 1.13 Screenshot of PLABDC01: Displaying selecting Control Panel from the Best match pop-up menu.

Step 2

In the **Control Panel** window, click the **View by** drop-down menu and select **Large icons.**

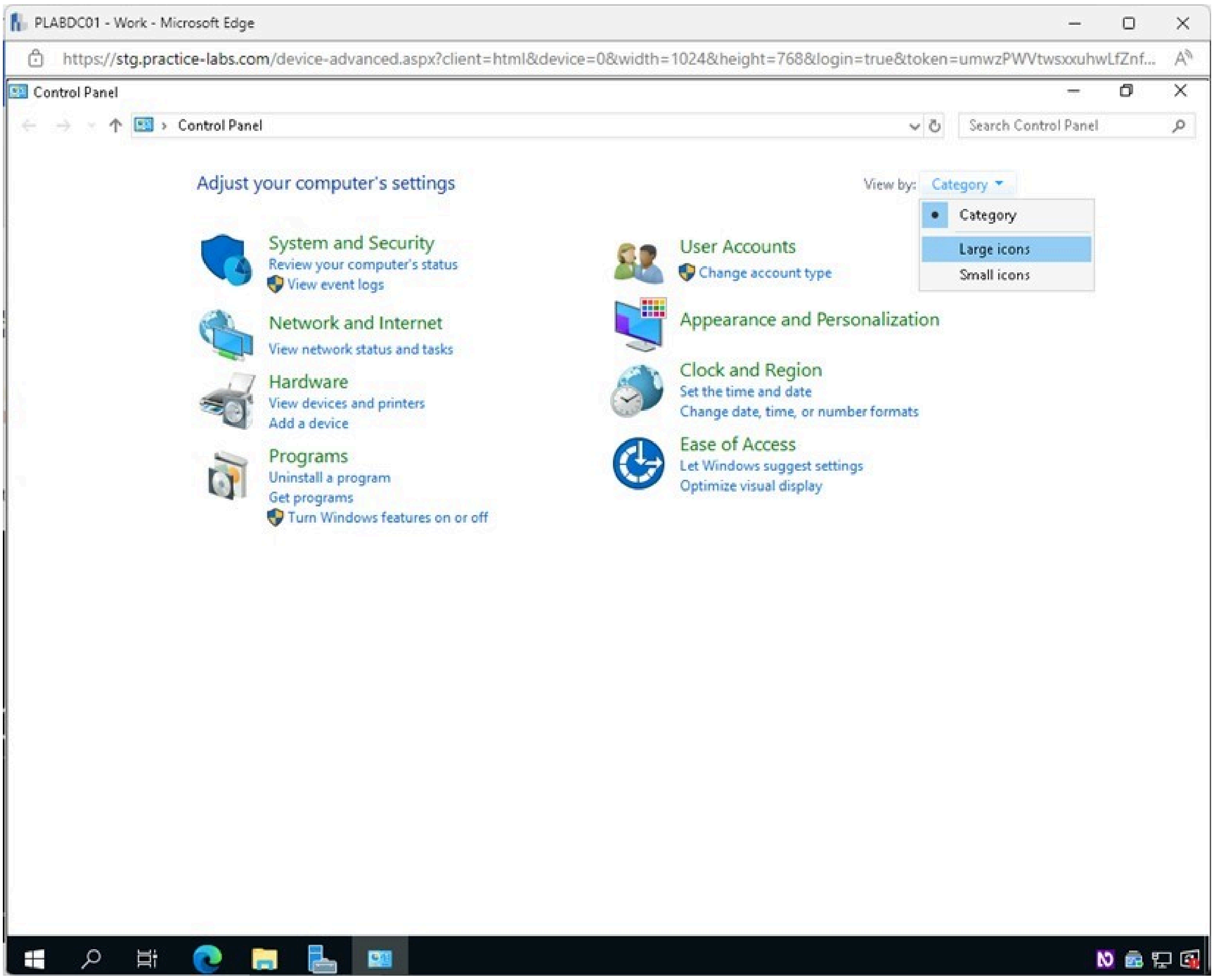

Figure 1.14 Screenshot of PLABDC01: Displaying the Control Panel window. View by drop-down menu is accessed, and Large icons is selected.

Step 3

In the **Control Panel** window, select **Internet Options**.

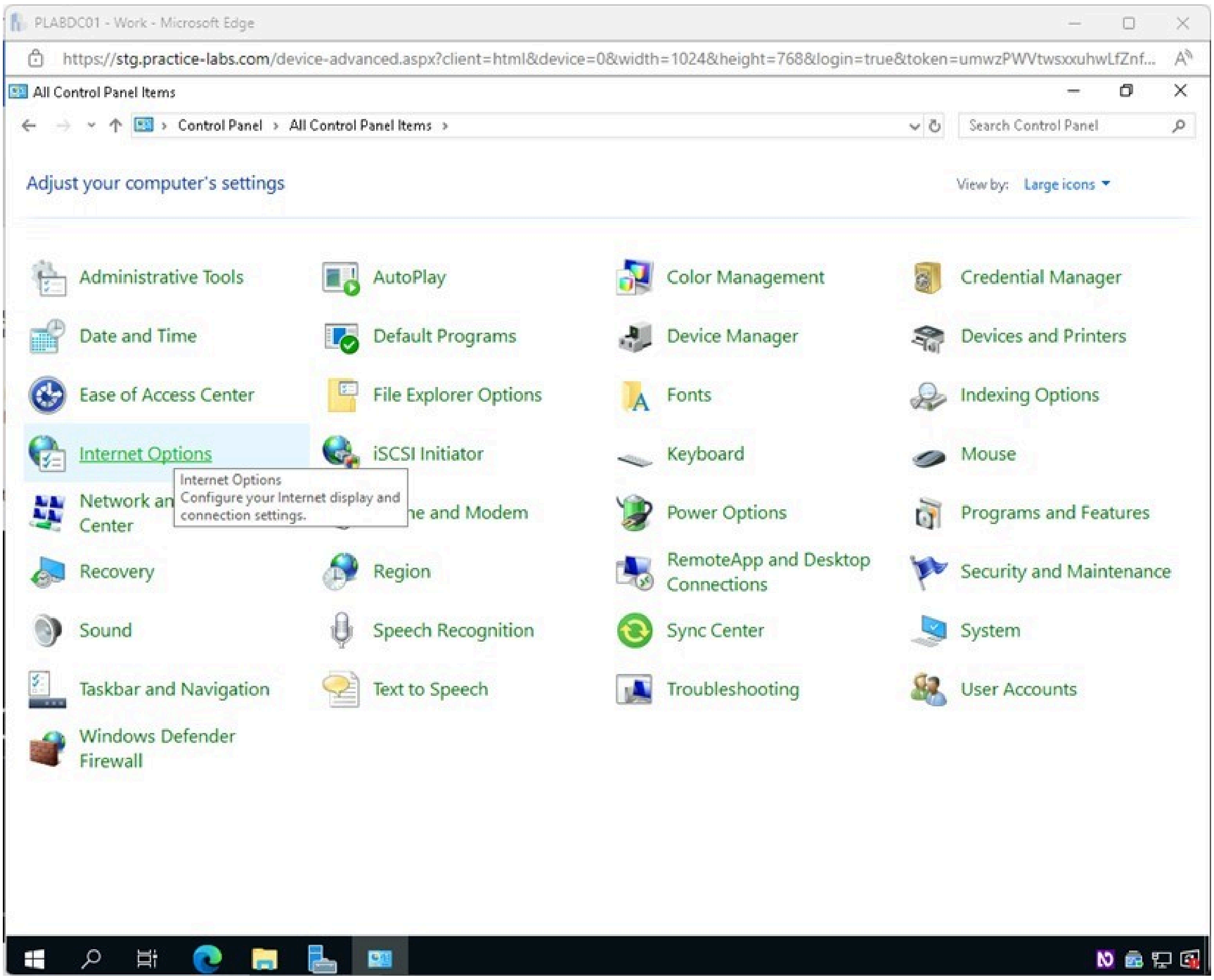

Figure 1.15 Screenshot of PLABDC01: Displaying selecting Internet Options in the Control Panel window.

Step 4

In the **Internet Properties** window, select the **Connections** tab.

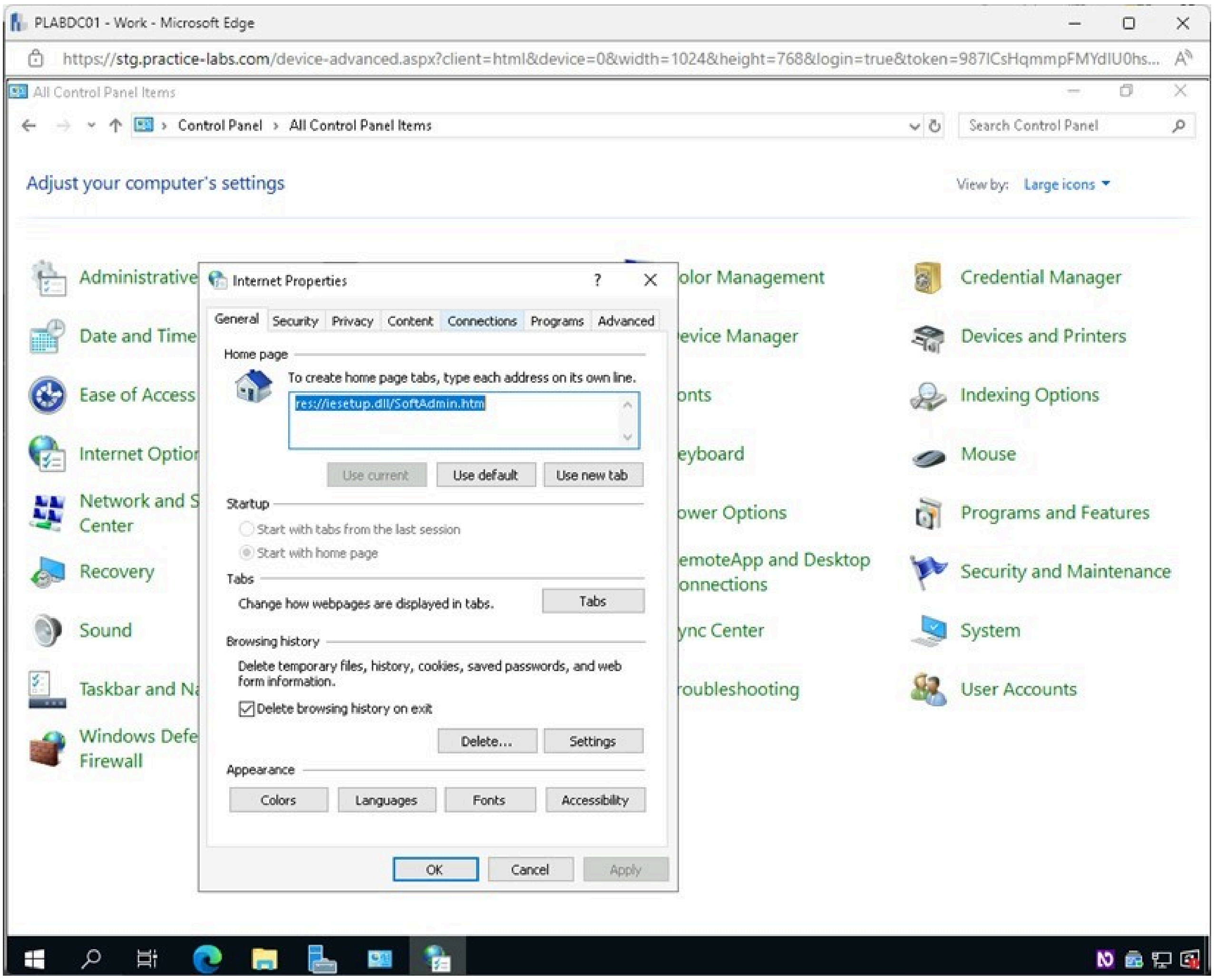

Figure 1.16 Screenshot of PLABDC01: Displaying selecting the Connections tab in the Internet Properties window.

Step 5

In the **Internet Properties - Connections** tab, click **Setup**.

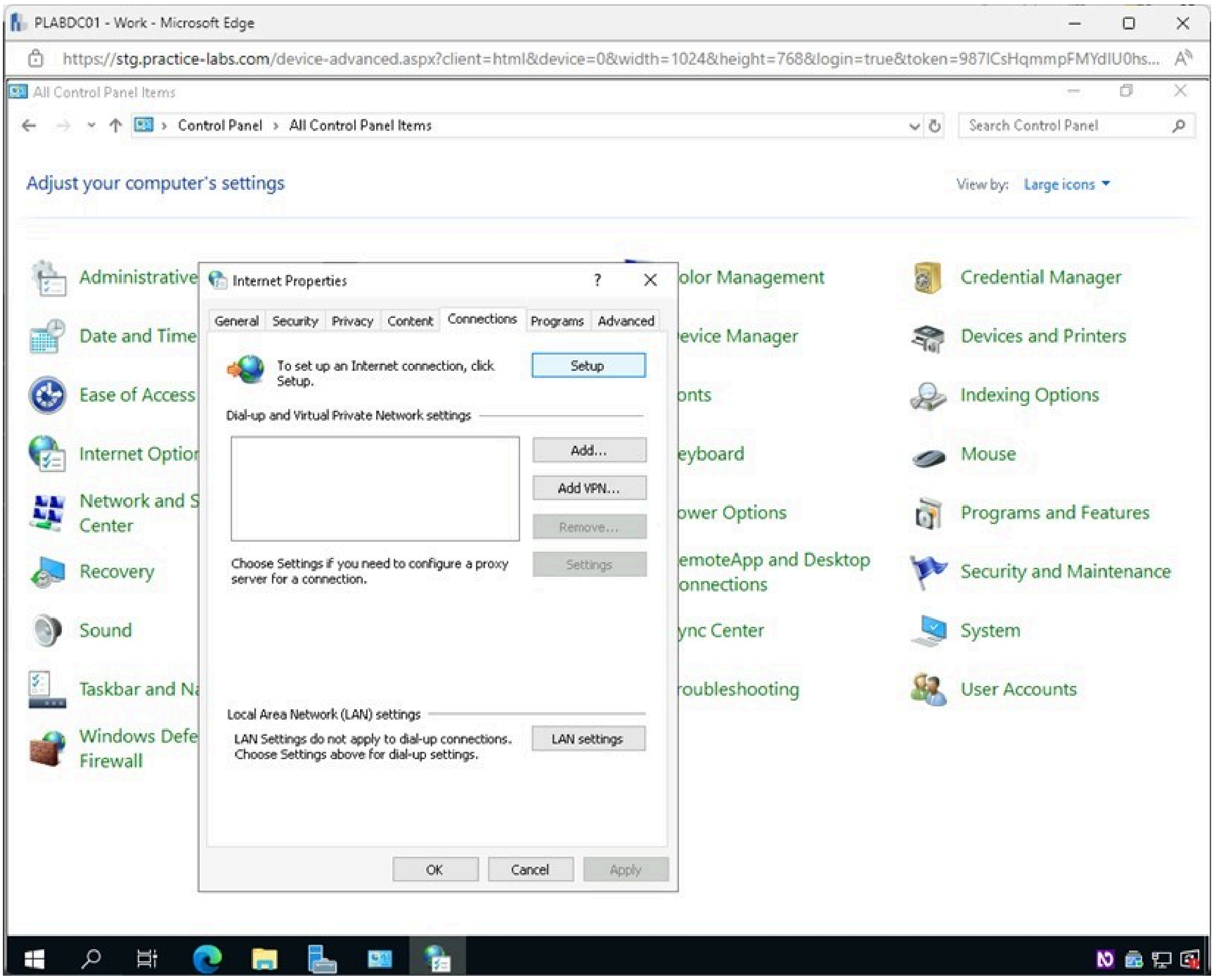

Figure 1.17 Screenshot of PLABDC01: Displaying selecting Setup in the Internet Properties - Connections tab.

Step 6

In the **Connect to the internet** window, notice you are already connected to the internet, so no further troubleshooting is required.

You can, however, see what can be configured for a new connection.

Click **Set up a new connection anyway**.

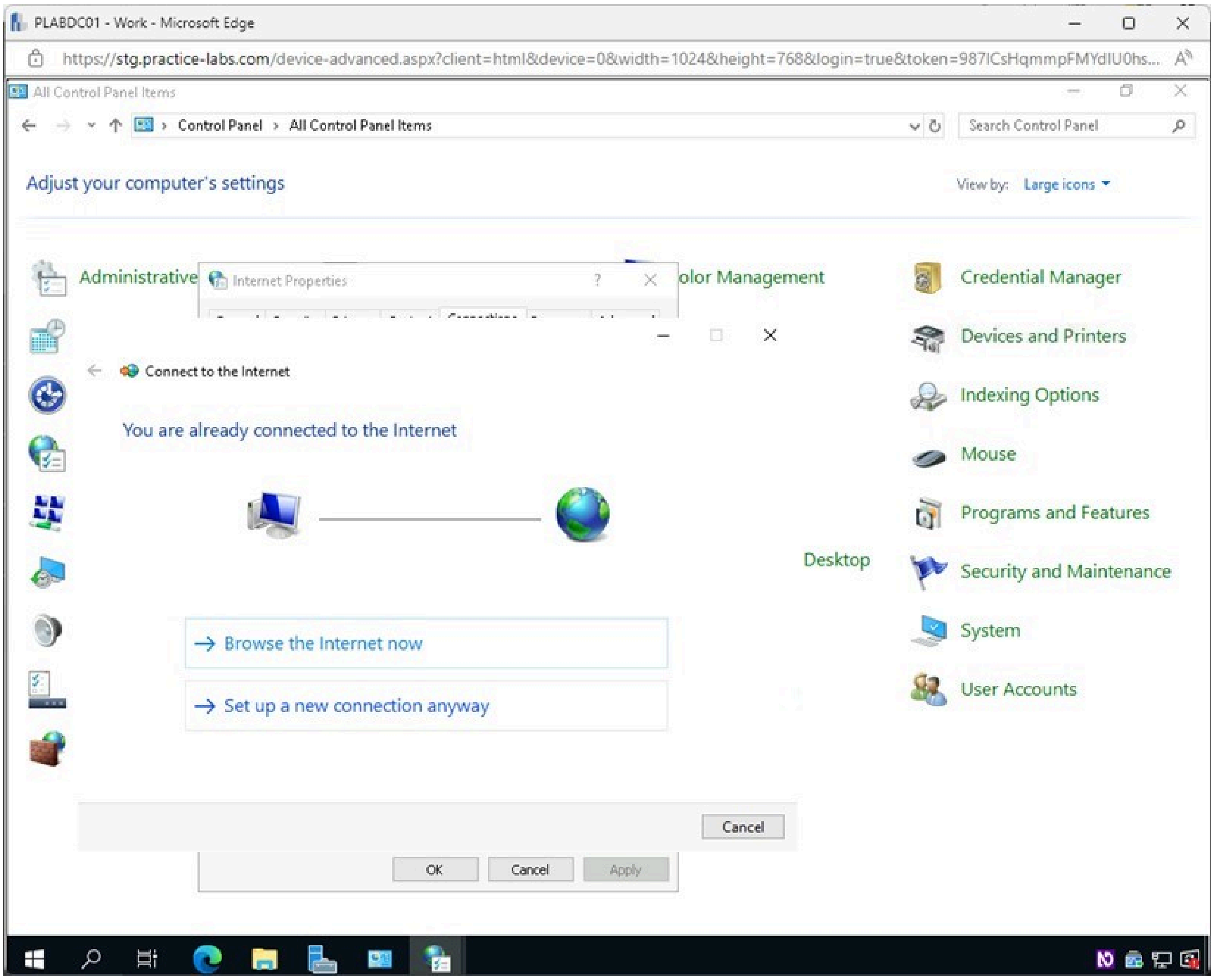

Figure 1.18 Screenshot of PLABDC01: Displaying selecting Set up a new connection anyway in the Connect to the Internet window.

Step 7

In the **Connect to the Internet - How do you want to connect?** page, select **Broadband (PPOE)**.

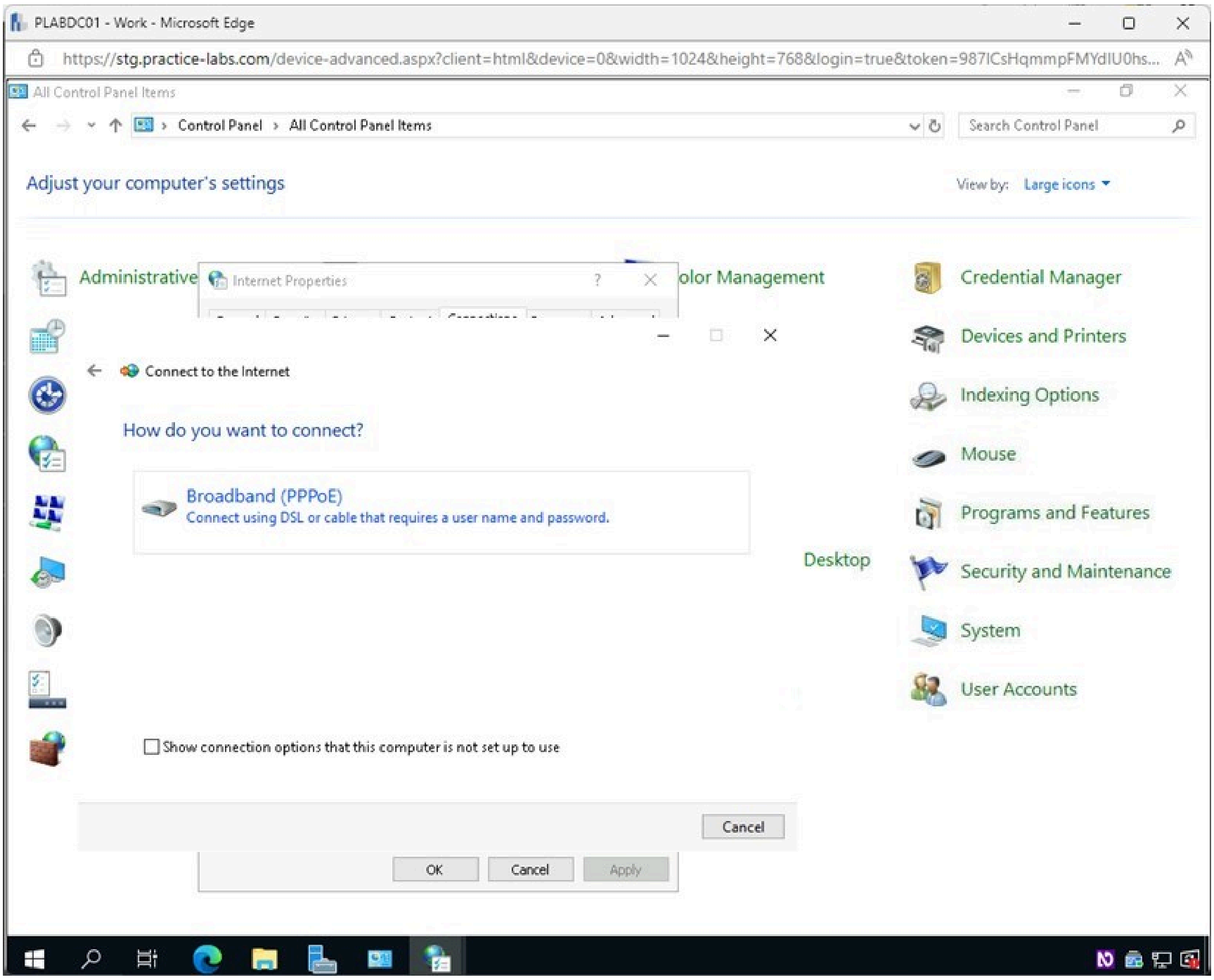

Figure 1.19 Screenshot of PLABDC01: Displaying selecting Broadband (PPOE) in the Connect to the Internet - How do you want to connect? page.

Step 8

In the **Connect to the Internet - Type the information from your Internet service provider (ISP)** page, users can configure the network access.

You will not be making any changes in this window.

Click **Cancel**.

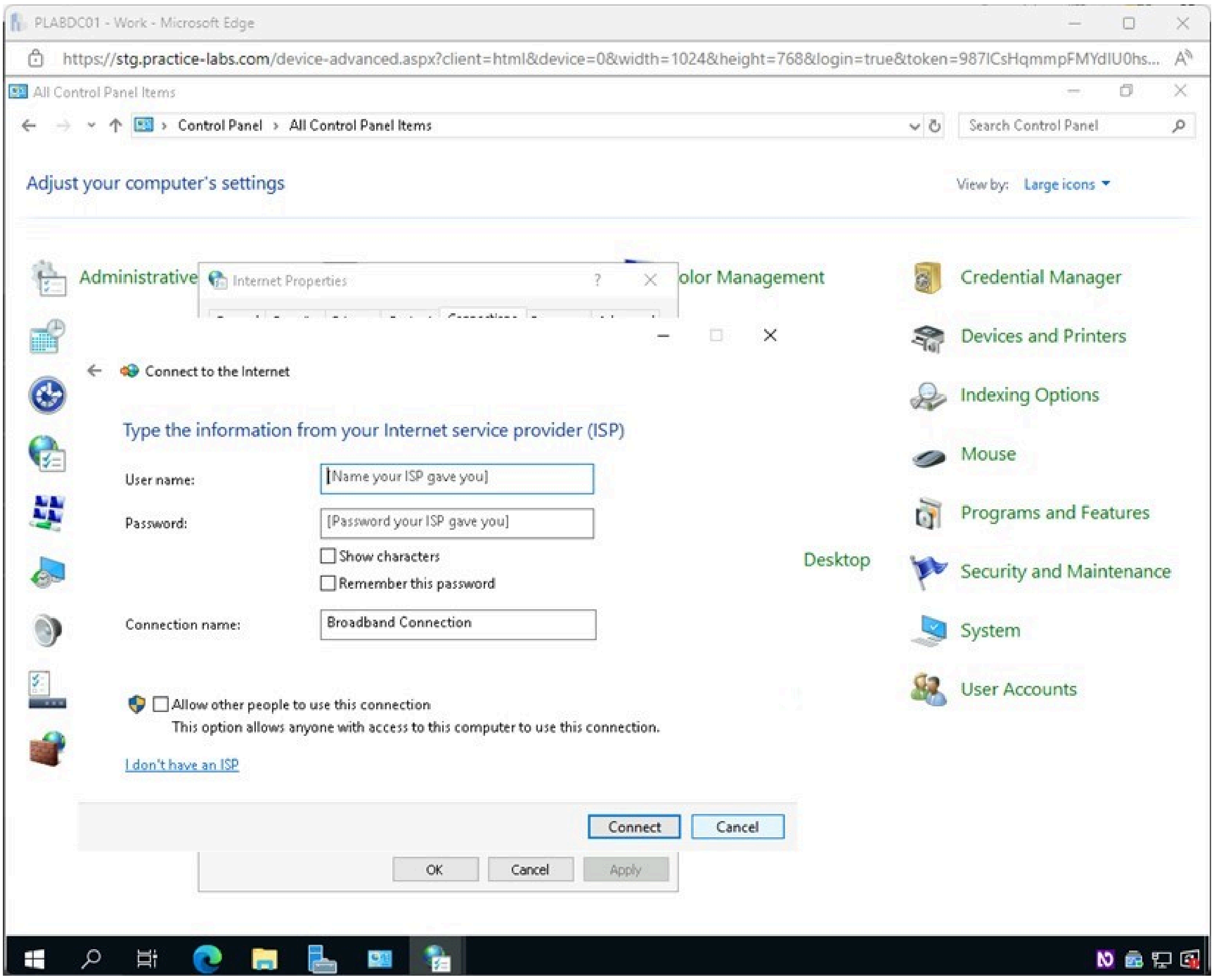

Figure 1.20 Screenshot of PLABDC01: Displaying selecting Cancel in the Connect to the Internet - Type the information from your Internet service provider (ISP) page.

Keep the **Control Panel** window open.

Task 4 - Configure Network Adapters

Network adapters are one of the numerous components used to connect computing devices to the Internet. Devices that can receive data wirelessly can have an antenna or card built into them, or they can use external antennas or USB dongles.

In this task, you will access the interface to configure the network adapter and make configuration changes.

Step 1

In **PLABDC01**, the **Control Panel** window is open.

Click **Network and Sharing Center.**

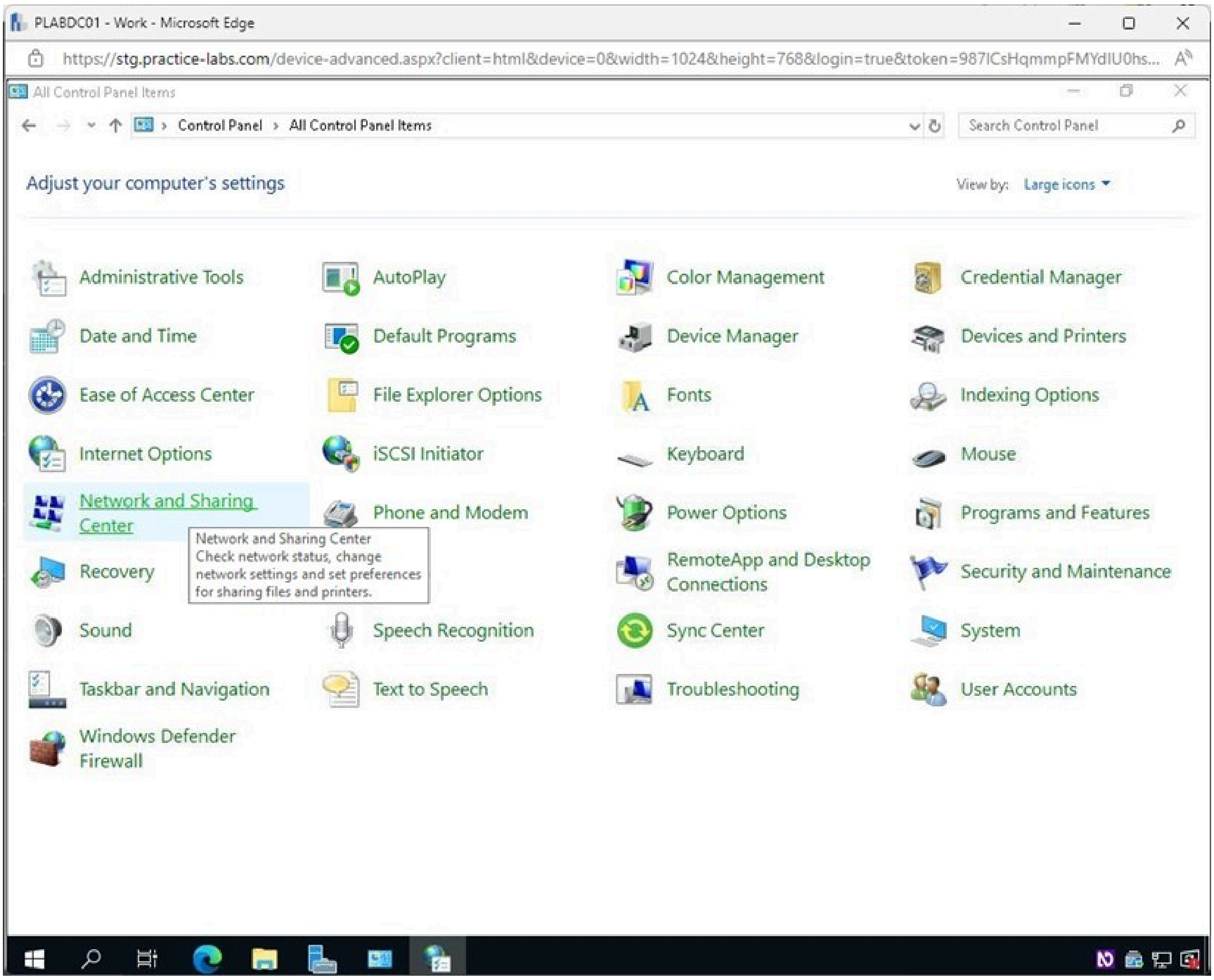

Figure 1.21 Screenshot of PLABDC01: Displaying selecting Network and Sharing Center in the Control Panel window.

Step 2

In the **Network and Sharing Center** window, click the **Change adapter settings** link on the left pane.

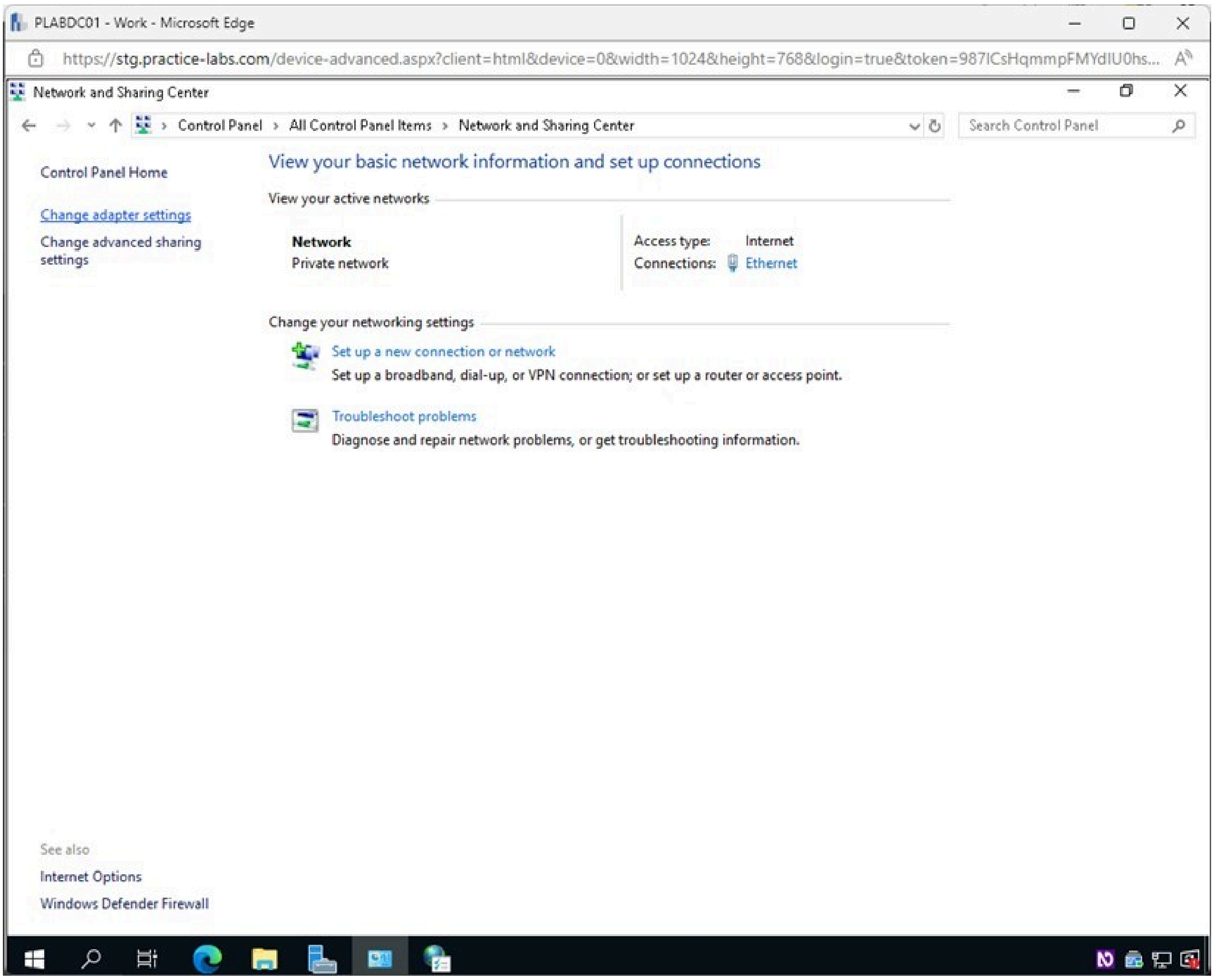

Figure 1.22 Screenshot of PLABDC01: Displaying selecting the Change adapter settings link in the Network and Sharing Center window.

Step 3

In the **Network Connections** window, right-click on **Ethernet** and select **Properties**.

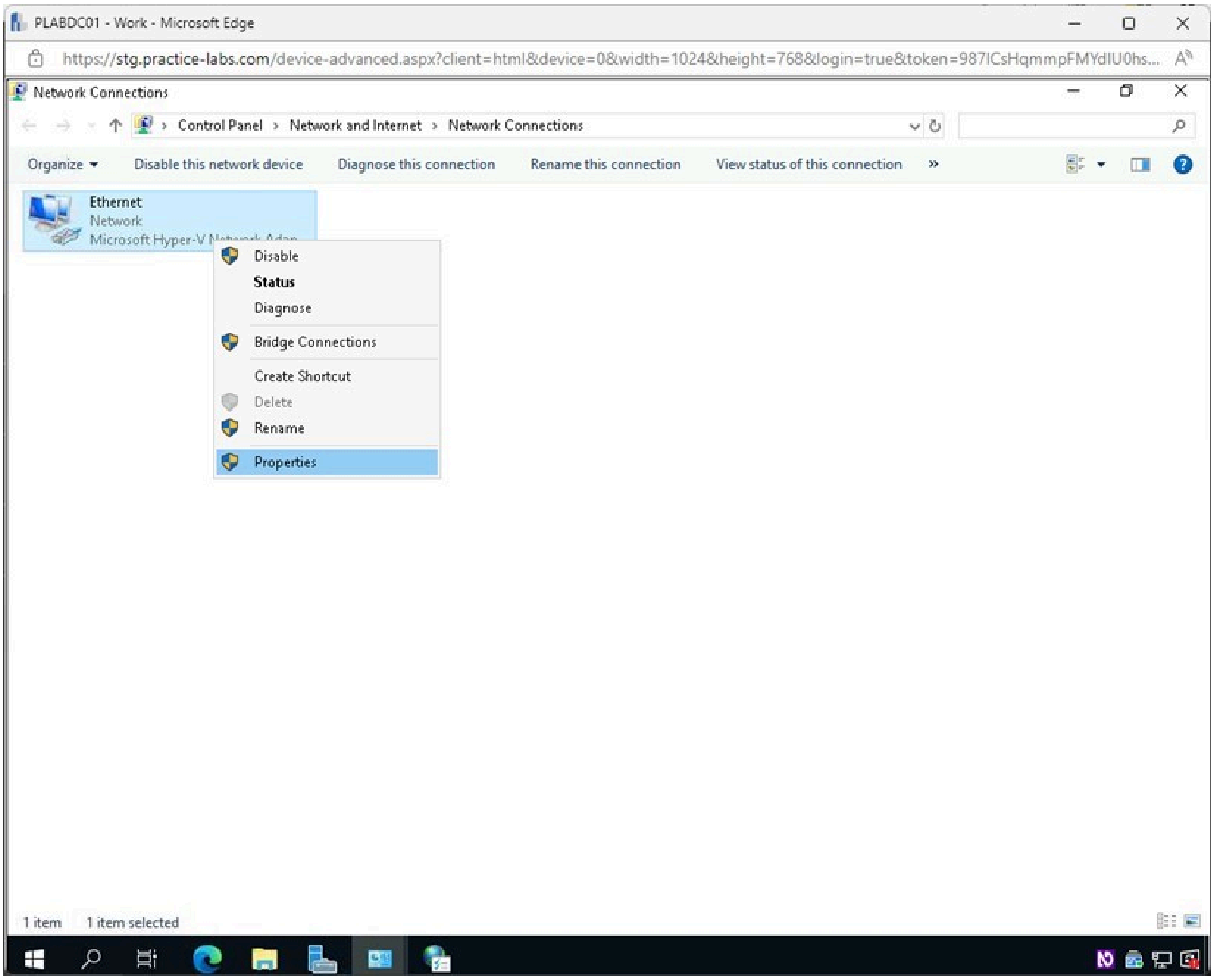

Figure 1.23 Screenshot of PLABDC01: Displaying the Network Connections window. Ethernet is right-clicked, and Properties is selected.

Step 4

In the **Ethernet Properties** window, you can make configuration changes to the network adapter.

Click **Configure**.

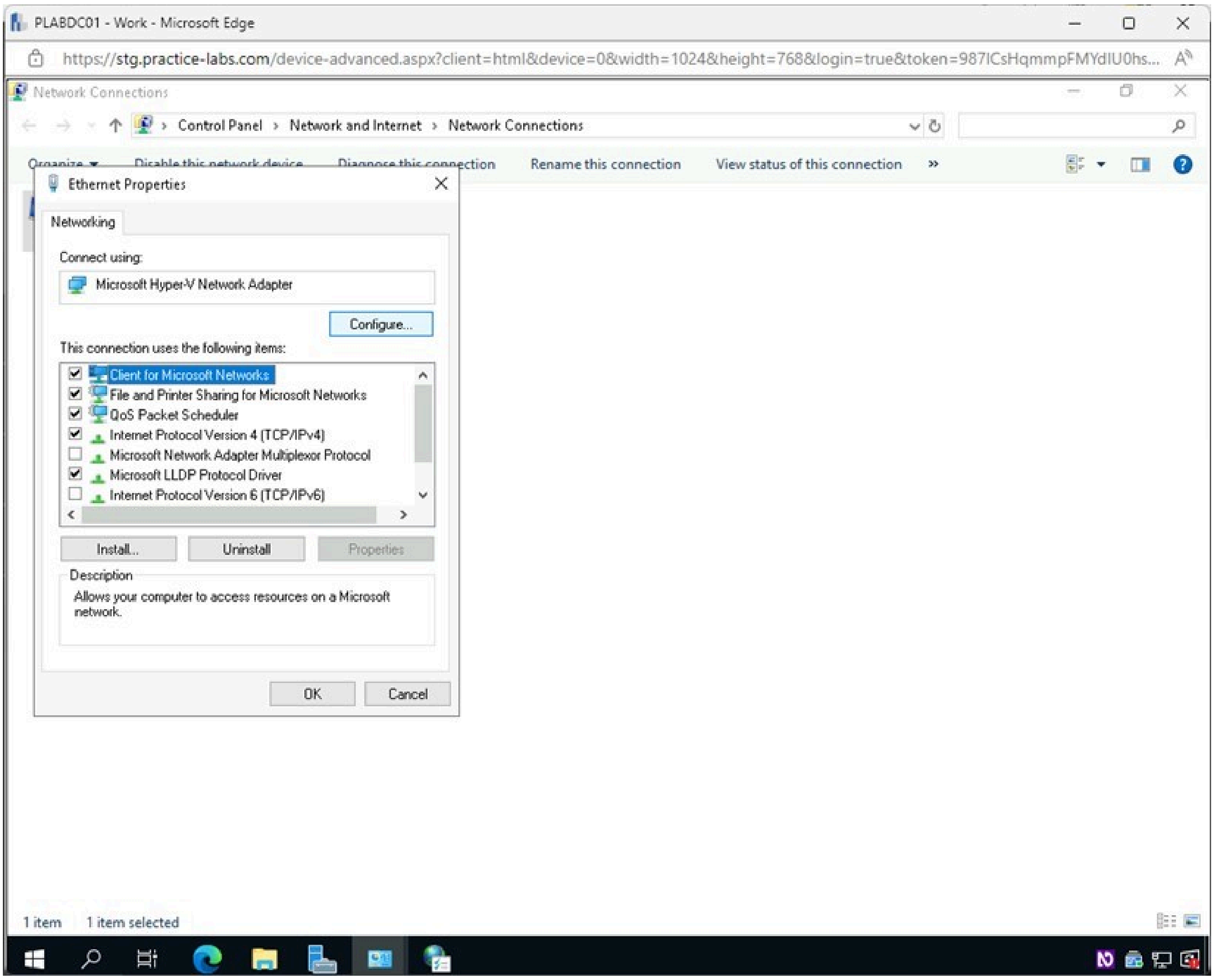

Figure 1.24 Screenshot of PLABDC01: Displaying selecting Configure in the Ethernet Properties window.

Step 5

In the **Microsoft Hyper-V Network Adapter Properties** window, select the **Advanced** tab.

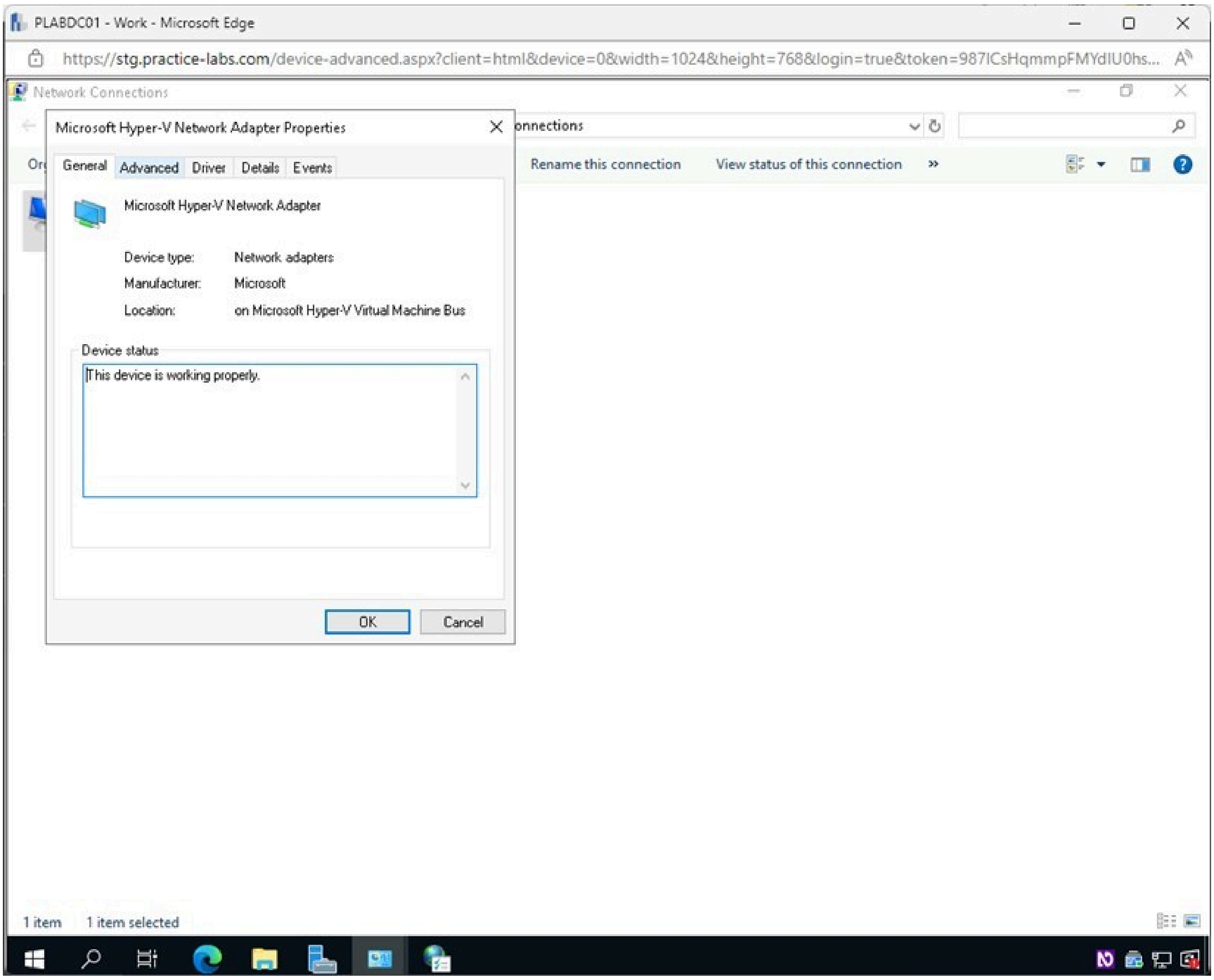

Figure 1.25 Screenshot of PLABDC01: Displaying selecting the Advanced tab in the Microsoft Hyper-V Network Adapter Properties window.

Step 6

In the **Microsoft Hyper-V Network Adapter Properties - Advanced** tab, users can adjust advanced options on the network adapter.

> ***Note:*** *Users should think about making certain adjustments that might boost the network's efficiency. Shut down Microsoft Networks' ability to share files and printers. Increase data transfer using* ***Jumbo Frames****, improve traffic management using* ***Flow Control****, enable* ***Side Scaling*** *to distribute data processing across processors or CPU cores, choose the* ***speed*** *and* ***duplex*** *of the connection, choose* ***buffer sizes*** *used for receiving and transmitting data on the adapter, modify the adapter's* ***interrupt moderation*** *when handling packets.*

Select the **Driver** tab.

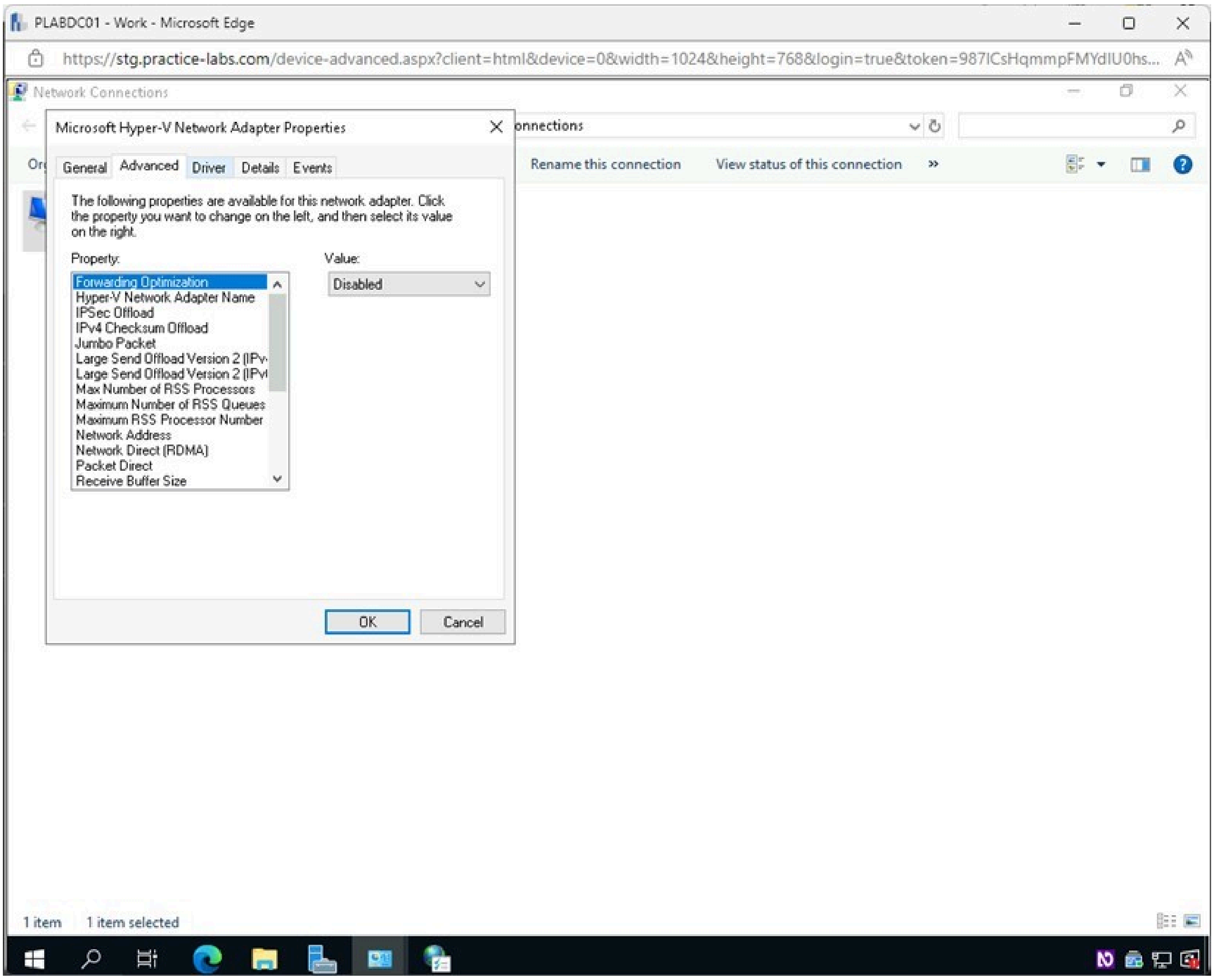

Figure 1.26 Screenshot of PLABDC01: Displaying selecting the Driver tab in the Microsoft Hyper-V Network Adapter Properties window.

Step 7

In the **Driver** tab, click **Update Driver**.

> ***Note:*** *Having up to date drivers is essential for the proper operation of the device, as well as making sure any additional security can be applied.*

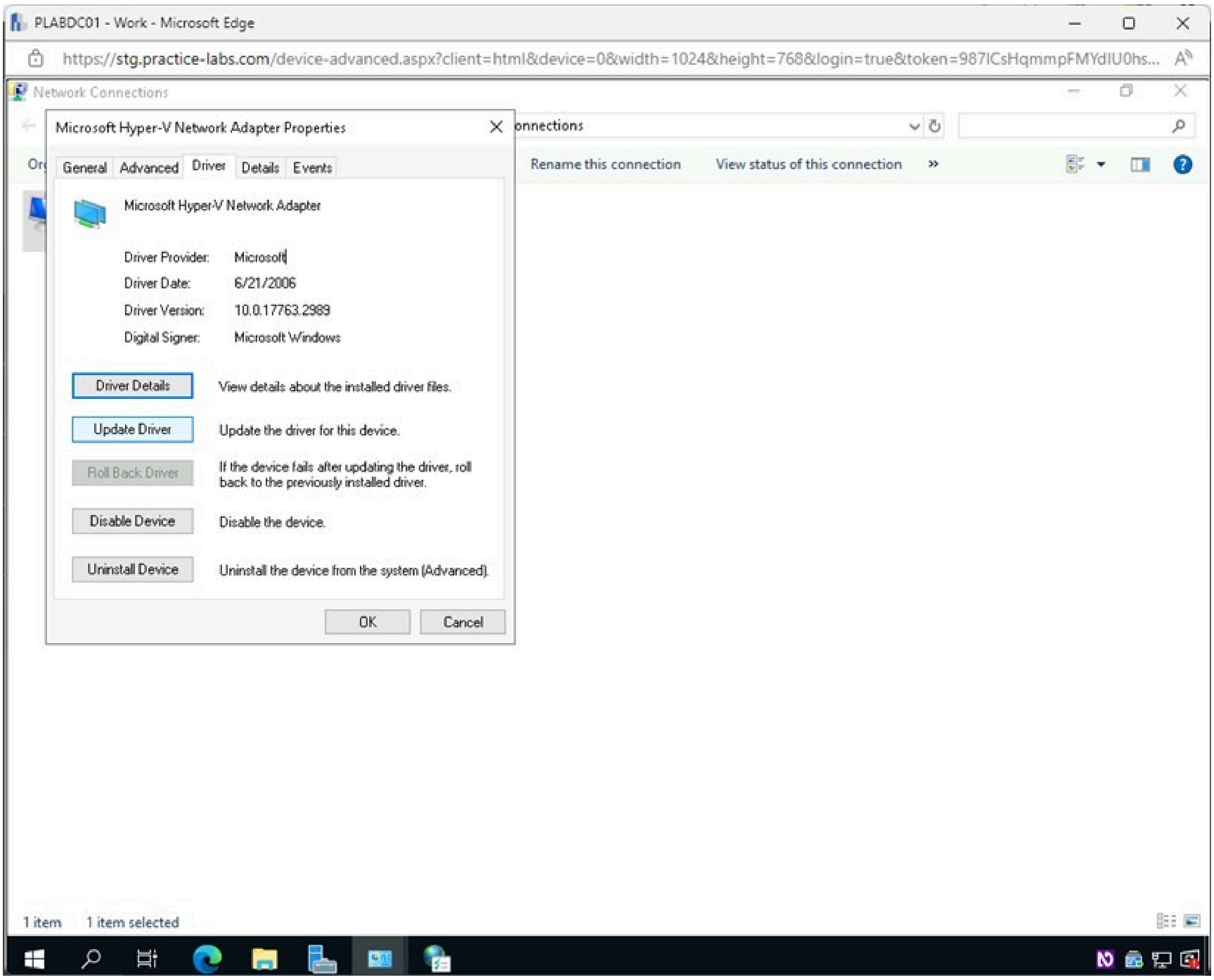

Figure 1.27 Screenshot of PLABDC01: Displaying selecting the Update Driver button in the Microsoft Hyper-V Network Adapter Properties - Driver tab.

Step 8

In the **Update Drivers - Microsoft Hyper-V Network Adapter - How do you want to search for drivers?** window, select **Search automatically for updated driver software**.

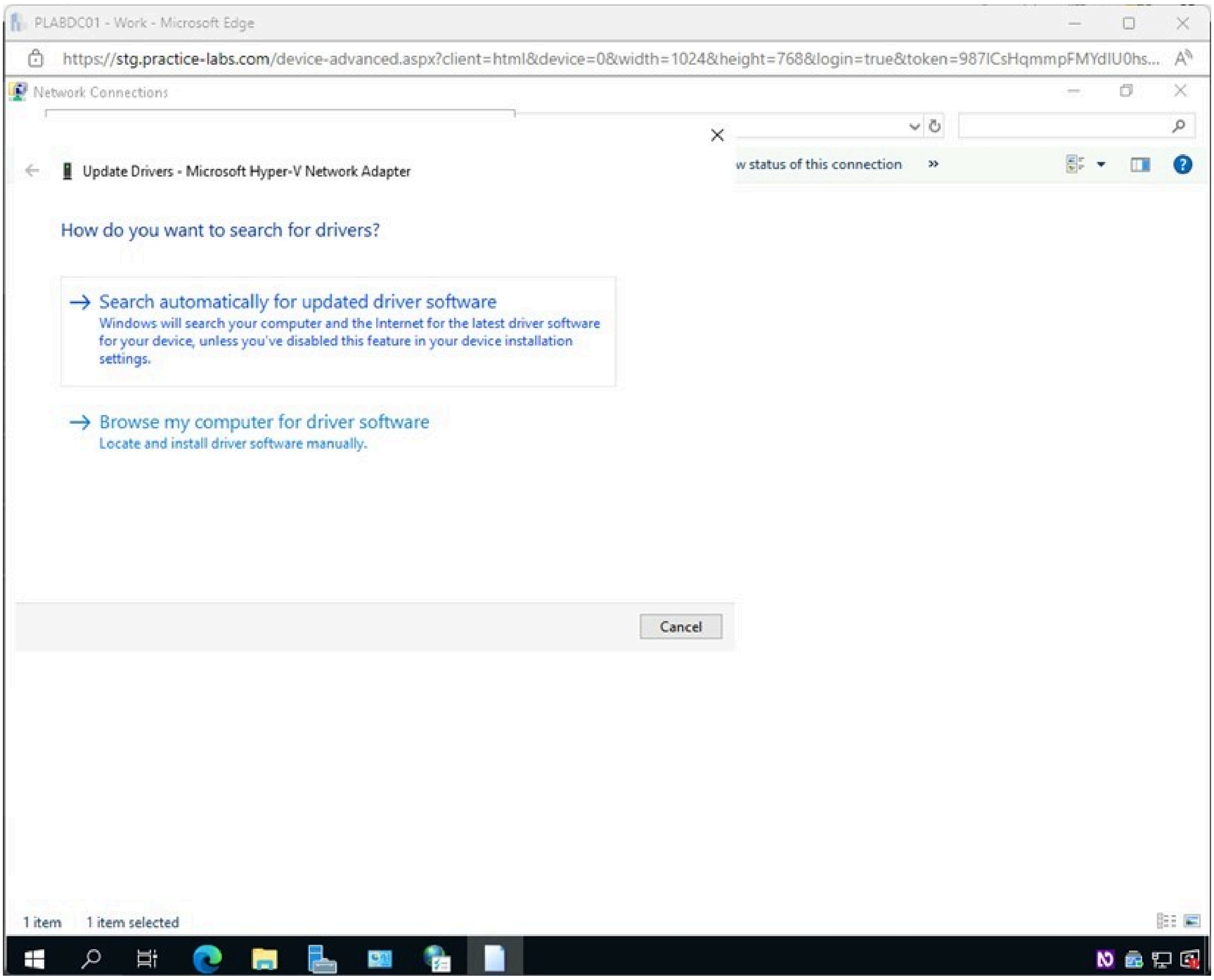

Figure 1.28 Screenshot of PLABDC01: Displaying selecting Search automatically for updated driver software in the Update Drivers - Microsoft Hyper-V Network Adapter - How do you want to search for drivers? window.

Step 9

In the **Update Drivers - Microsoft Hyper-V Network Adapter** window, notice that **The best drivers for your drivers are already installed** message.

Click **Close**.

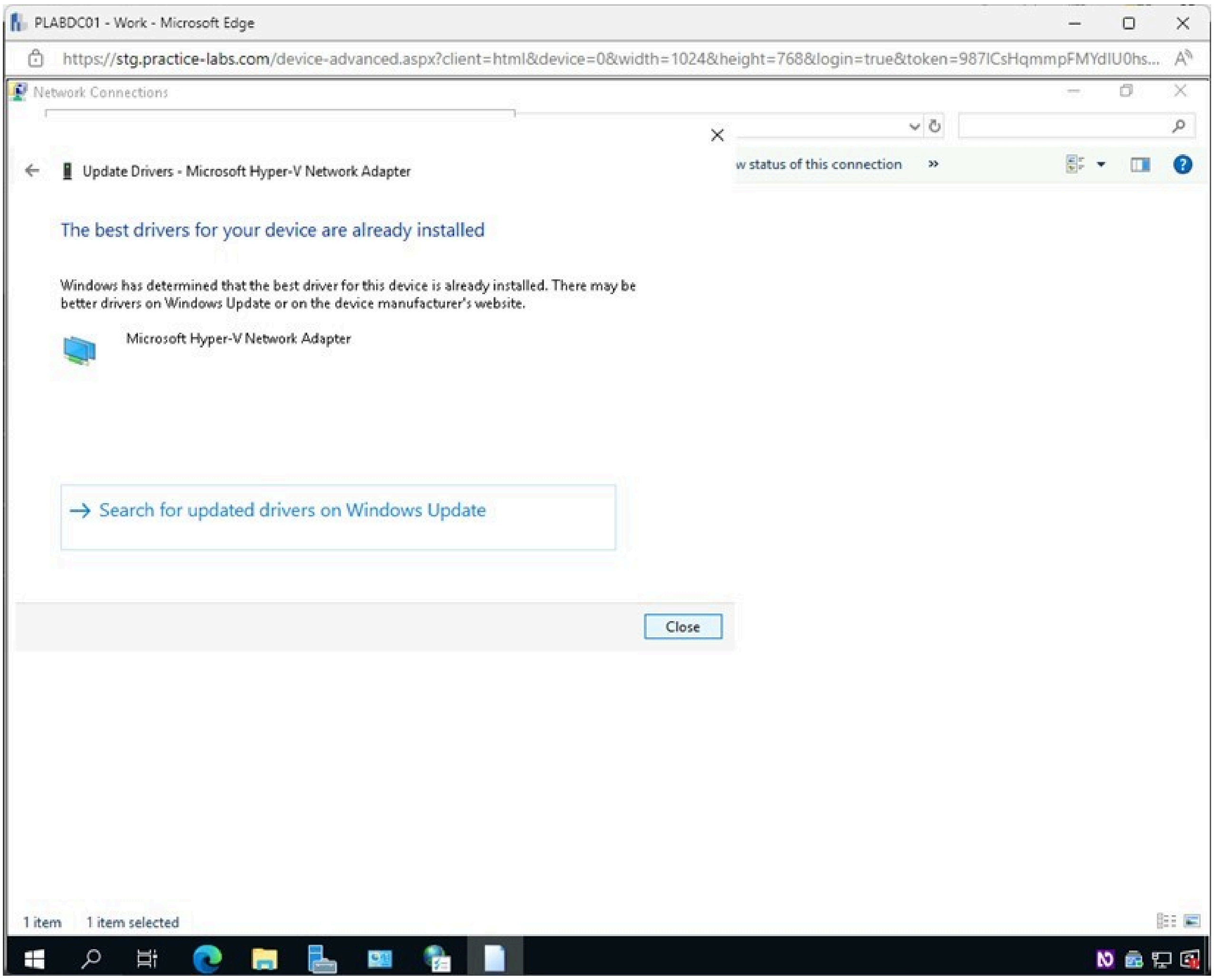

Figure 1.29 Screenshot of PLABDC01: Displaying selecting Close in the Update Drivers - Microsoft Hyper-V Network Adapter window.

Step 10

Back on the **Microsoft Hyper-V Network Adapter Properties - Driver** tab, click **Close**.

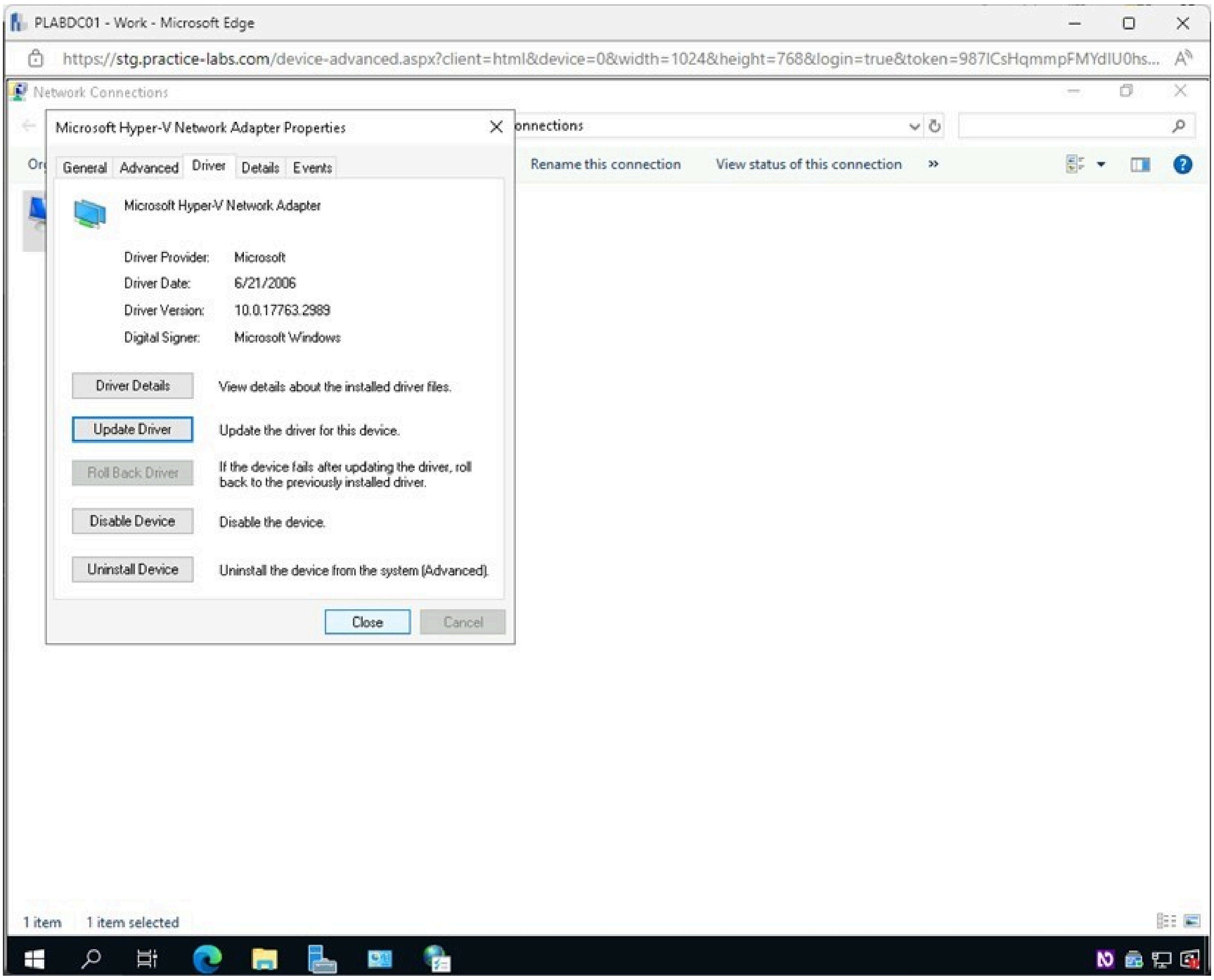

Figure 1.30 Screenshot of PLABDC01: Displaying selecting Close in the Microsoft Hyper-V Network Adapter Properties - Driver tab.

Keep the **Network Connections** window open.

Task 5 - Configure Network Adapters IPv4 and IPv6 Properties

The network adapter allows users to configure settings for IPv4 and IPv6, like using a DHCP Server or statically setting the device's IP address. You can also set the DNS and WINS server's address and configure backup servers.

In this task, you will add a new DNS server and remove the loopback address. This will also allow you to use tools like nslookup in the next exercise.

Step 1

Ensure you are connected to **PLABDC01** and the **Network Connections** window is open.

Right-click on **Ethernet** and select **Properties**.

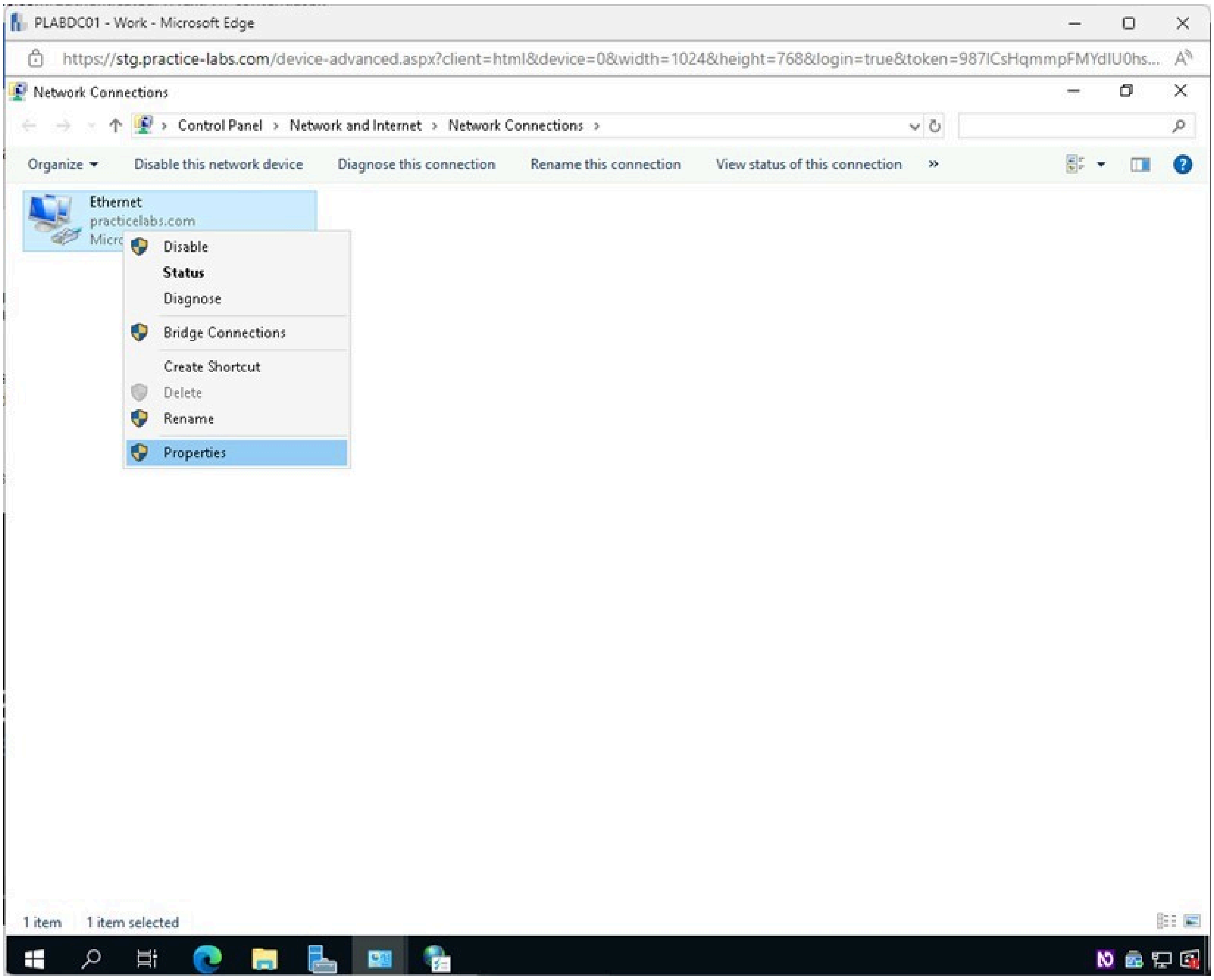

Figure 1.31 Screenshot of PLABDC01: Displaying the Network Connections window. Ethernet is right-clicked, and Properties is selected.

Step 2

In the **Ethernet Properties** window, select **Internet Protocol Version 4 (TCP/IPv4)** and then click **Properties**.

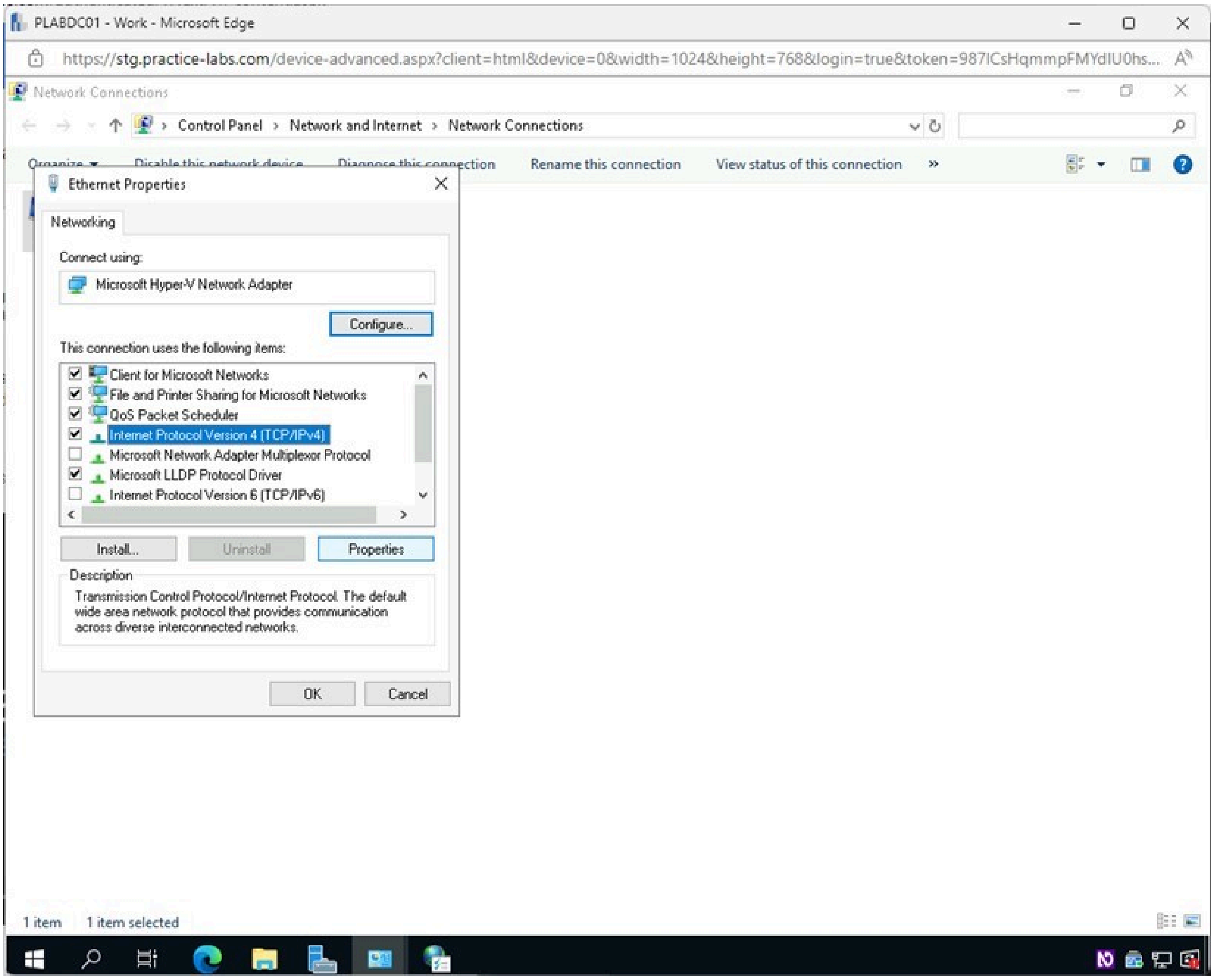

Figure 1.32 Screenshot of PLABDC01: Displaying the Ethernet Properties window. Internet Protocol Version 4 (TCP/IPv4) is selected, and the Properties button is highlighted.

Step 3

In the **Internet Protocol Version 4 (TCP/IPv4) Properties** window, click **Advanced**.

> ***Note:*** *Here, you can see a **Static IP address** of **192.168.0.1** is being used for this machine. The **Subnet mask** is **255.255.255.0,** which will provide the network with **256** total IP addresses. The **Default gateway** is **192.168.0.250,** which is the address of the router that will let you out of the LAN. To get the computer's IP address from the **DHCP Server**, you would select the **Obtain an IP address automatically** option. Notice the **IP address** of **127.0.0.1** for the **DNS server.** This is a loopback address and will not return any results for your queries.*

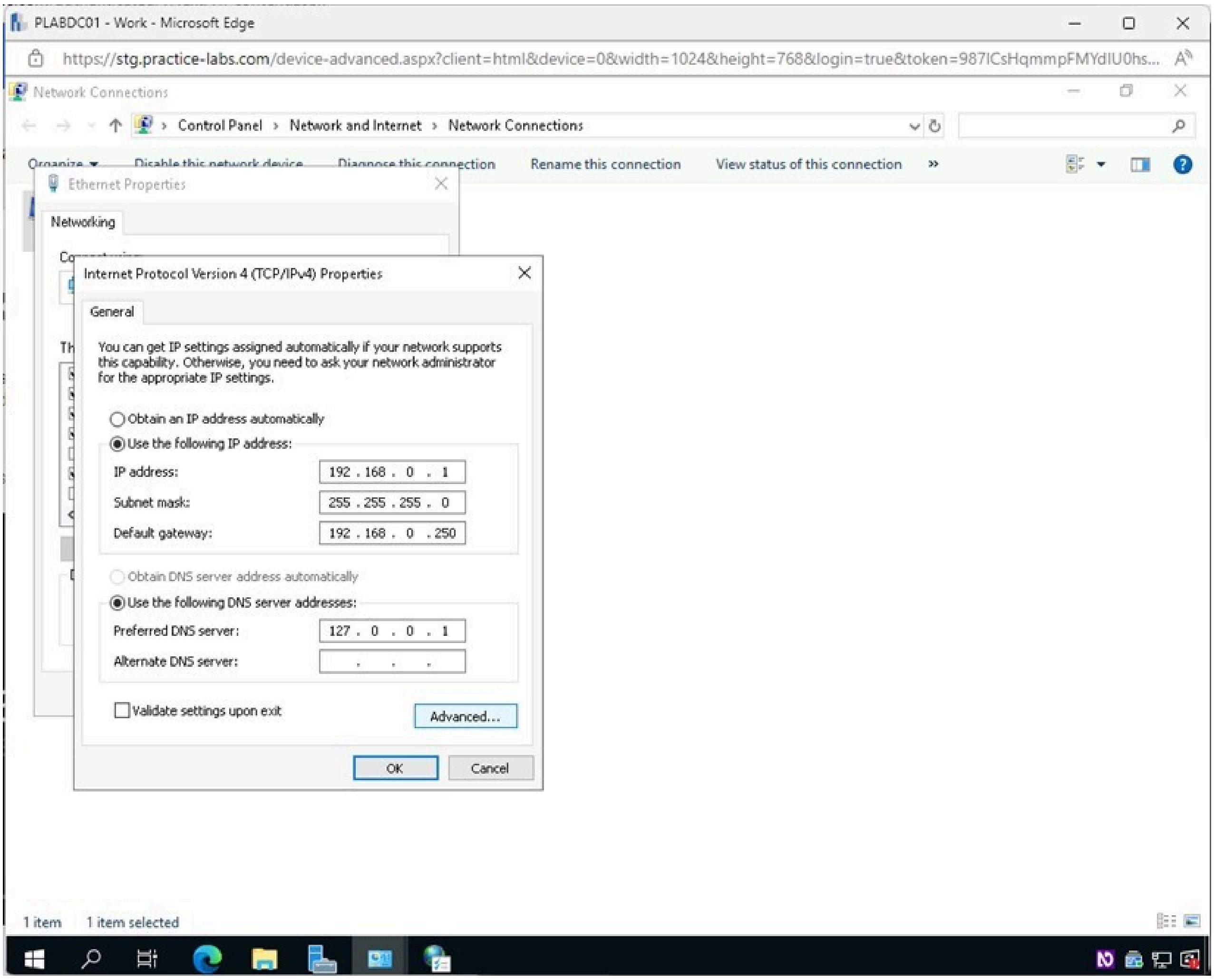

Figure 1.33 Screenshot of PLABDC01: Displaying selecting Advanced in the Internet Protocol Version 4 (TCP/IPv4) Properties window.

Step 4

In the **Advanced TCP/IP Settings** window, select the **DNS** tab.

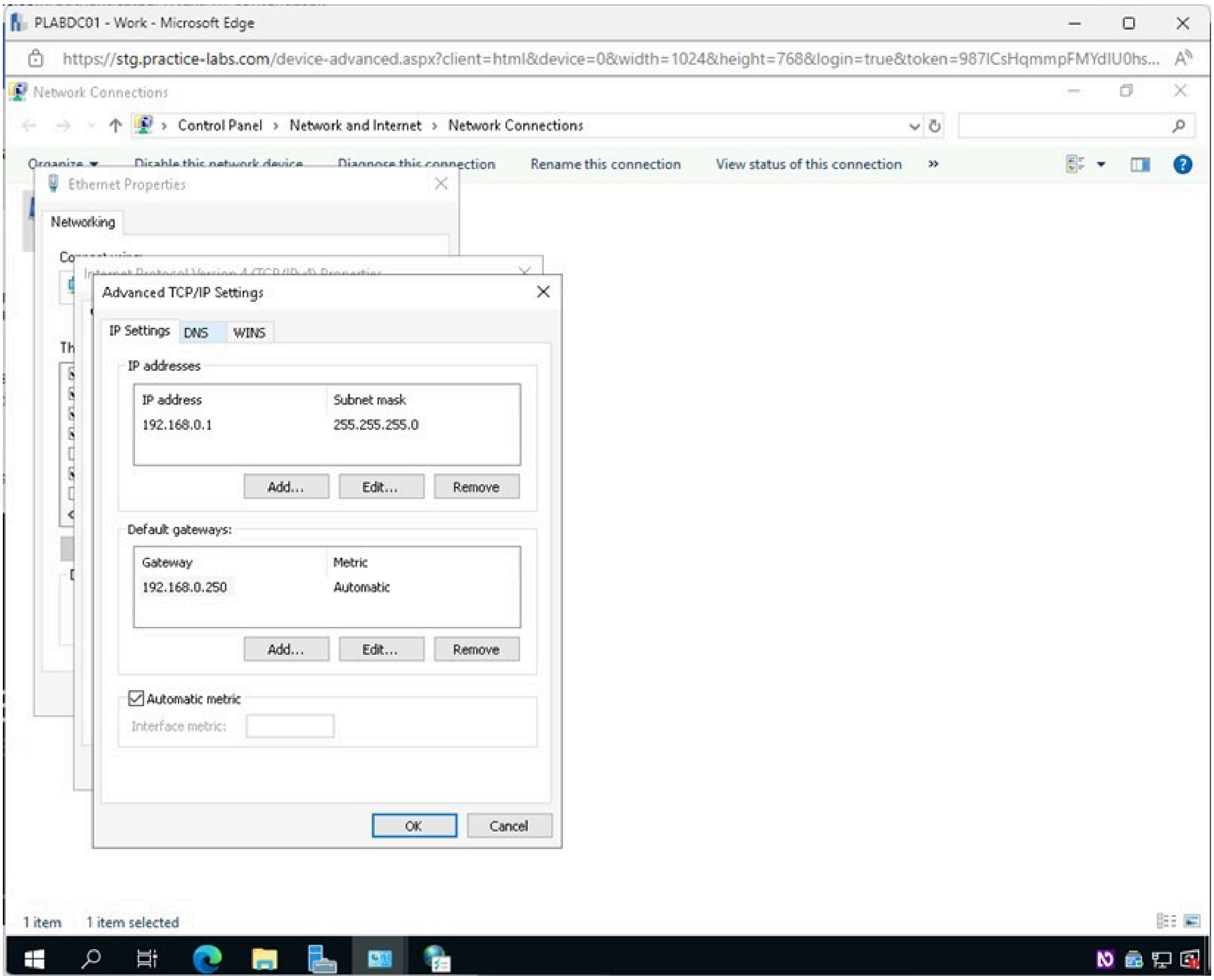

Figure 1.34 Screenshot of PLABDC01: Displaying selecting the DNS tab in the Advanced TCP/IP Settings window.

Step 5

In the **Advanced TCP/IP Settings - DNS** tab, click **Add**.

> ***Note:*** *Here, you can add and remove DNS Servers and create alternate servers to be used when the preferred is not functioning.*

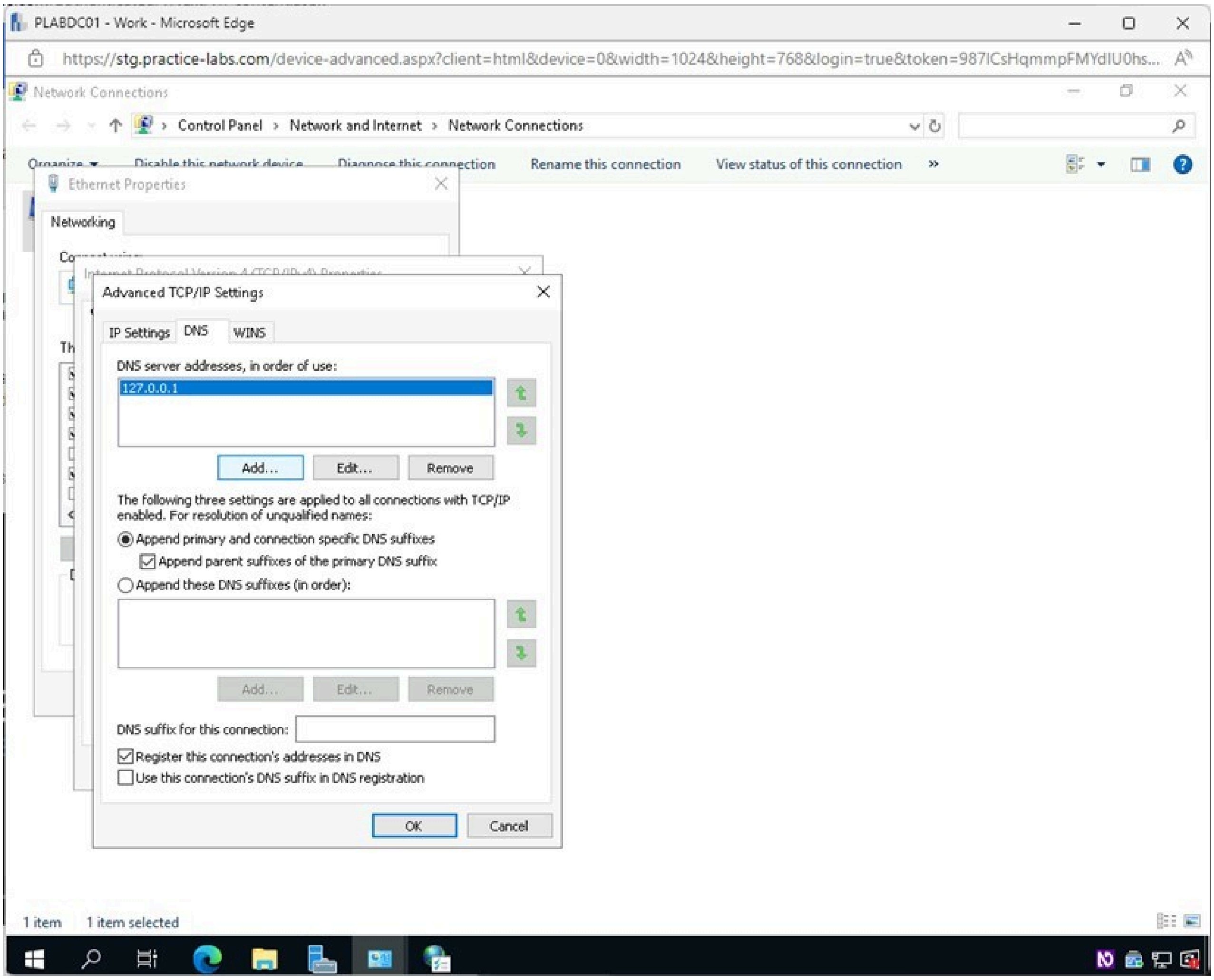

Figure 1.35 Screenshot of PLABDC01: Displaying selecting Add in the Advanced TCP/IP Settings - DNS tab.

Step 6

In the **TCP/IP DNS Server** pop-up window, type the following for the **DNS server** field:

```
8.8.8.8
```

Click **Add**.

> ***Note:*** *The above address entered is Google's Public DNS Server address.*

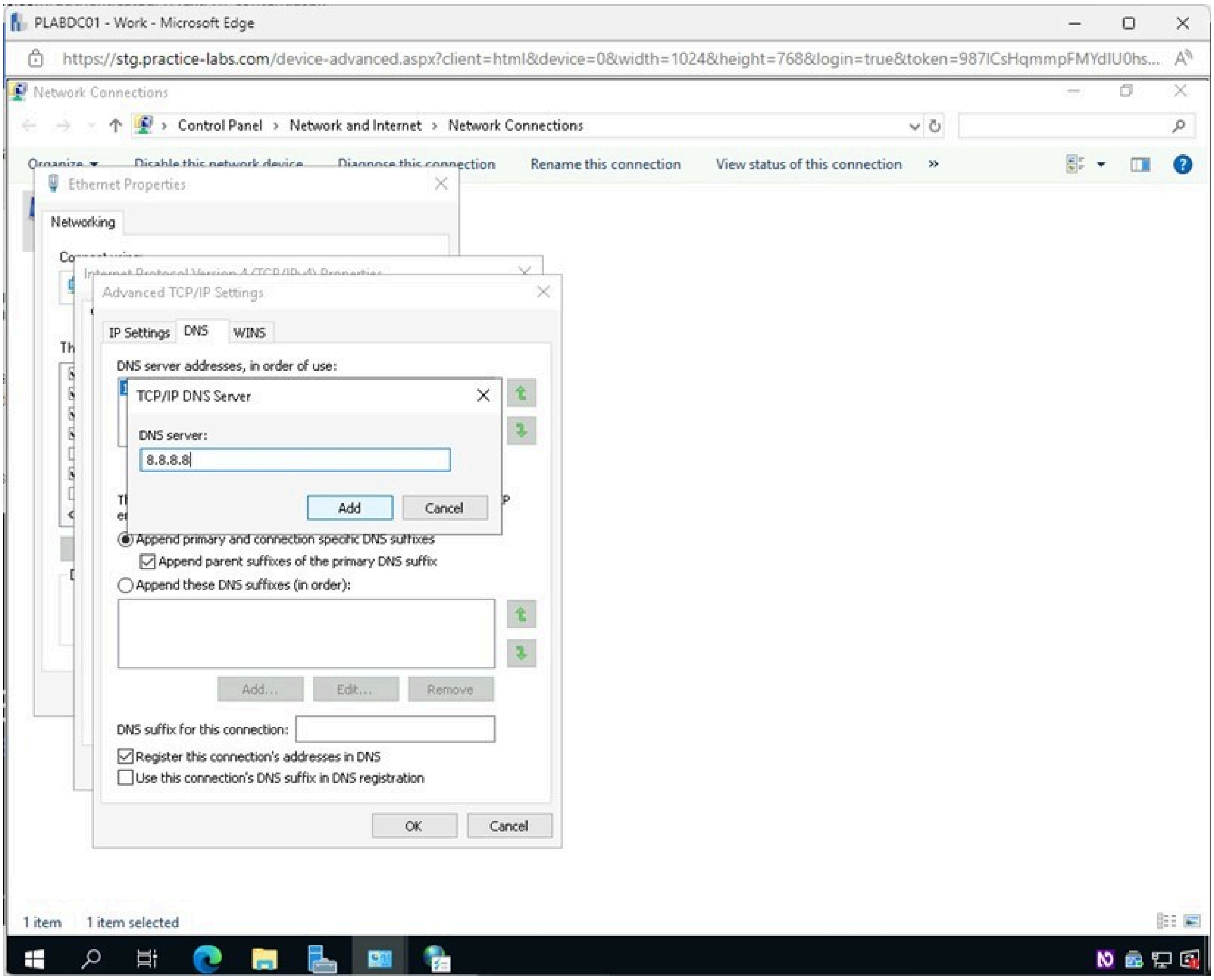

Figure 1.36 Screenshot of PLABDC01: Displaying the TCP/IP DNS Server pop-up window, showing the required settings performed and Add selected.

Step 7

In the **Advanced TCP/IP Settings** window, notice that Google's Public DNS Server address is not the preferred DNS Server. It can act as an alternate DNS Server in case of failure.

You will now remove the **loopback address** that has been acting as the **DNS Server** and make Google's Public DNS Server address the primary DNS Server.

Select **127.0.0.1** and then click **Remove.**

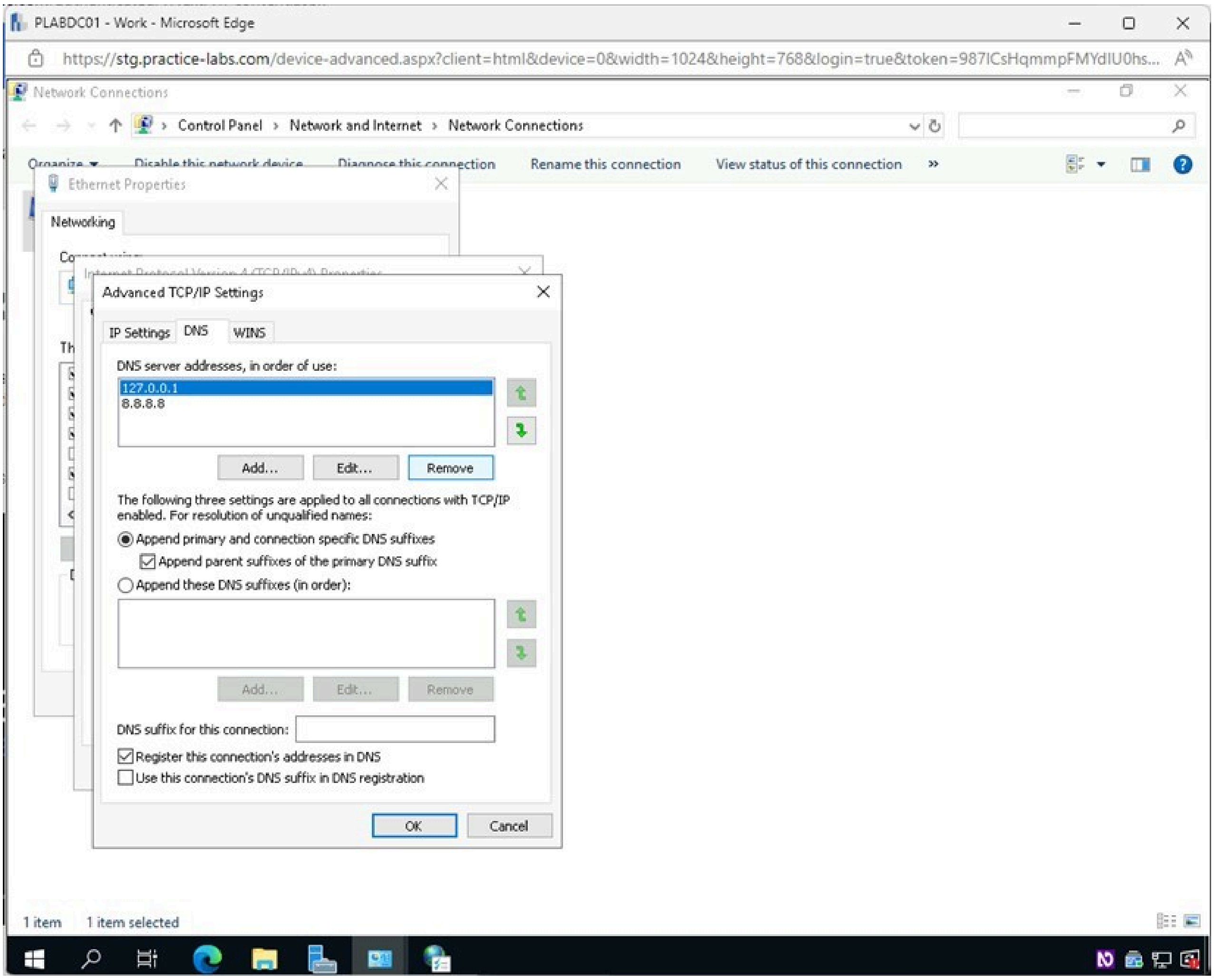

Figure 1.37 Screenshot of PLABDC01: Displaying the Advanced TCP/IP Settings window. The required DNS Server address is selected and Remove highlighted.

Step 8

In the **Advanced TCP/IP Settings** window, Google’s Public DNS Server is now the primary DNS Server.

Click **OK.**

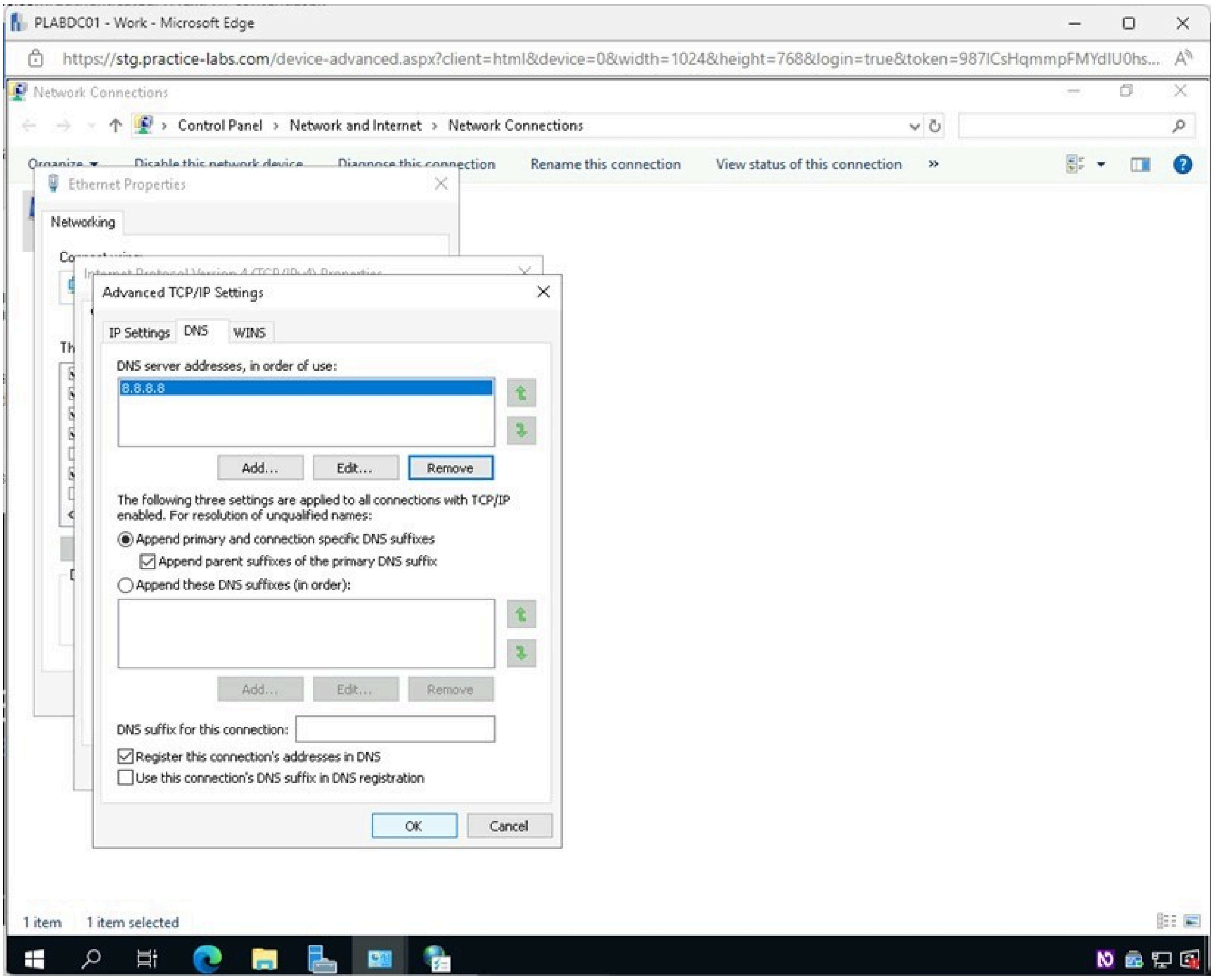

Figure 1.38 Screenshot of PLABDC01: Displaying selecting OK in the Advanced TCP/IP Settings window.

Step 9

Back on the **Internet Protocol Version 4 (TCP/IPv4) Properties** window, click **OK**.

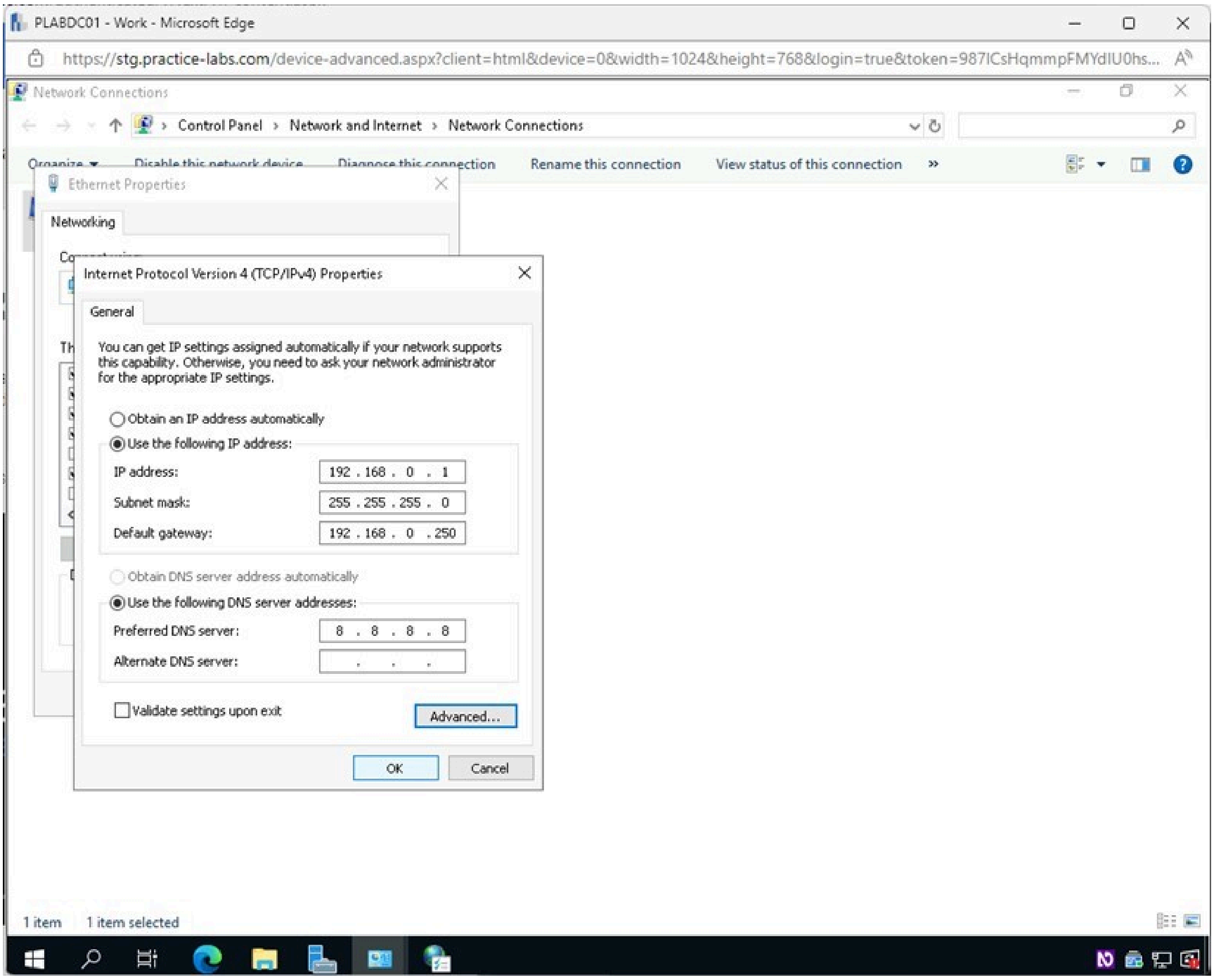

Figure 1.39 Screenshot of PLABDC01: Displaying selecting OK in the Internet Protocol Version 4 (TCP/IPv4) Properties window.

Step 10

Back on the **Ethernet Properties** window, tick the checkbox next to **Internet Protocol Version 6 (TCP/IPv6).**

Click **Properties**.

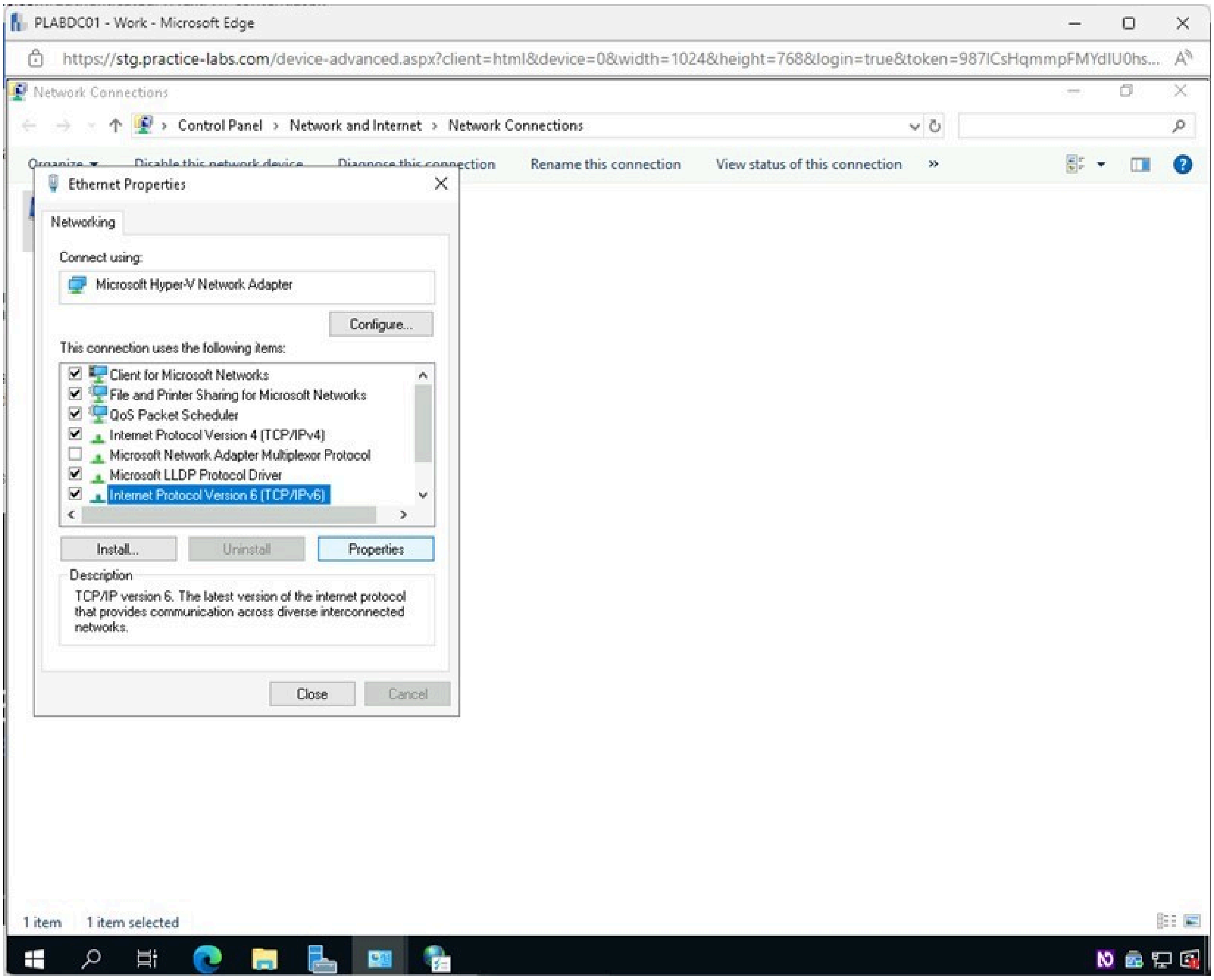

Figure 1.40 Screenshot of PLABDC01: Displaying the Ethernet Properties window. Internet Protocol Version 6 (TCP/IPv6) is selected, and Properties is highlighted.

Step 11

In the **Internet Protocol Version 6 (TCP/IPv6) Properties** window, notice the same settings that were configured for IPv4 can be configured for IPv6.

Click **Cancel.**

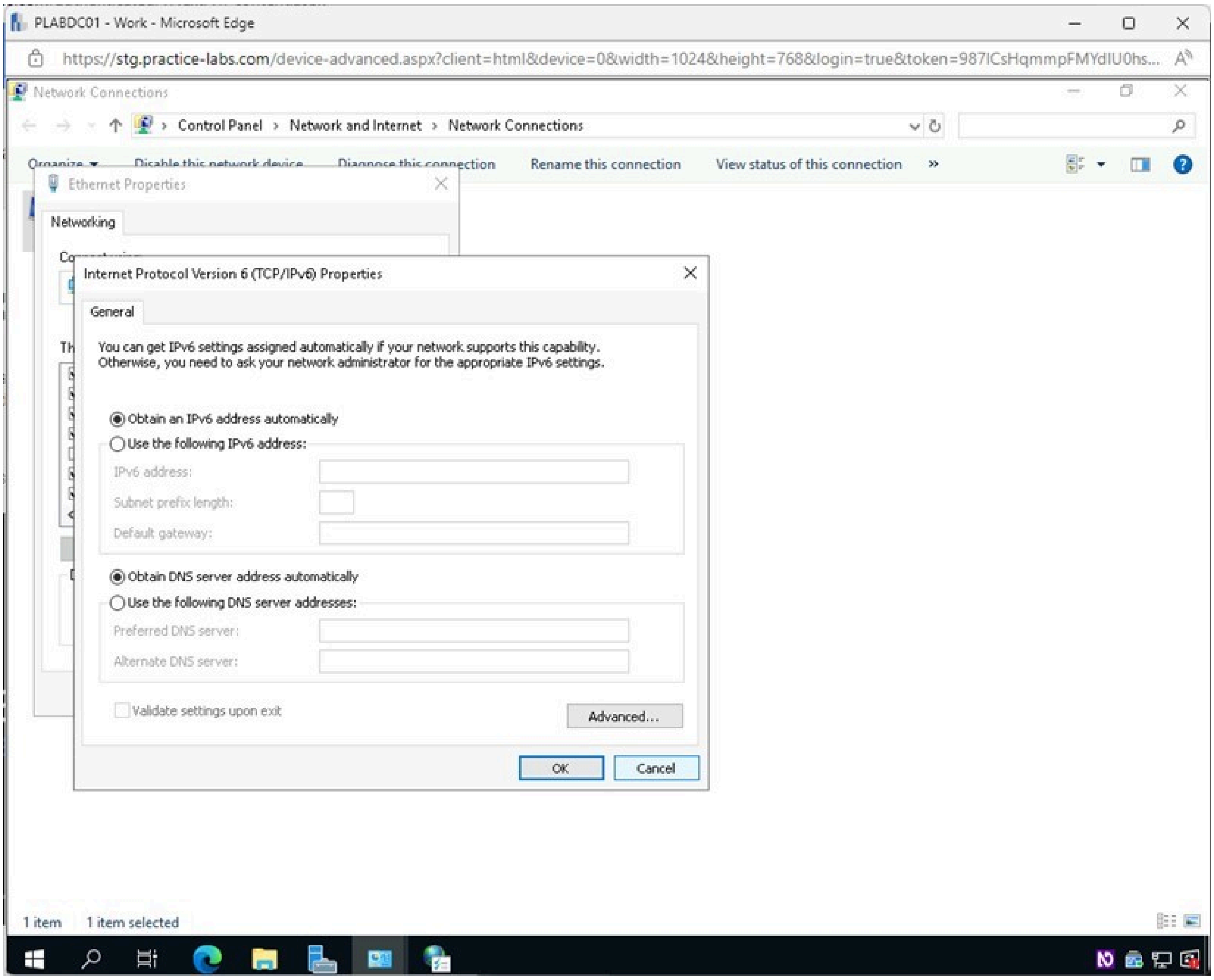

Figure 1.41 Screenshot of PLABDC01: Displaying selecting Cancel in the Internet Protocol Version 6 (TCP/IPv6) Properties window.

Step 12

Click **Close** on the **Ethernet Properties** window.

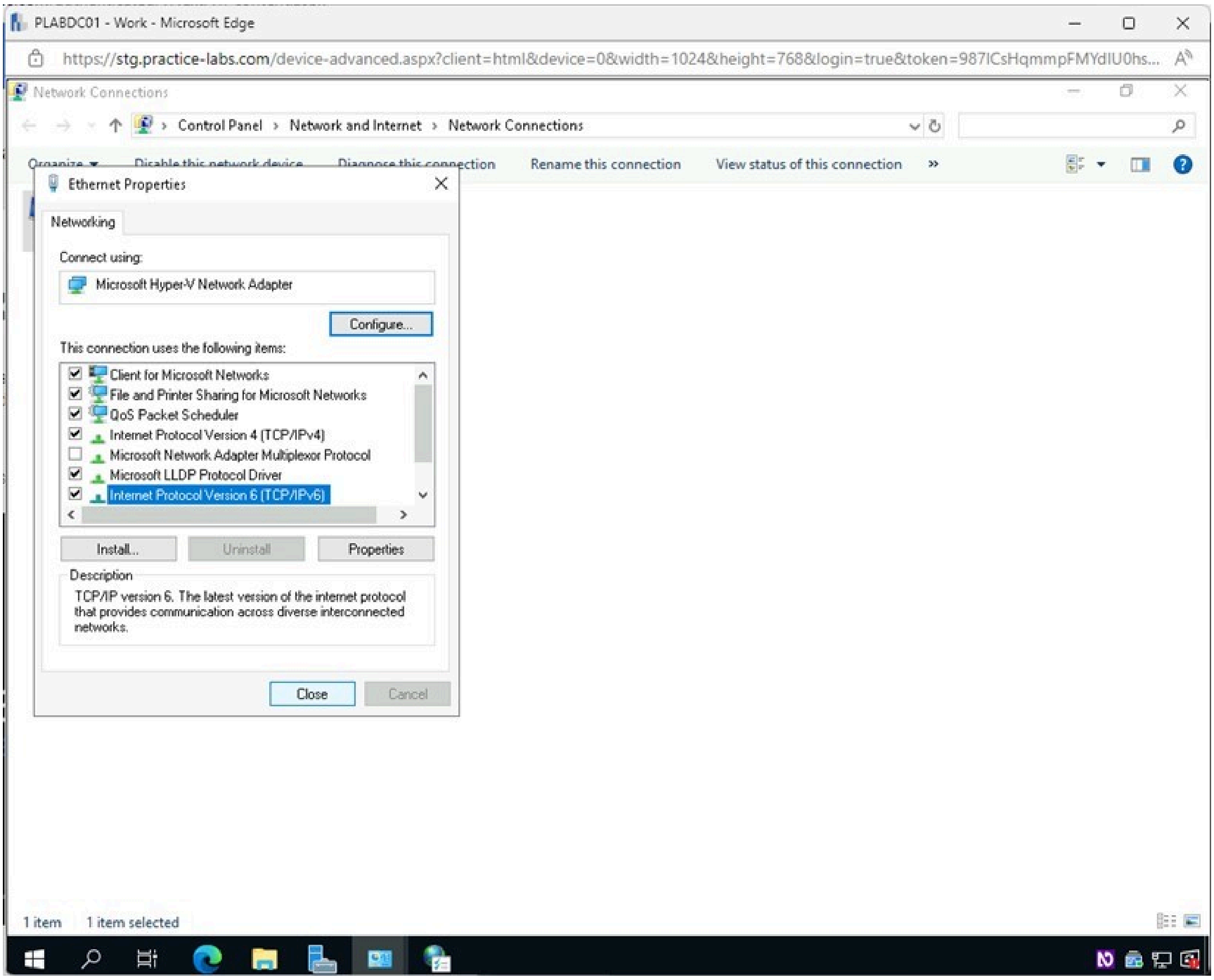

Figure 1.42 Screenshot of PLABDC01: Displaying selecting Close in the Ethernet Properties window.

Step 13

Close the **Network Connections** and all other open windows.

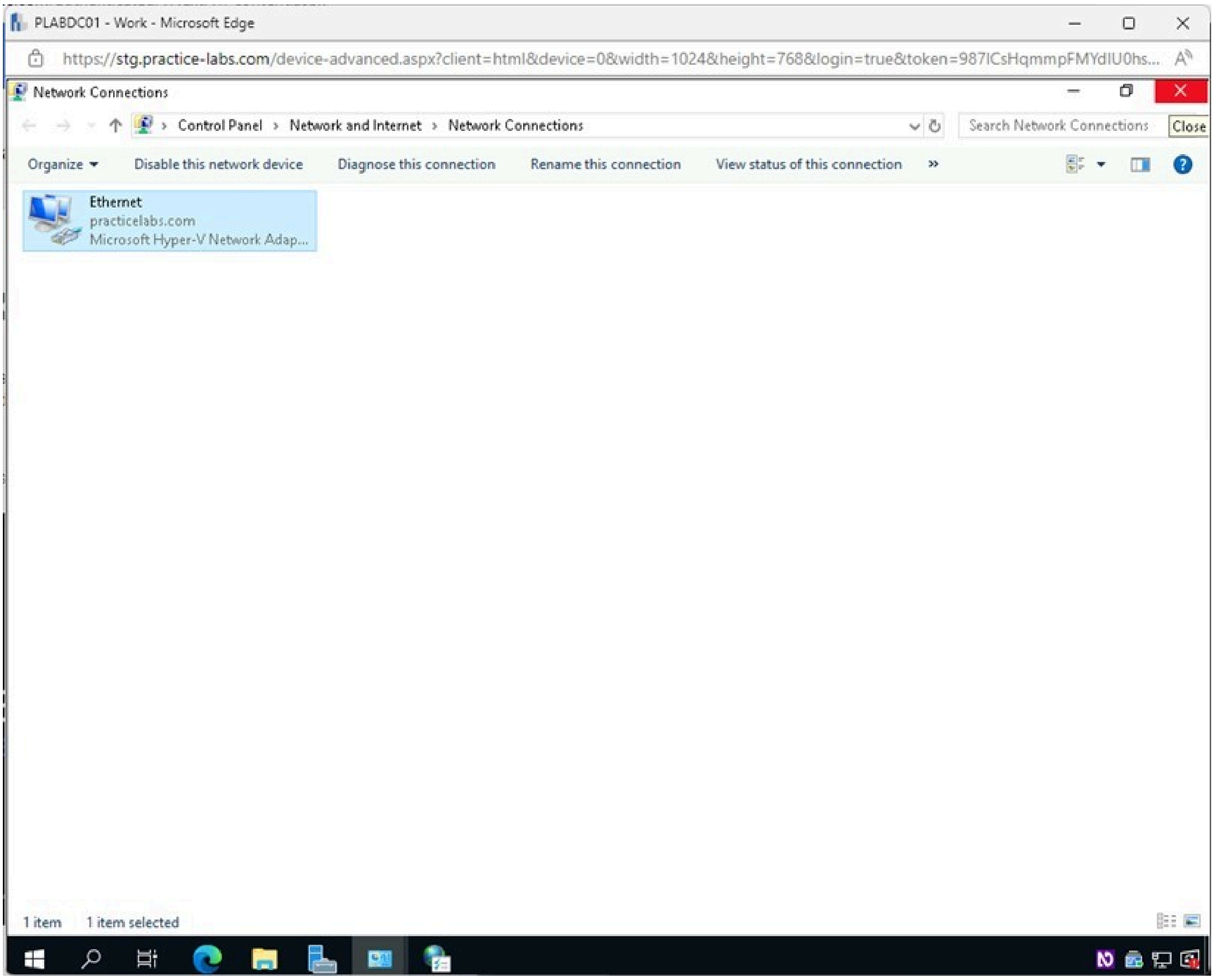

Figure 1.43 Screenshot of PLABDC01: Displaying closing the Network Connections window.

Task 6 - Install and Configure DHCP Server

To access the network, computers must get an IP address compatible with the network. The DHCP Server manages this. Misconfigured users can run into issues where they can’t get a valid IP address and get a replacement address known as an APIPA IP address that will allow access only to local resources.

When a DHCP server is implemented on the network, it will prevent the misconfiguration of client IP addresses which can be caused by human error.

In this task, you will configure a DHCP Server and make DHCP reservations for static settings.

Step 1

Ensure you are connected to **PLABDC01.**

Restore the **Server Manager** window from the **Taskbar**.

Click the **Manage** menu and then select **Add Roles and Features.**

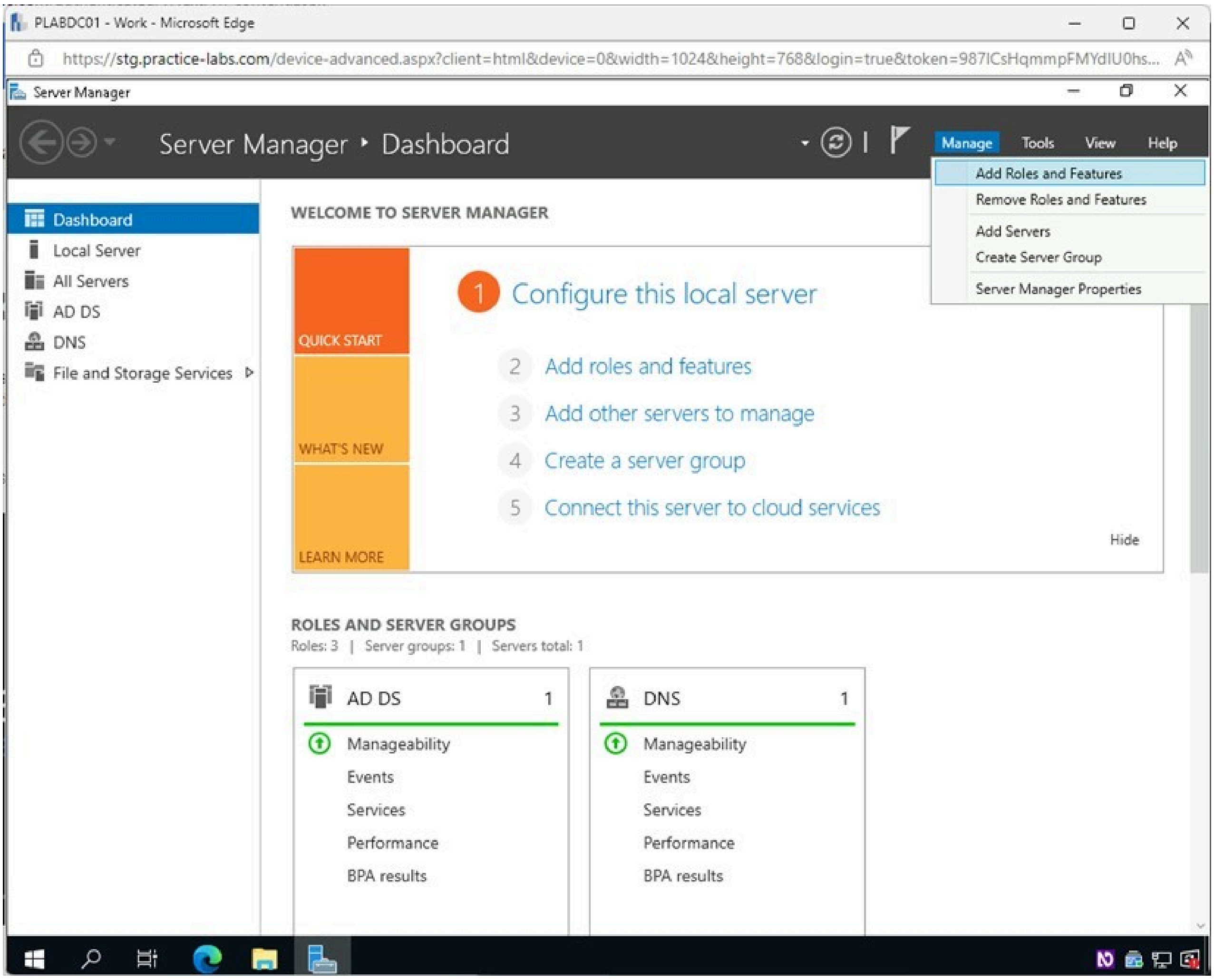

Figure 1.44 Screenshot of PLABDC01: Displaying the Server Manager window. The Manage menu is accessed, and Add Roles and Features is selected.

Step 2

In the **Add Roles and Features Wizard - Before you begin** window, read the information and click **Next.**

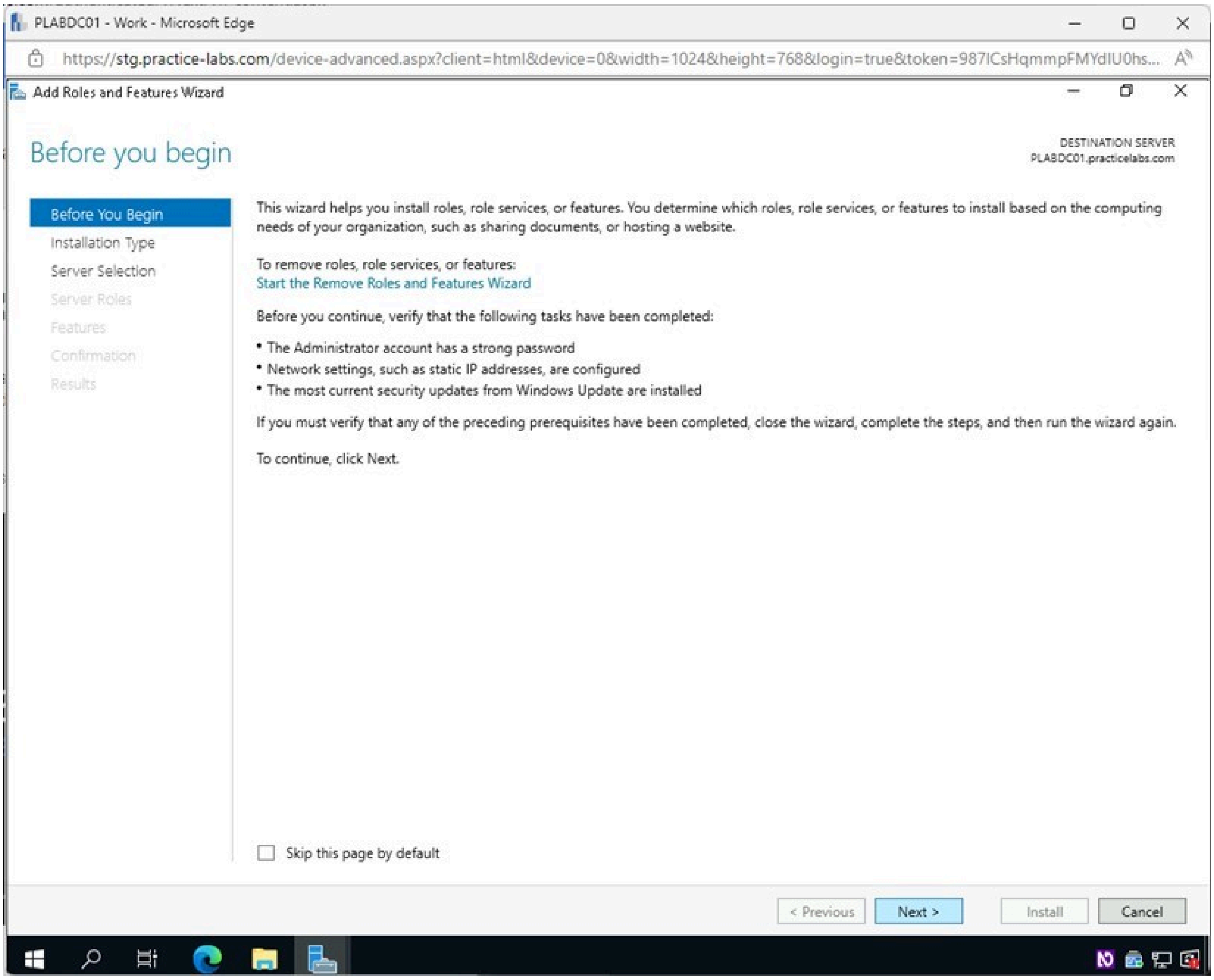

Figure 1.45 Screenshot of PLABDC01: Displaying selecting Next in the Add Roles and Features Wizard - Before you begin window.

Step 3

In the **Select installation type** page, leave **Role-based or feature-based installation** selected and click **Next.**

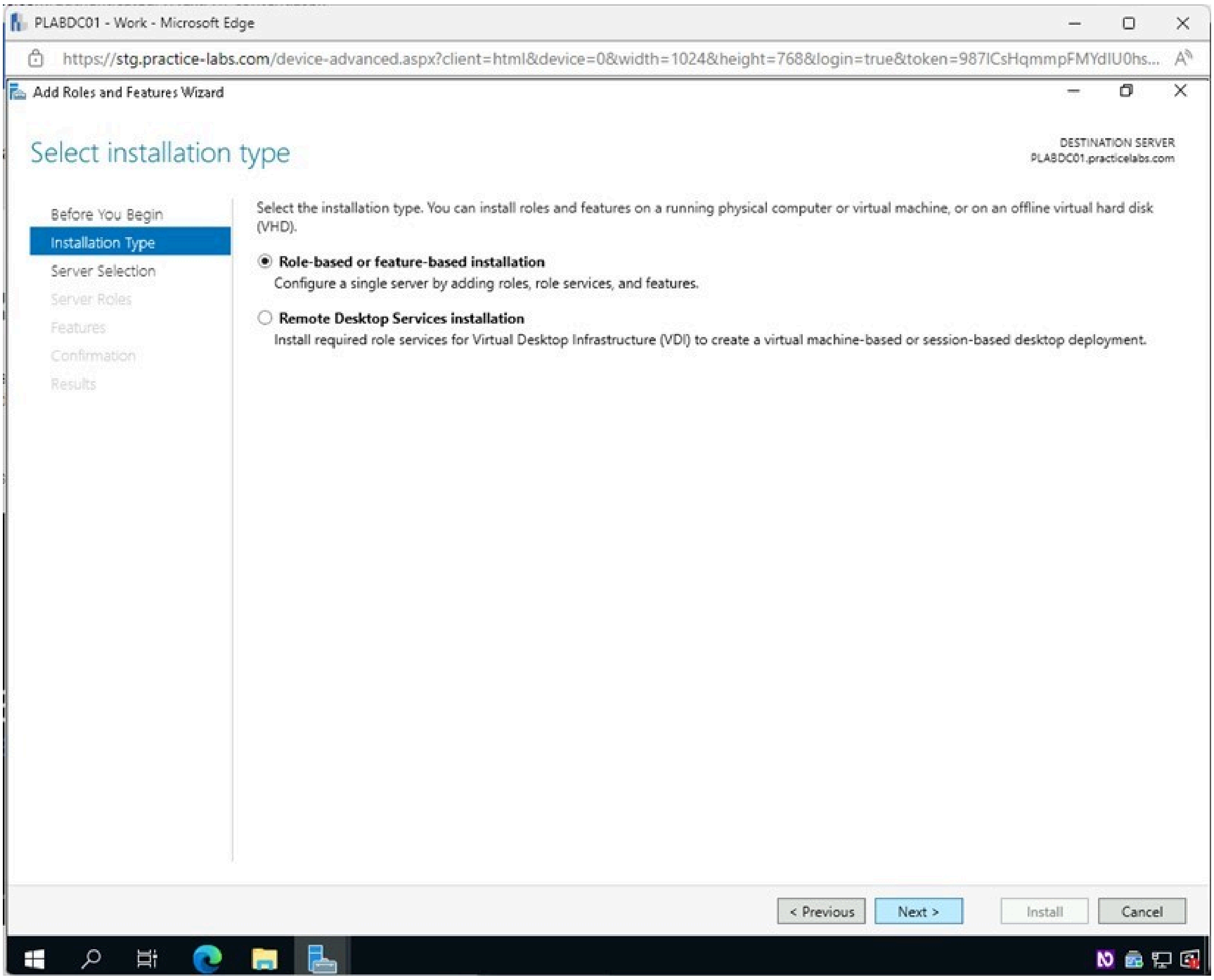

Figure 1.46 Screenshot of PLABDC01: Displaying selecting Next in the Select installation type page.

Step 4

In the **Select destination server** page, click **Next.**

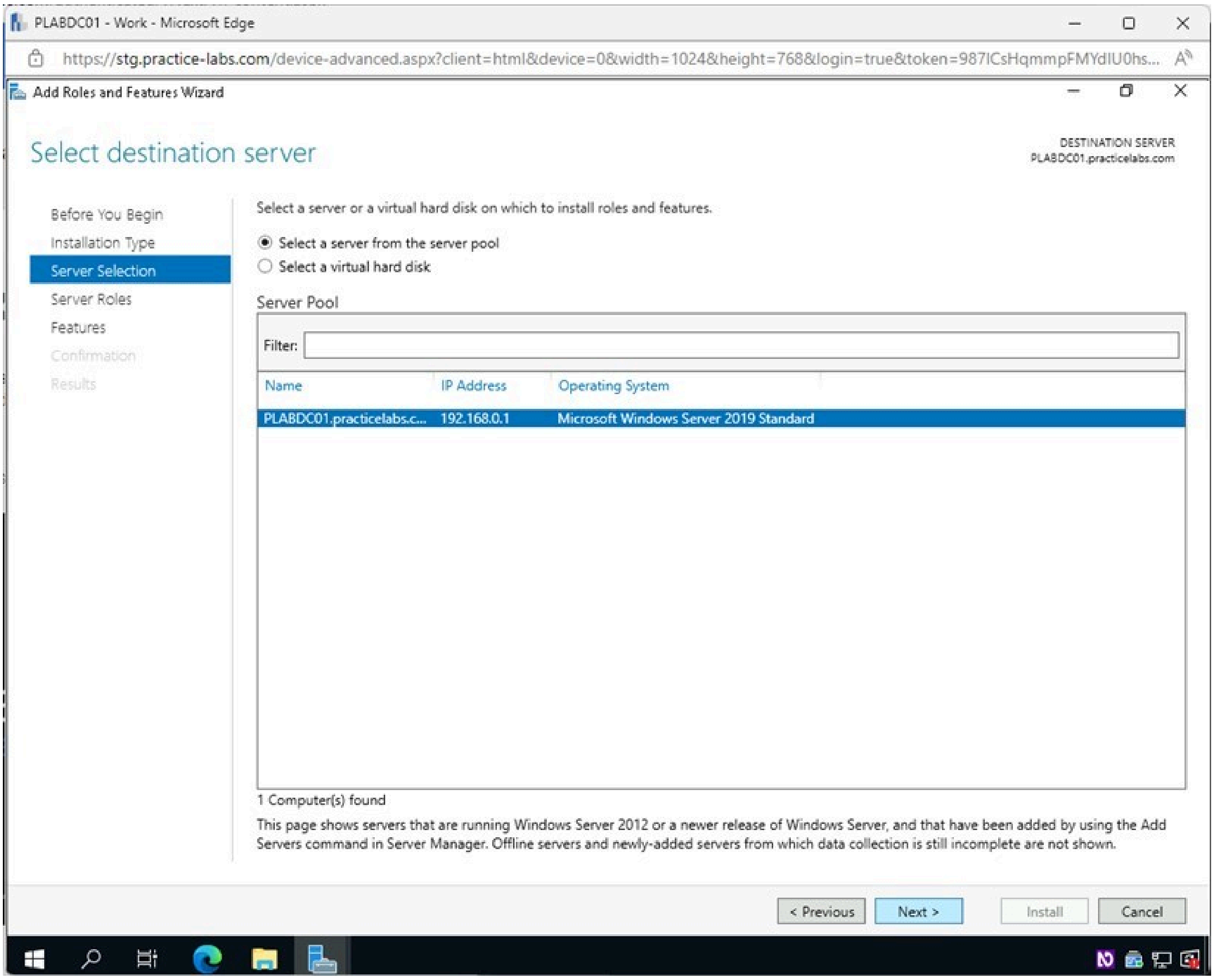

Figure 1.47 Screenshot of PLABDC01: Displaying selecting Next in the Select destination server page.

Step 5

In the **Select server roles** page, tick the **DHCP Server** checkbox.

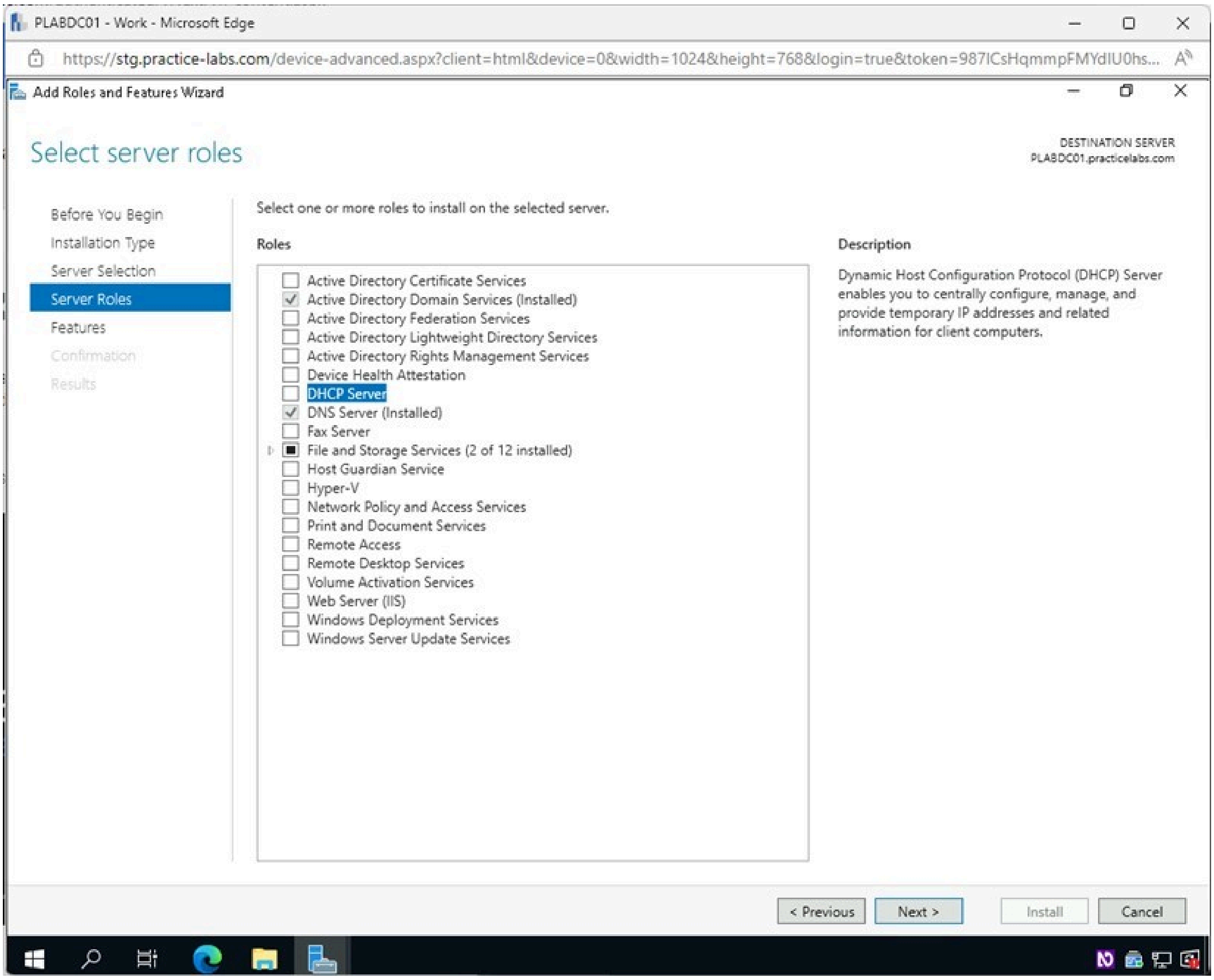

Figure 1.48 Screenshot of PLABDC01: Displaying selecting DHCP Server in the Select server roles page.

Step 6

In the **Add features that are required for DHCP Server?** pop-up window, click **Add Features**.

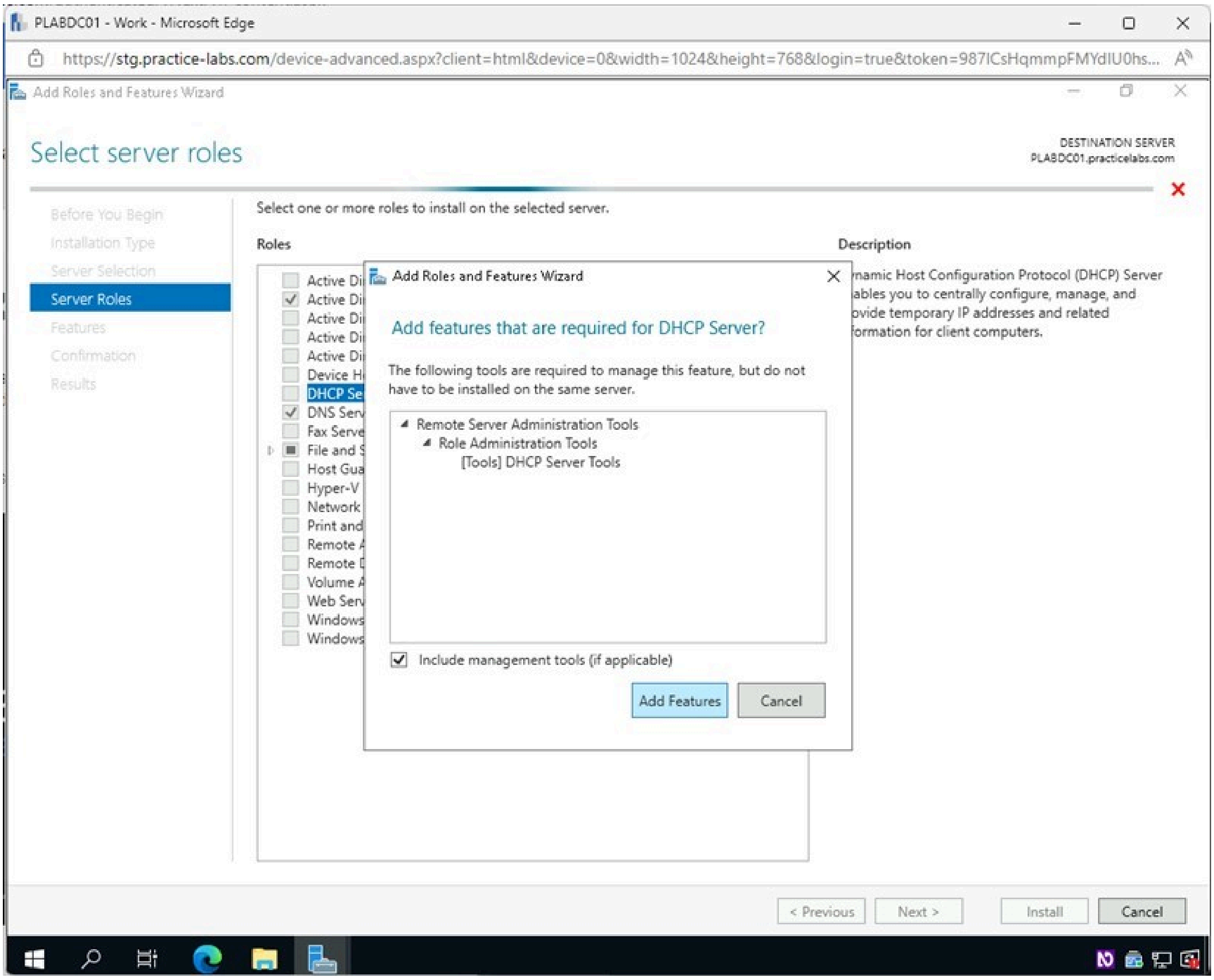

Figure 1.49 Screenshot of PLABDC01: Displaying selecting Add Features in the Add features that are required for DHCP Server? pop-up window.

Step 7

Back on the **Select server roles** page, click **Next**.

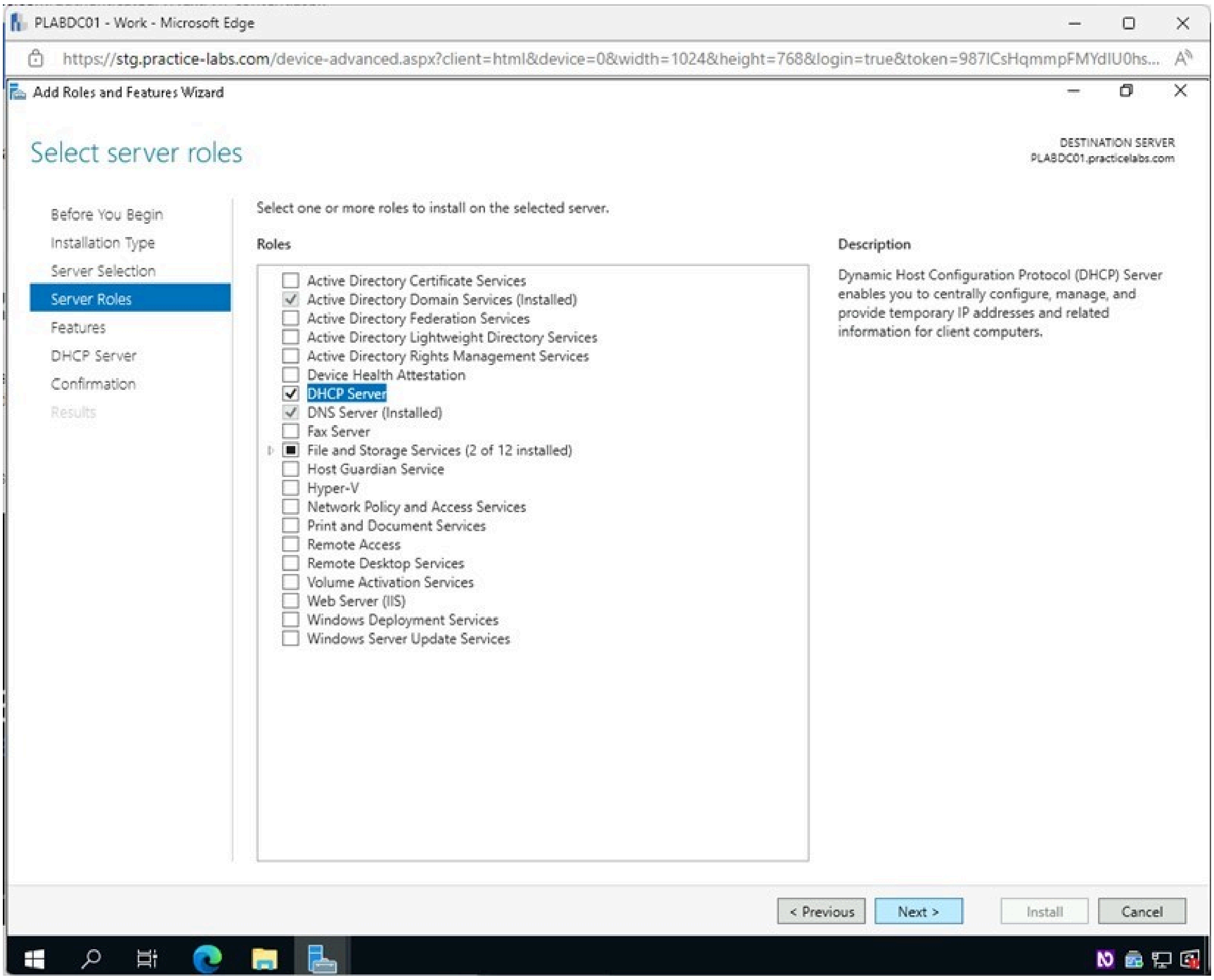

Figure 1.50 Screenshot of PLABDC01: Displaying selecting Next in the Select server roles page.

Step 8

Click **Next** on the **Select features** page.

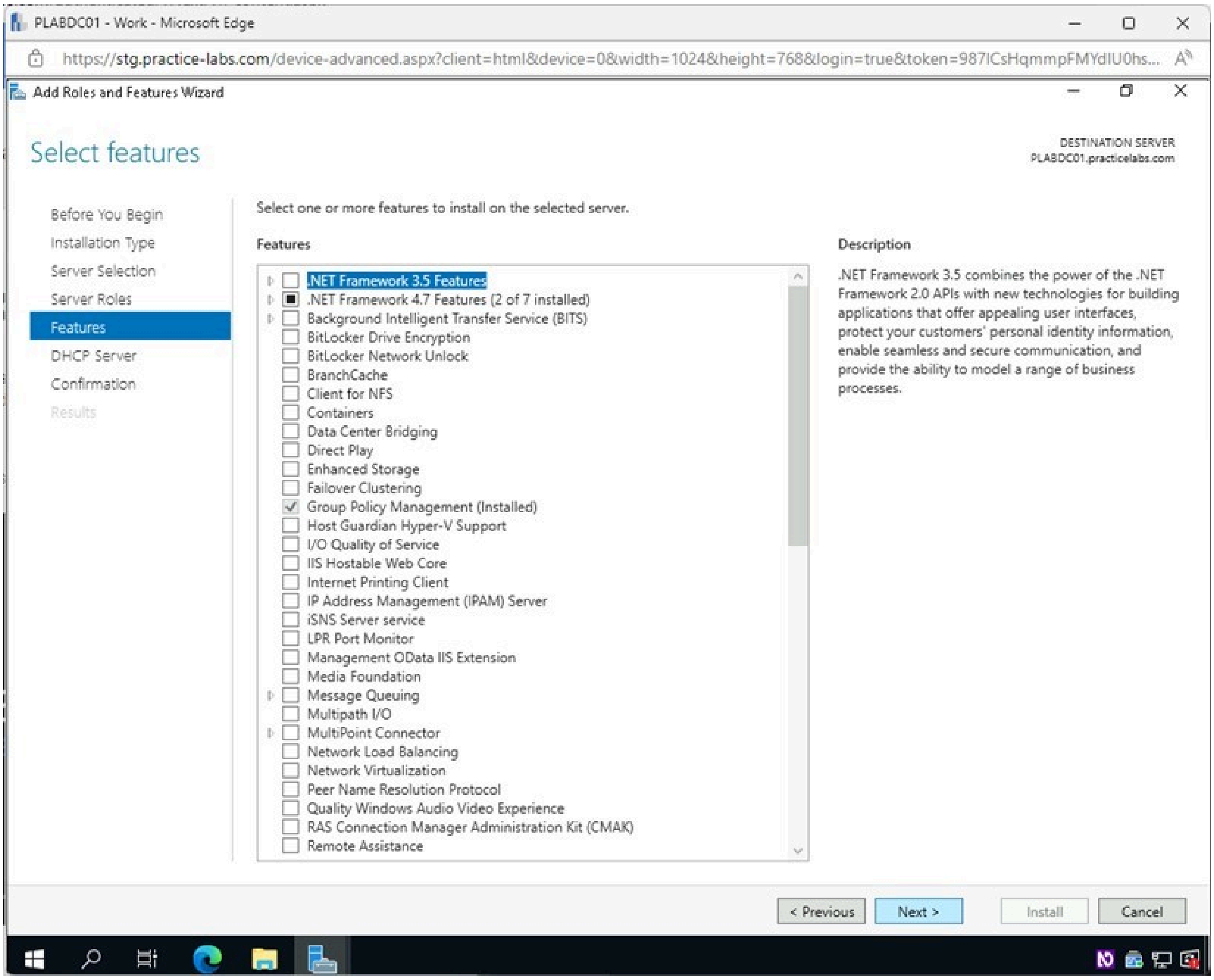

Figure 1.51 Screenshot of PLABDC01: Displaying selecting Next in the Select features page.

Step 9

On the **DHCP Server** page, click **Next.**

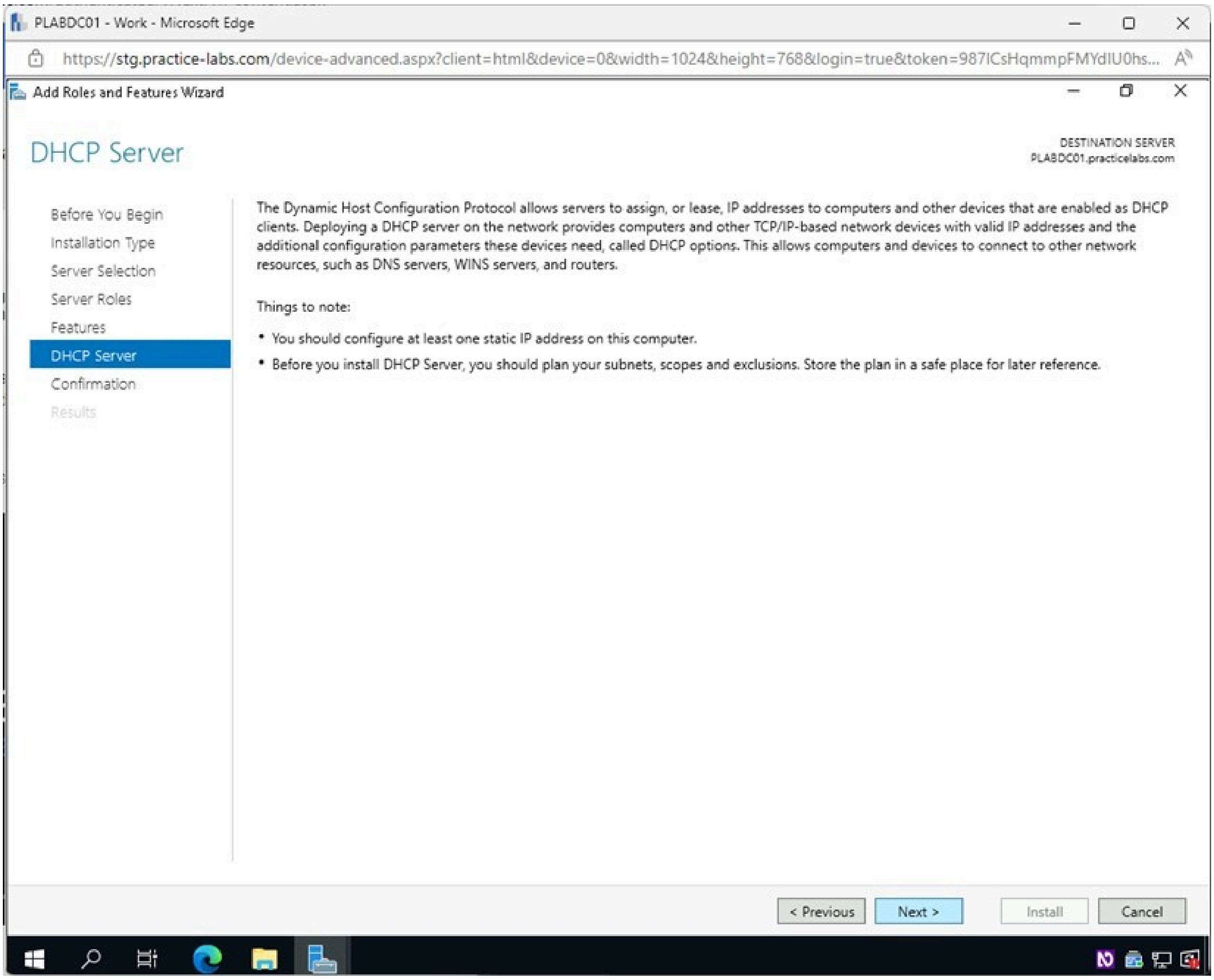

Figure 1.52 Screenshot of PLABDC01: Displaying selecting Next in the DHCP Server page.

Step 10

In the **Confirm installation selections** page, click **Install**.

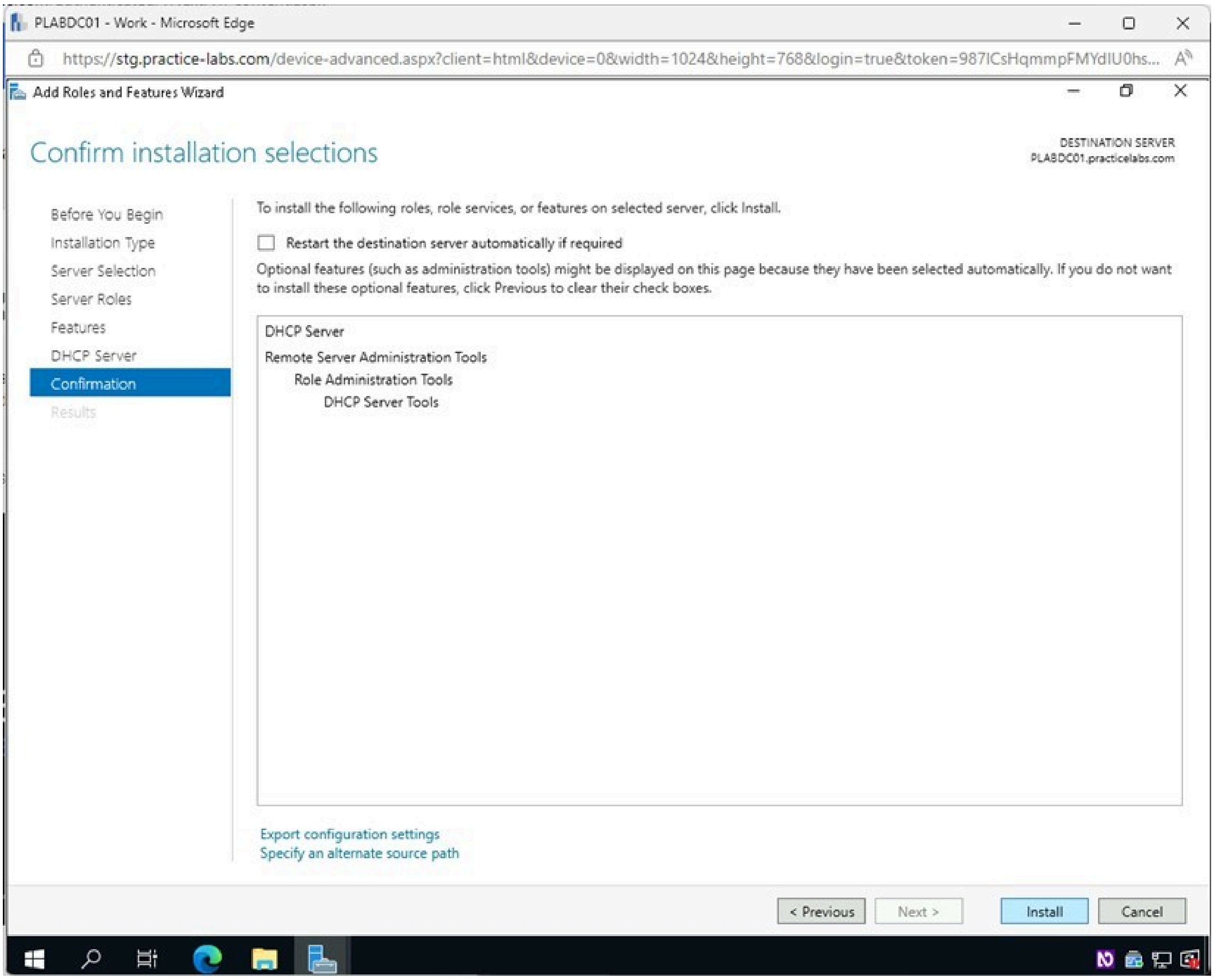

Figure 1.53 Screenshot of PLABDC01: Displaying selecting Install in the Confirm installation selections page.

Step 11

In the **Installation progress** page, click **Close** once the installation is complete.

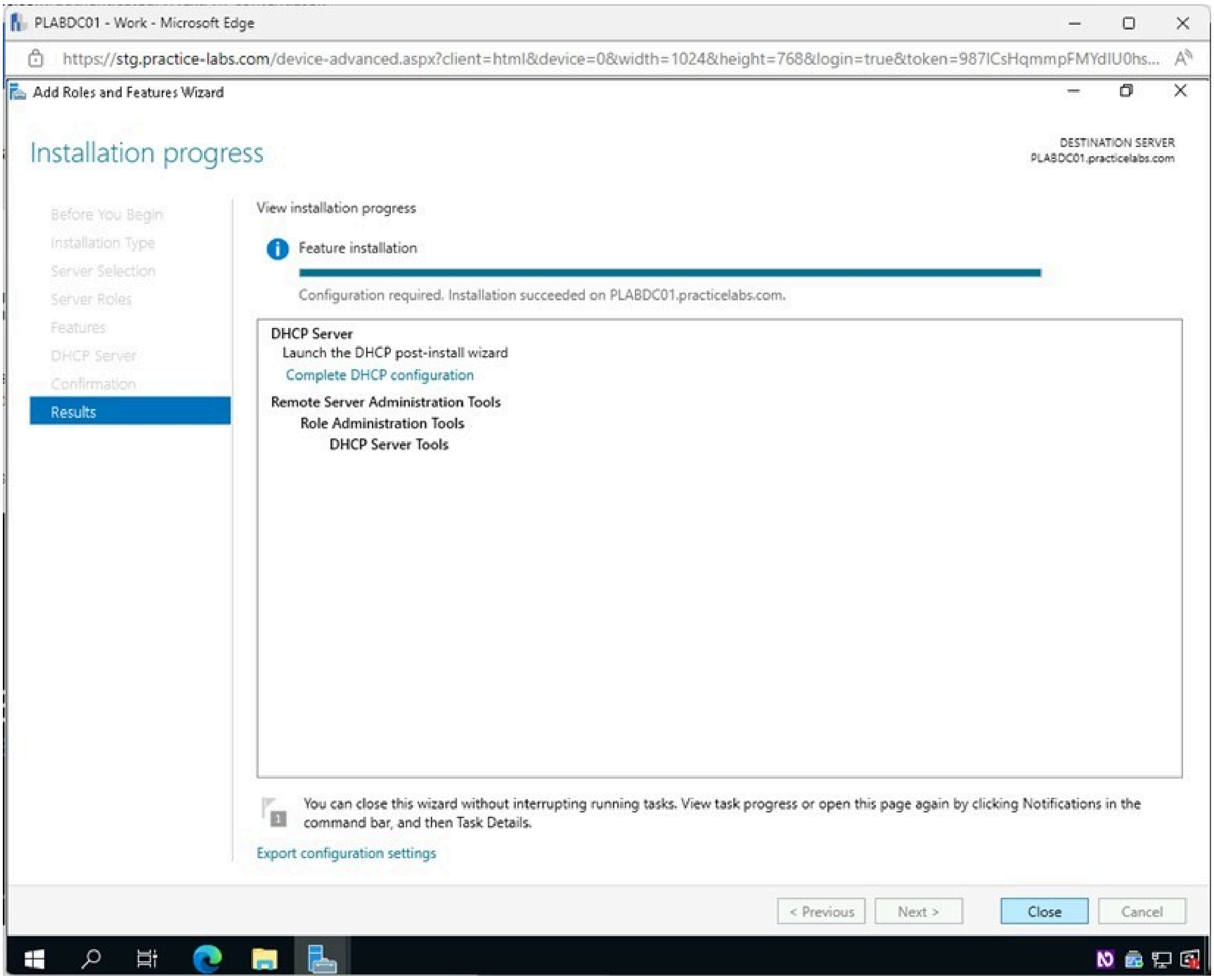

Figure 1.54 Screenshot of PLABDC01: Displaying selecting Close in the Installation progress page.

Step 12

Back on the **Server Manager** window, click the **yellow alert triangle** and then click the **Complete DHCP configuration** link.

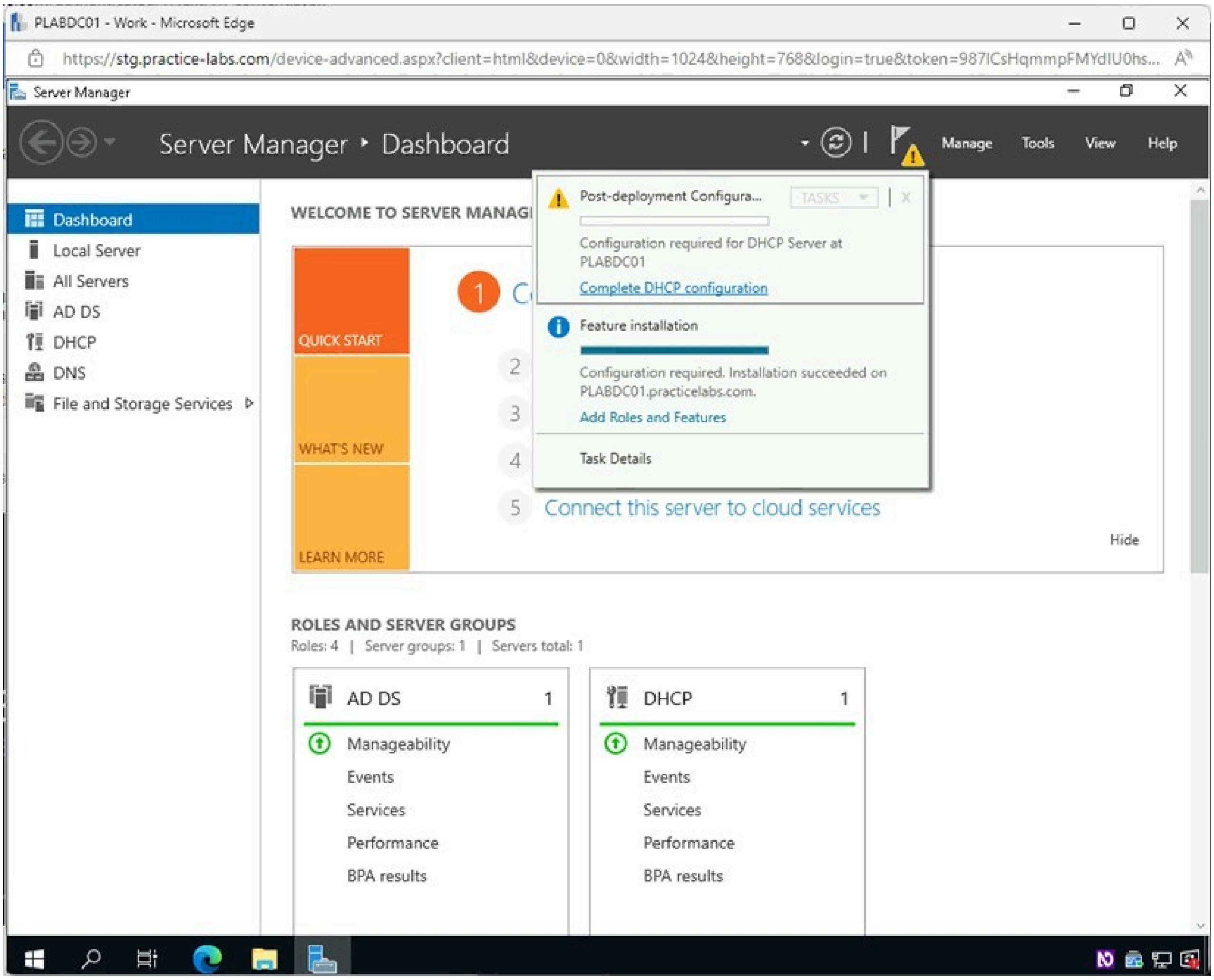

Figure 1.55 Screenshot of PLABDC01: Displaying the Server Manager window, showing the Complete DHCP configuration link selected.

Step 13

In the **DHCP Post-Install configuration wizard - Description** window, review the information and click **Next**.

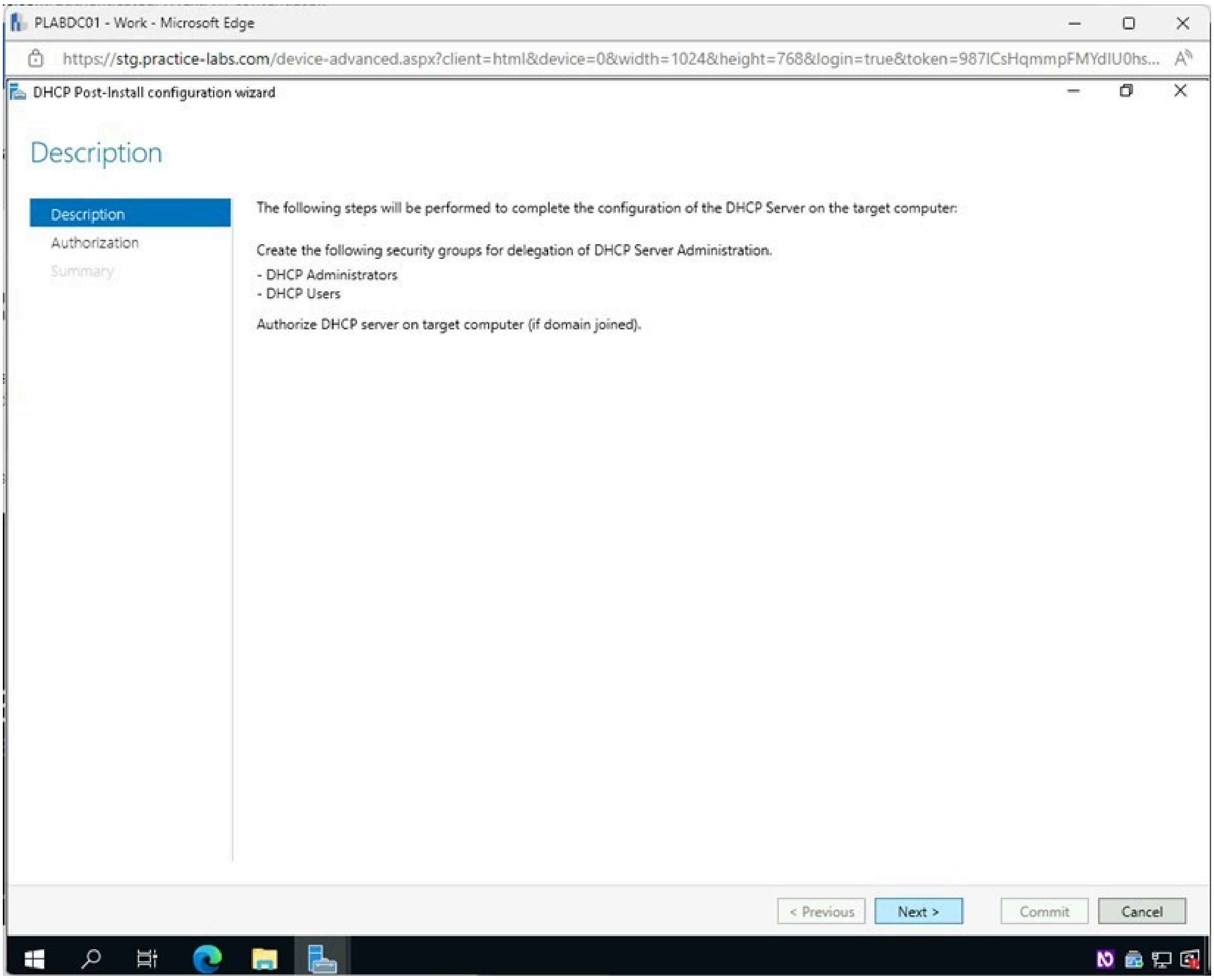

Figure 1.56 Screenshot of PLABDC01: Displaying selecting Next in the DHCP Post-Install configuration wizard - Description window.

Step 14

On the **Authorization** page, select the **Skip AD authorization** option.

Click **Commit**.

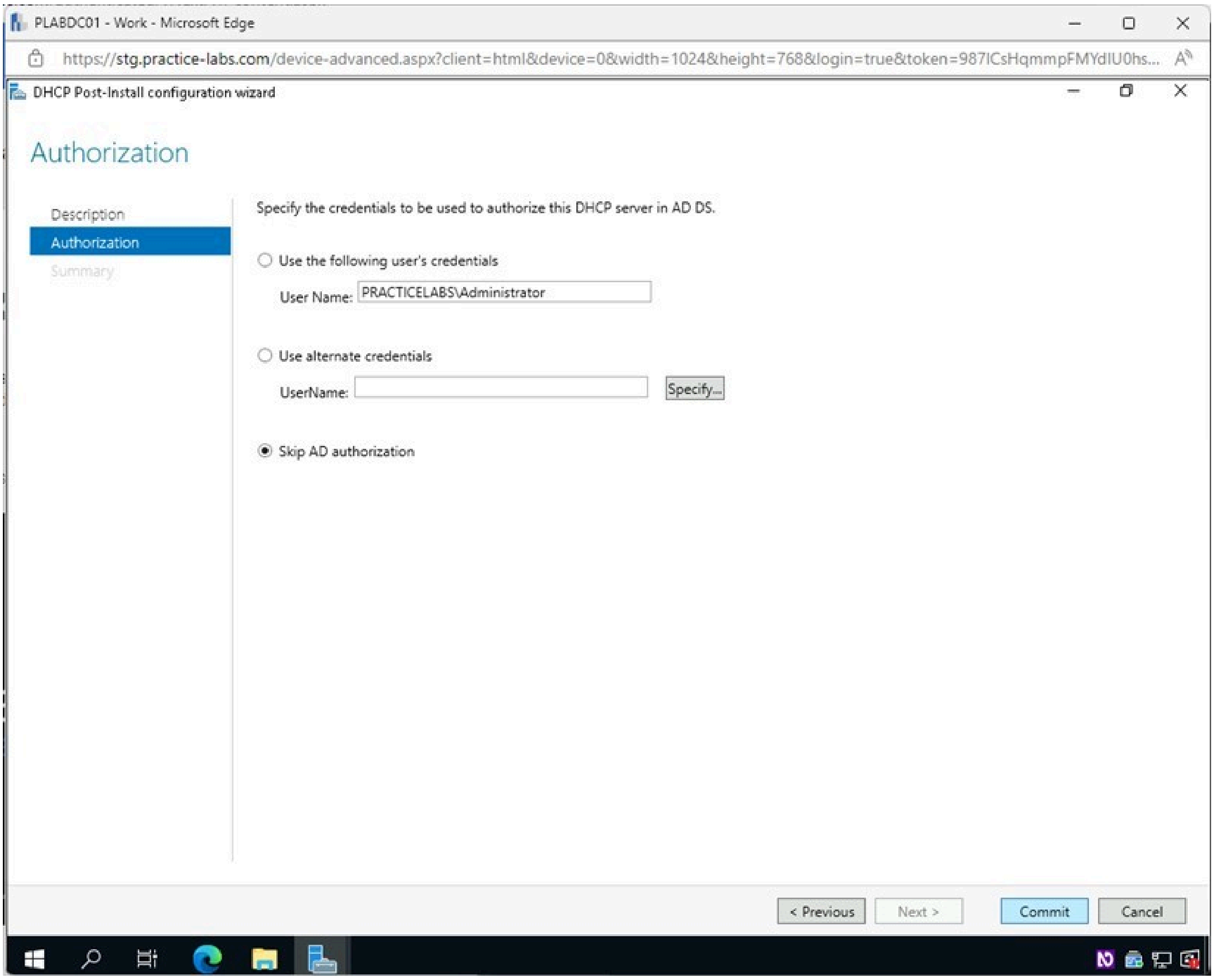

Figure 1.57 Screenshot of PLABDC01: Displaying the Authorization page, showing Skip AD authorization selected and the Commit button selected.

Step 15

Click **OK** on the **DHCP Post-Install configuration error** pop-up window.

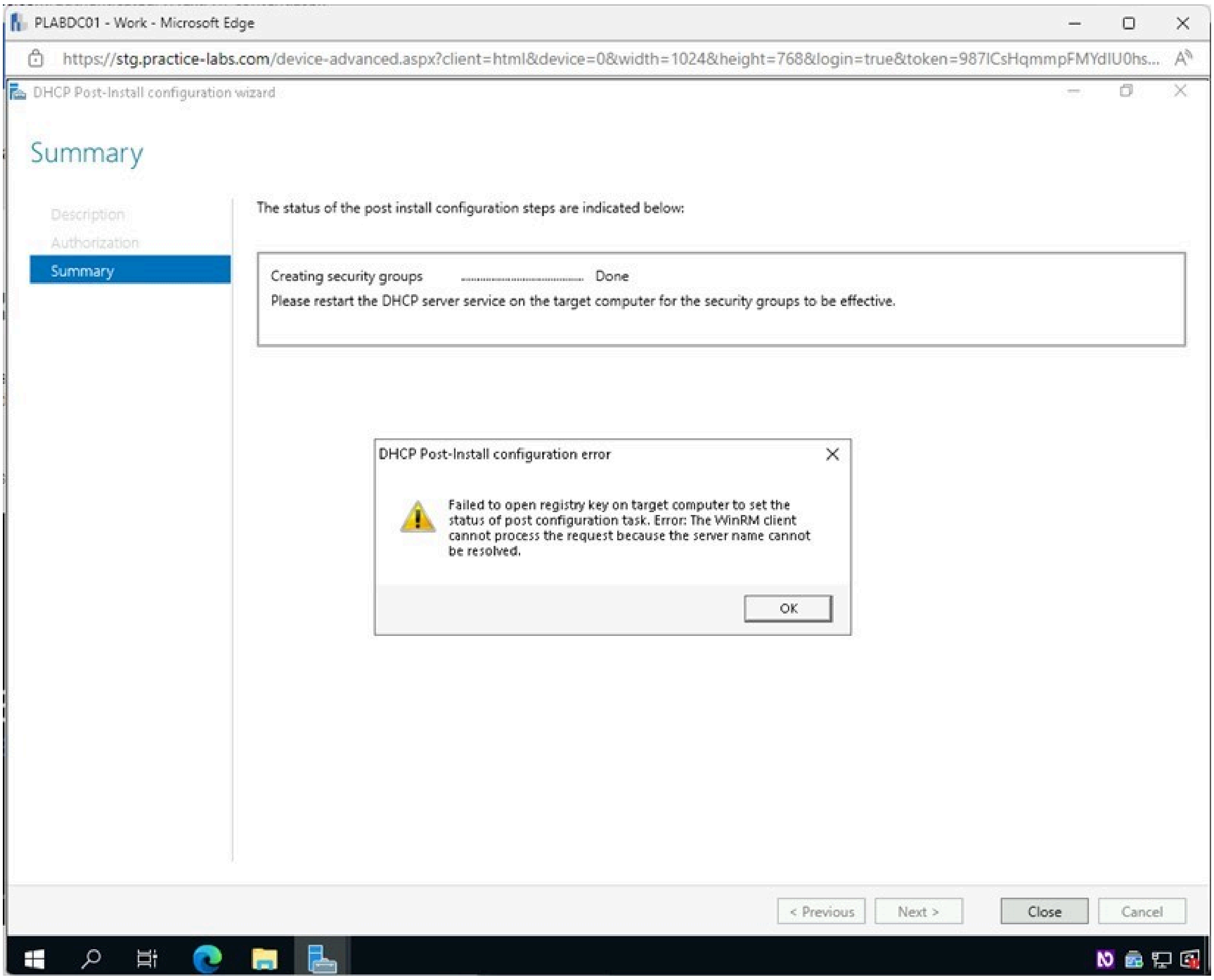

Figure 1.58 Screenshot of PLABDC01: Displaying selecting OK on the DHCP Post-Install configuration error pop-up window.

Step 16

On the **Summary** page, click **Close**.

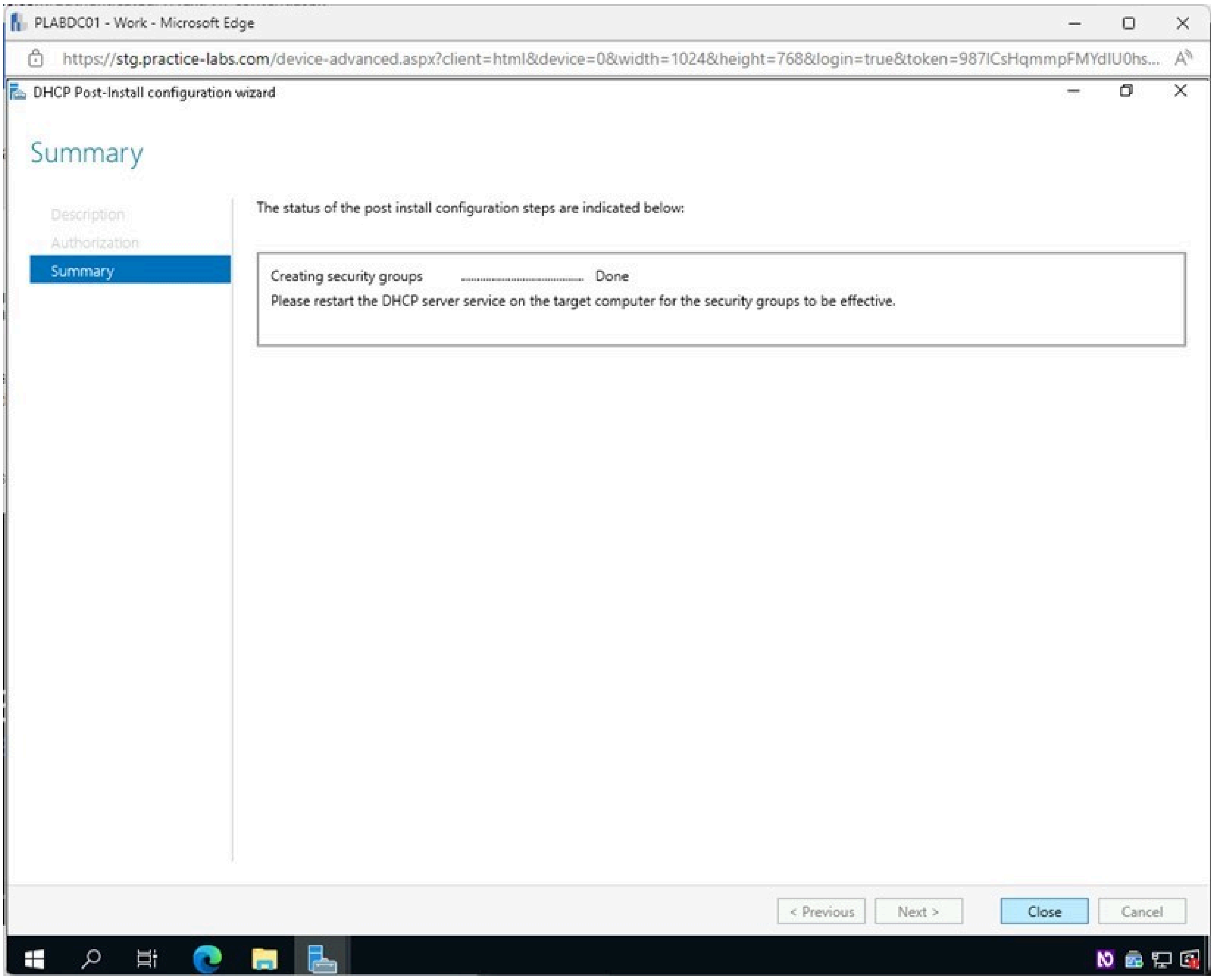

Figure 1.59 Screenshot of PLABDC01: Displaying selecting Close in the Summary page.

Step 17

Back on the **Server Manager** window, access the **Tools** menu and select **DHCP.**

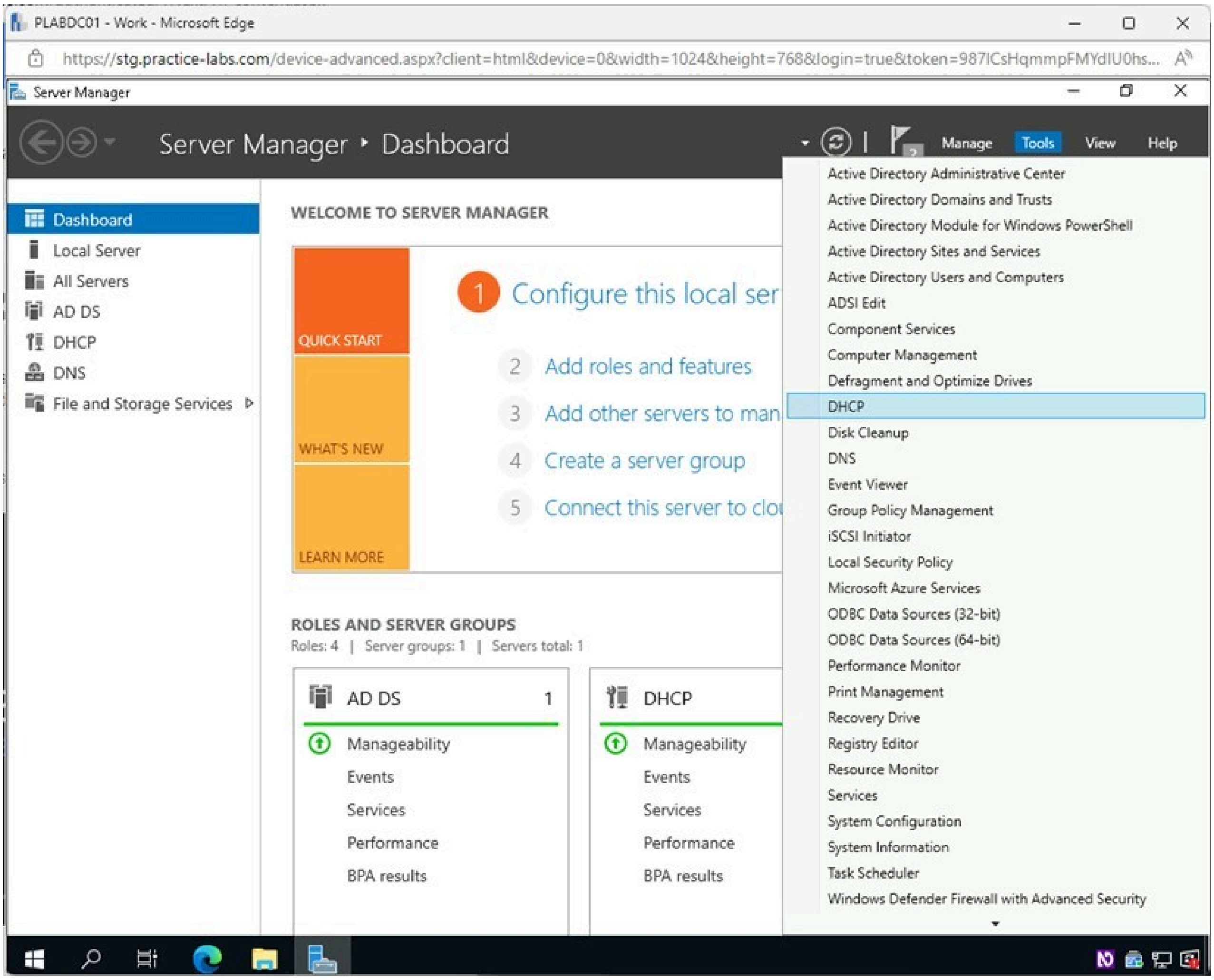

Figure 1.60 Screenshot of PLABDC01: Displaying the Server Manager window. The Tools menu is accessed, and DHCP is selected.

Step 18

In the **DHCP** window, expand **plabdc01.practicelabs.com** on the left pane.

Select and right-click on **IPv4** and then select **New Scope**.

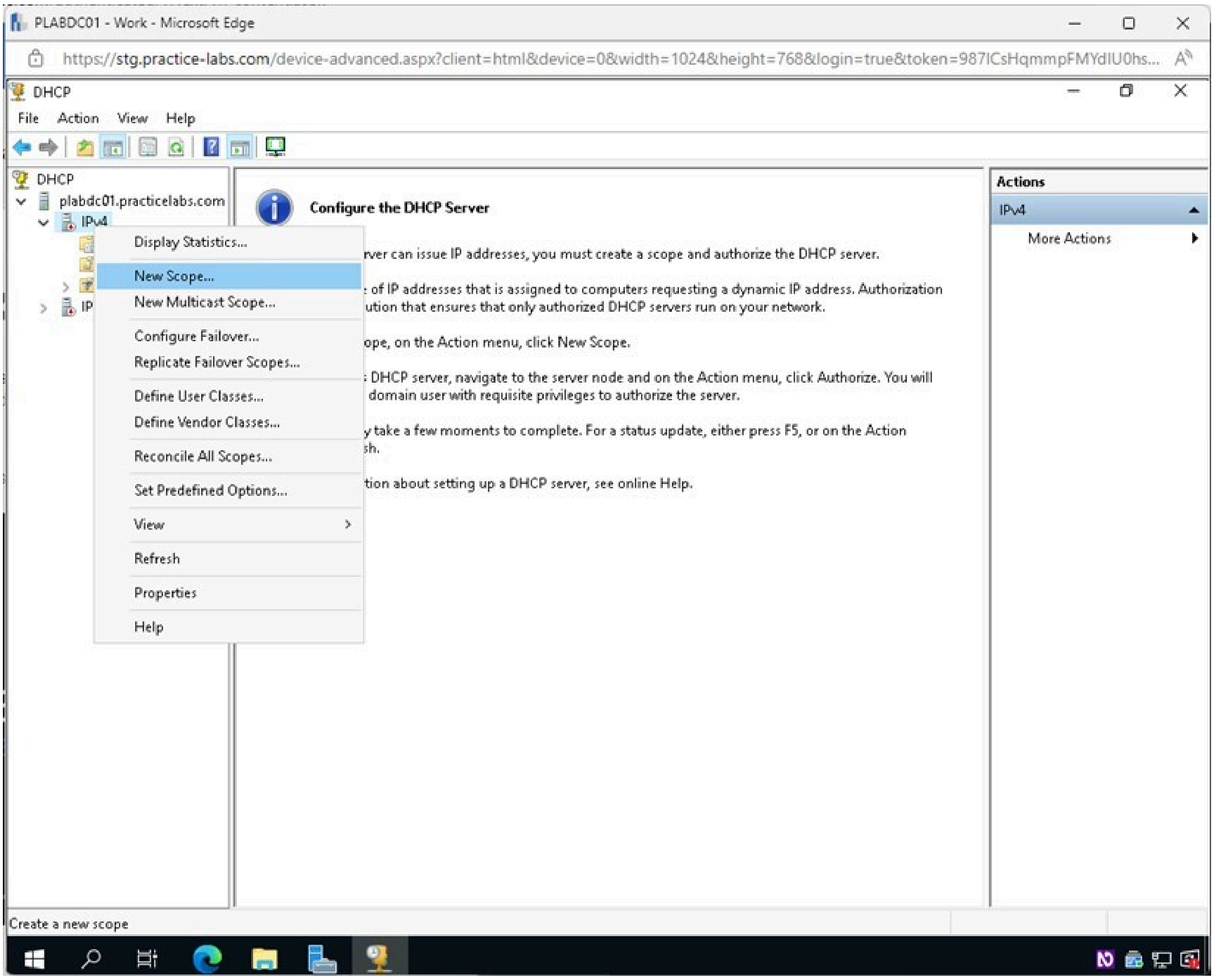

Figure 1.61 Screenshot of PLABDC01: Displaying the DHCP window. IPv4 is right-clicked, and New Scope is selected.

Step 19

In the **New Scope Wizard - Welcome** window, review the information and click **Next**.

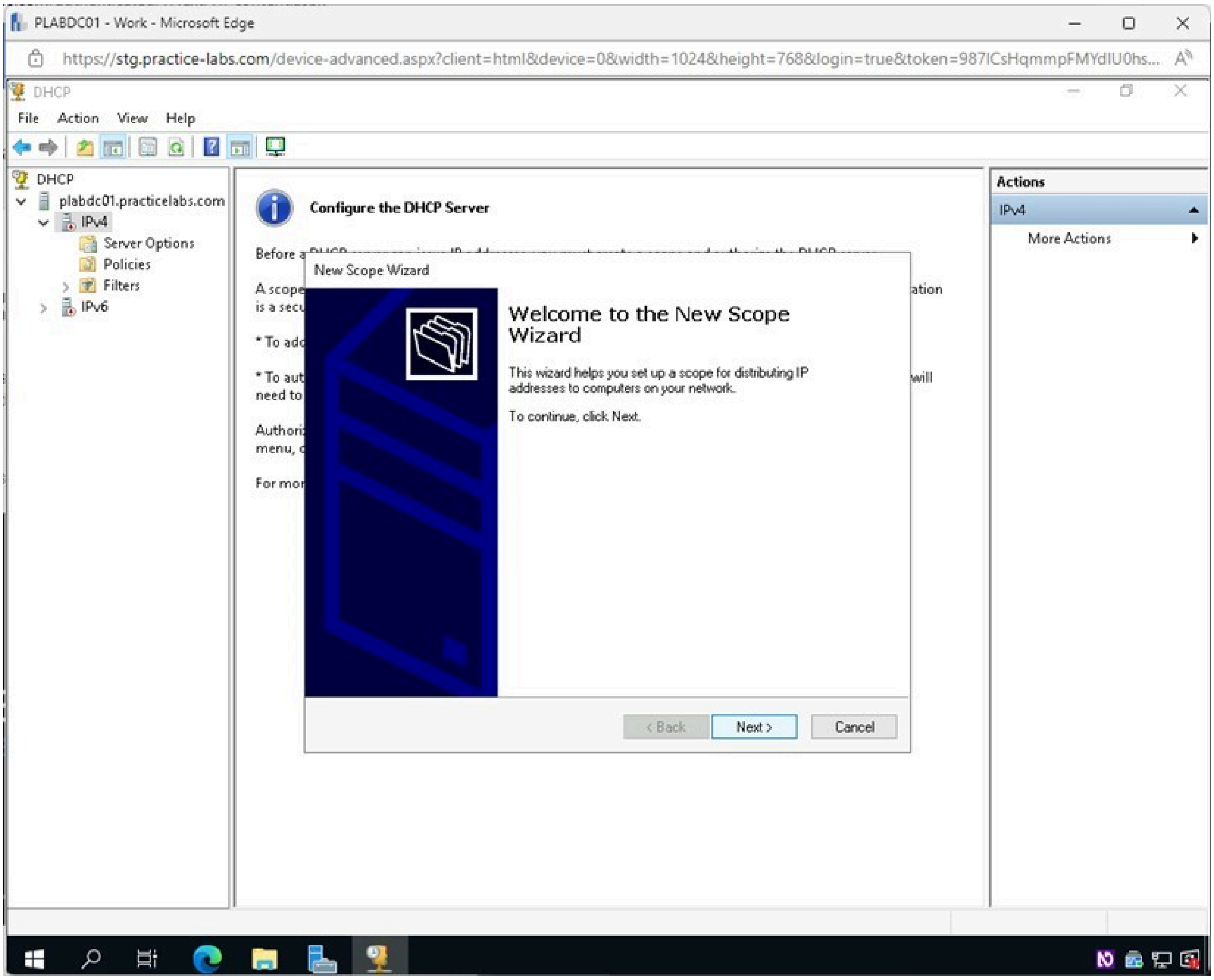

Figure 1.62 Screenshot of PLABDC01: Displaying selecting Next in the New Scope Wizard - Welcome window.

Step 20

In the **Scope Name** page, type the following for the **Name** field:

```
Test Network
```

Click **Next**.

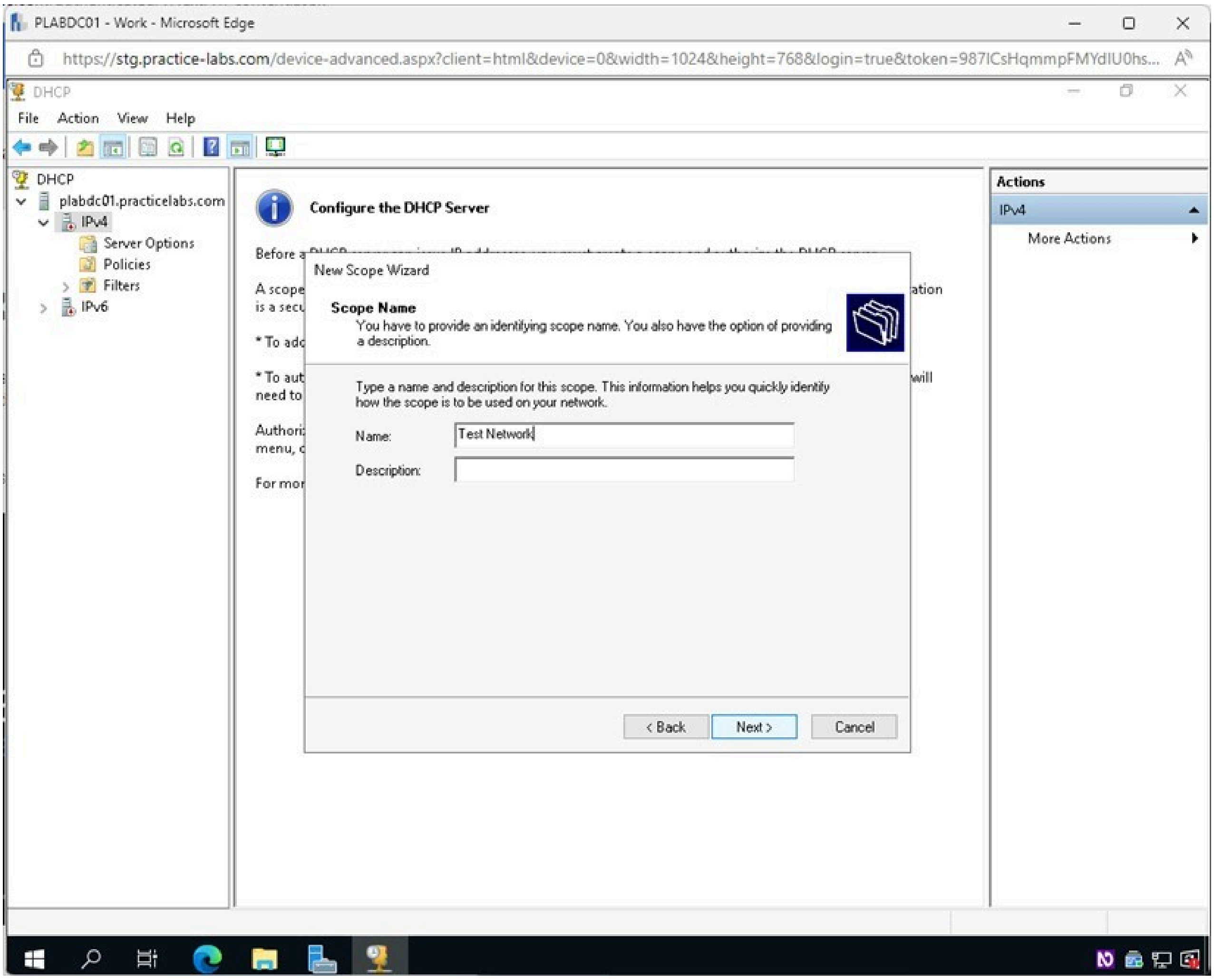

Figure 1.63 Screenshot of PLABDC01: Displaying the Scope Name page, showing the required Name typed in and Next selected.

Step 21

In the **IP Address Range** page, type the following for the corresponding fields:

```
Start IP address:  10.10.10.1
End IP address:    10.10.10.254
Subnet Mask:    255.255.255.0
```

Click **Next**.

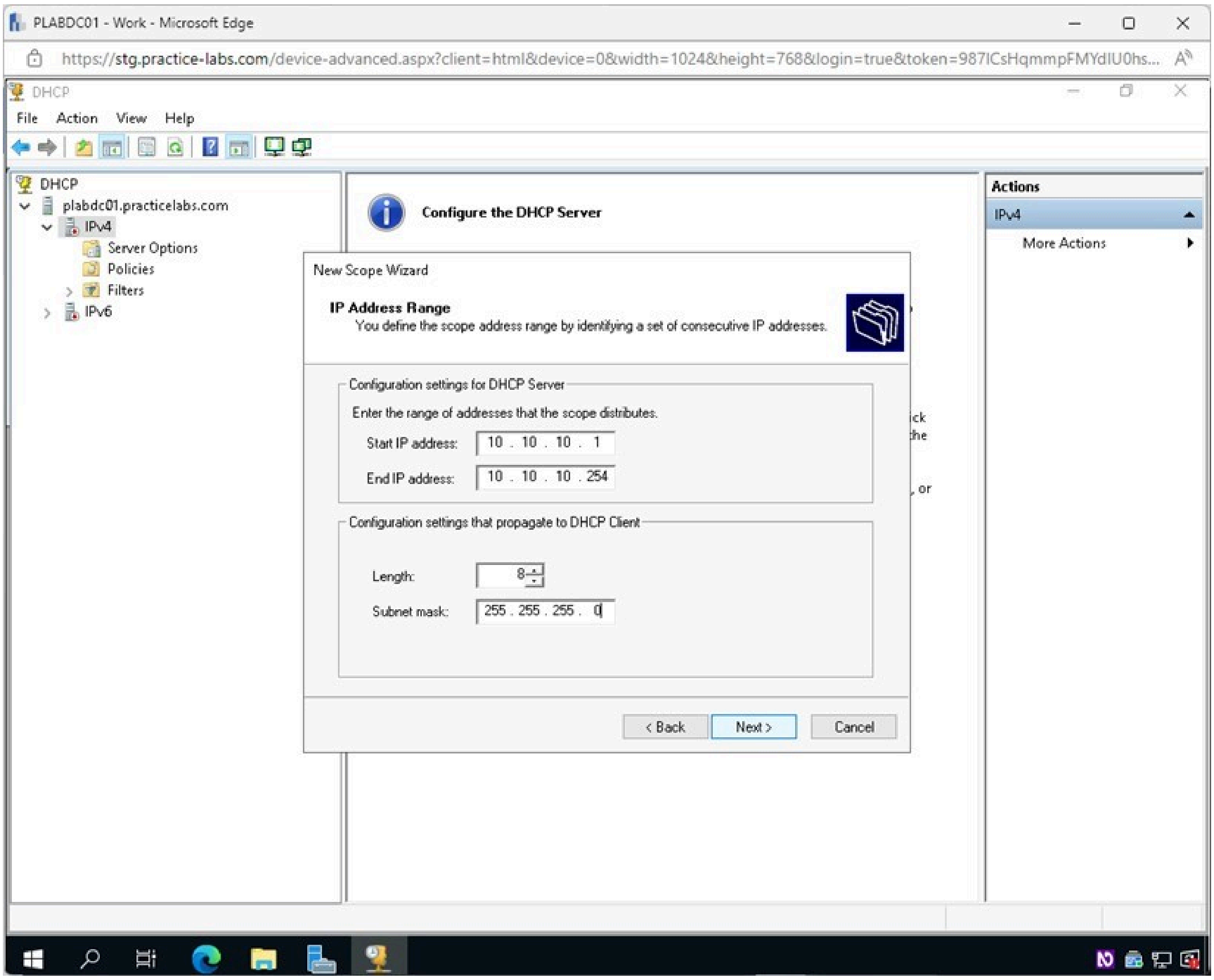

Figure 1.64 Screenshot of PLABDC01: Displaying the IP Address Range page, showing the required settings performed and Next selected.

Step 22

In the **Add Exclusions and Delay** page, you will create 2 scopes that the DHCP Server will not hand out and can be used for static addresses in the network.

Type the following for the corresponding fields:

```
Start IP address:  10.10.10.10
End IP address:    10.10.10.20
```

Click **Add**.

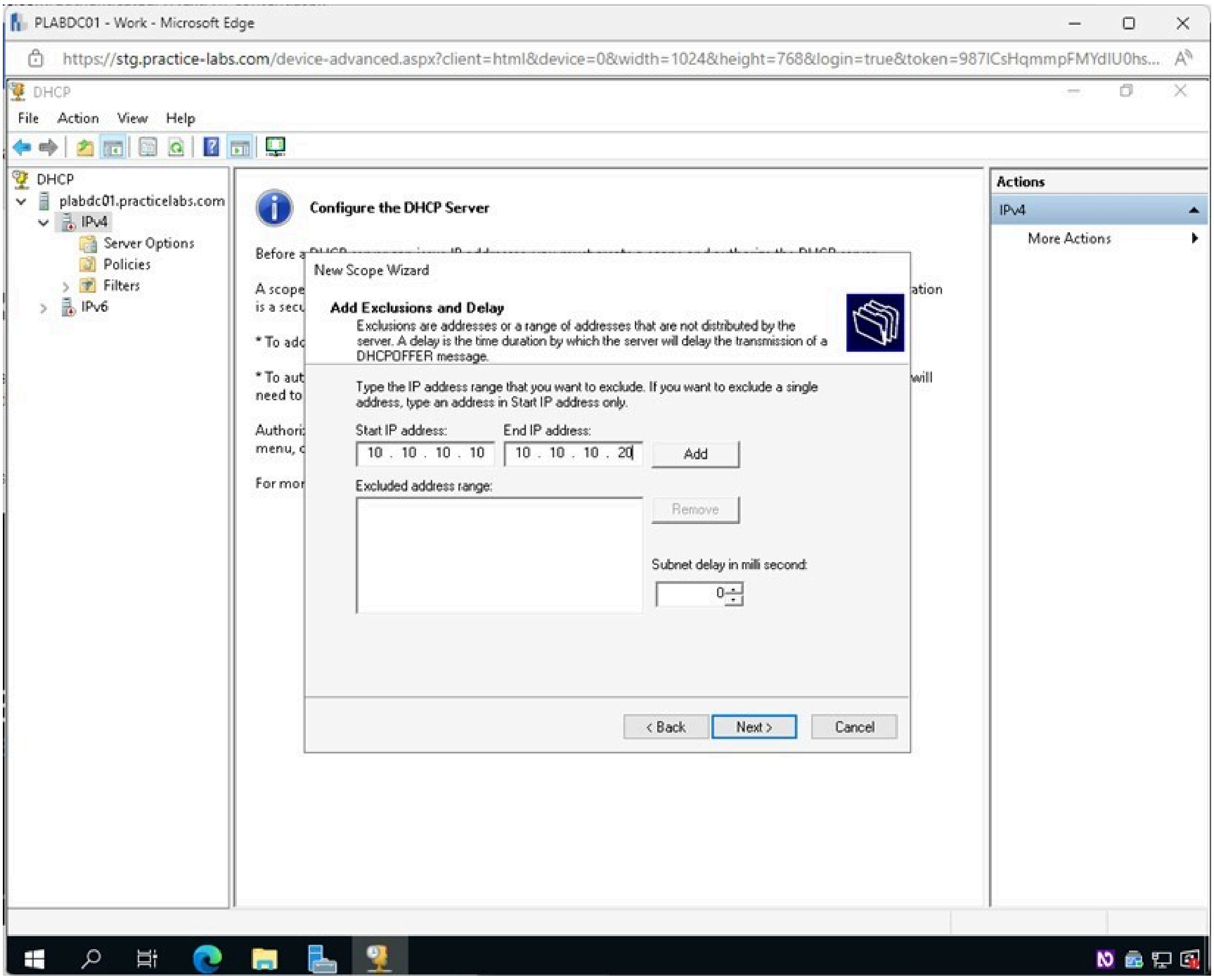

Figure 1.65 Screenshot of PLABDC01: Displaying the Add Exclusions and Delay page, showing the required settings performed and Add selected.

Step 23

The first scope to be excluded has been added. You will now add the second.

Type the following for the corresponding fields:

```
Start IP address:  10.10.10.120
End IP address:    10.10.10.140
```

Click **Add**.

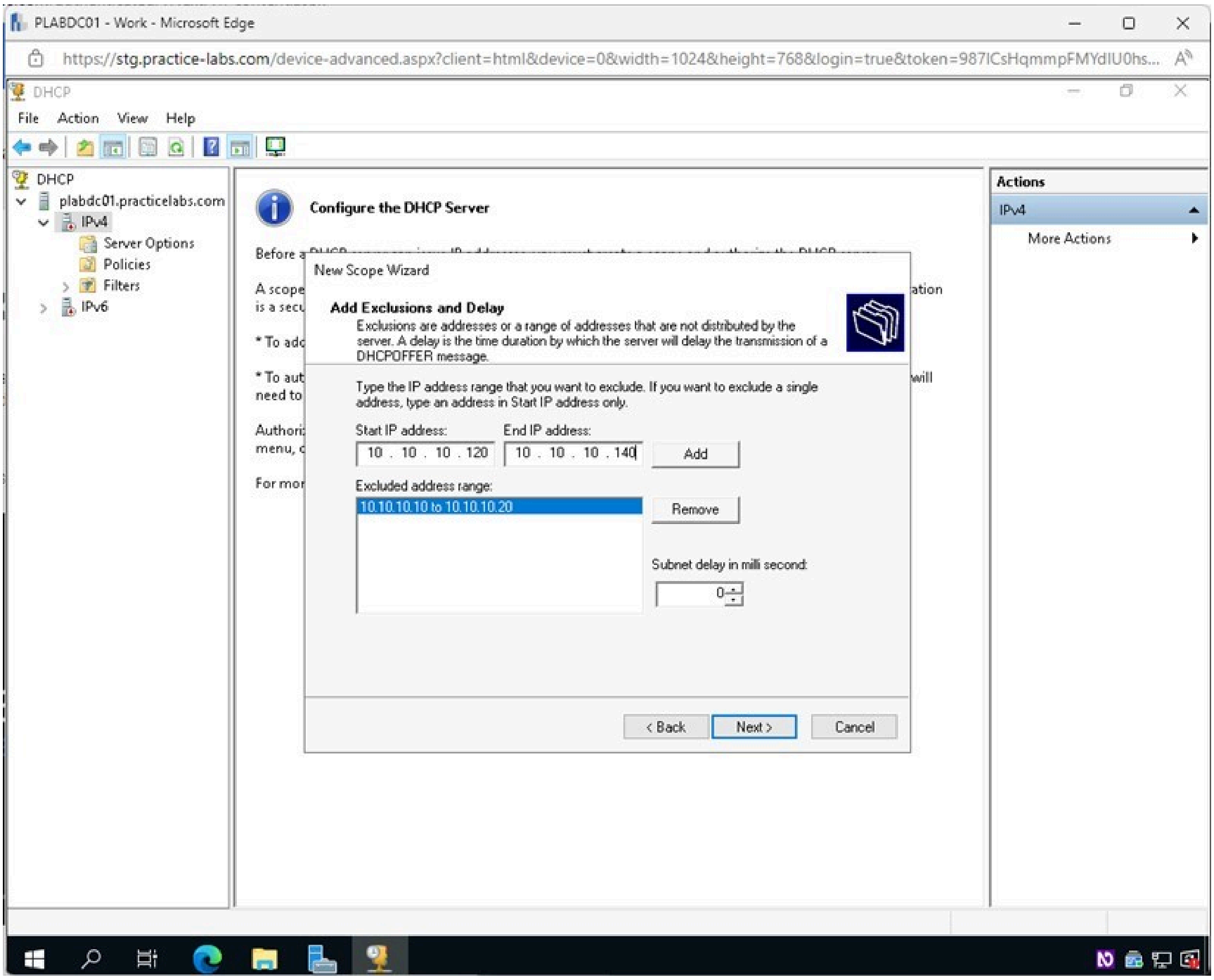

Figure 1.66 Screenshot of PLABDC01: Displaying the Add Exclusions and Delay page, showing the required settings performed and Add selected.

Step 24

Both desired scopes have now been created.

Click **Next** on the **Add Exclusions and Delay** page.

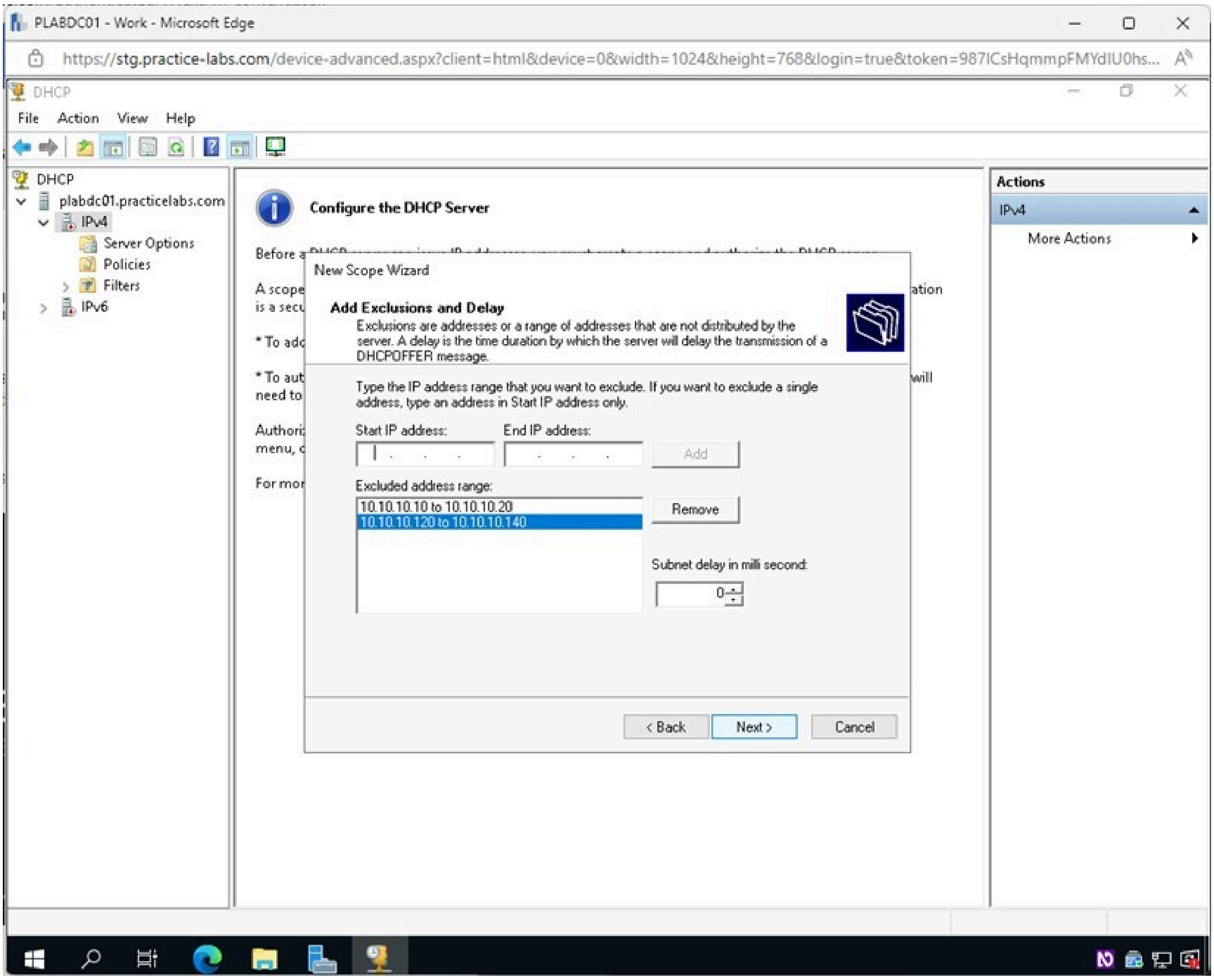

Figure 1.67 Screenshot of PLABDC01: Displaying selecting Next on the Add Exclusions and Delay page.

Step 25

In the **Lease Duration** page, click **Next**.

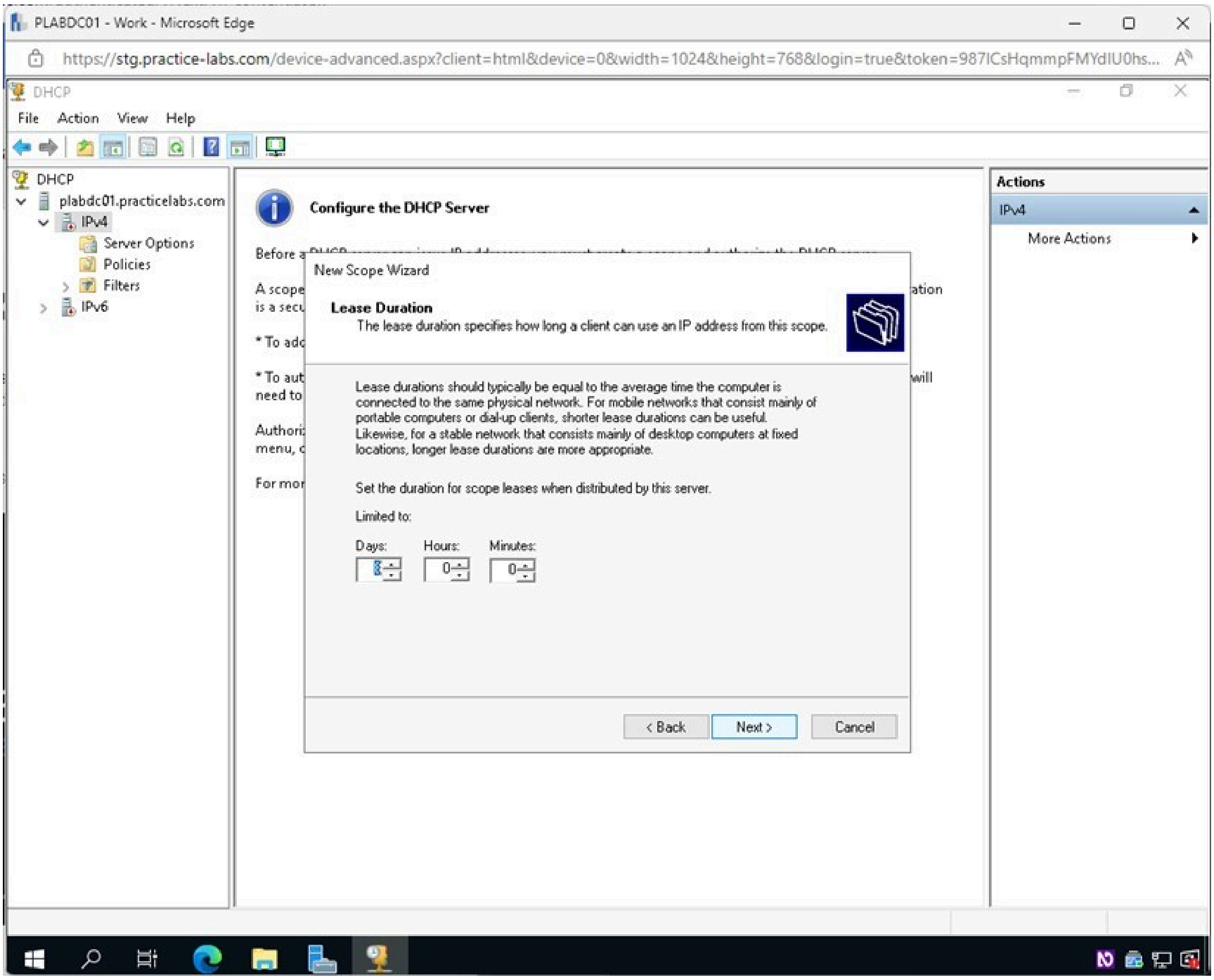

Figure 1.68 Screenshot of PLABDC01: Displaying selecting Next on the Lease Duration page.

Step 26

In the **Configure DHCP Options** page, select the **Yes, I want to configure these options now** option.

Click **Next**.

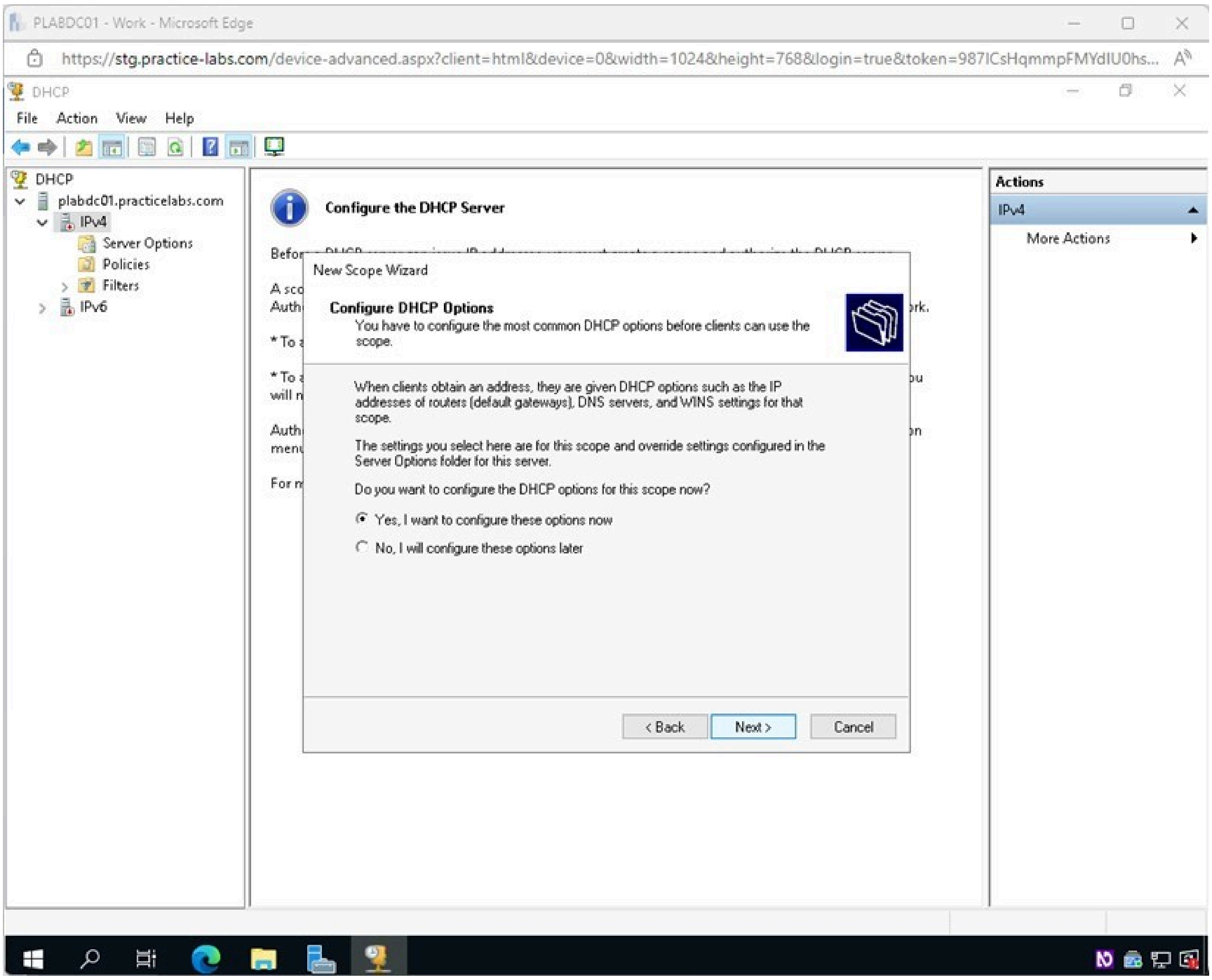

Figure 1.69 Screenshot of PLABDC01: Displaying the Configure DHCP page, showing the required option selected and the Next button highlighted.

Step 27

In the **Router (Default Gateway)** page, click **Next.**

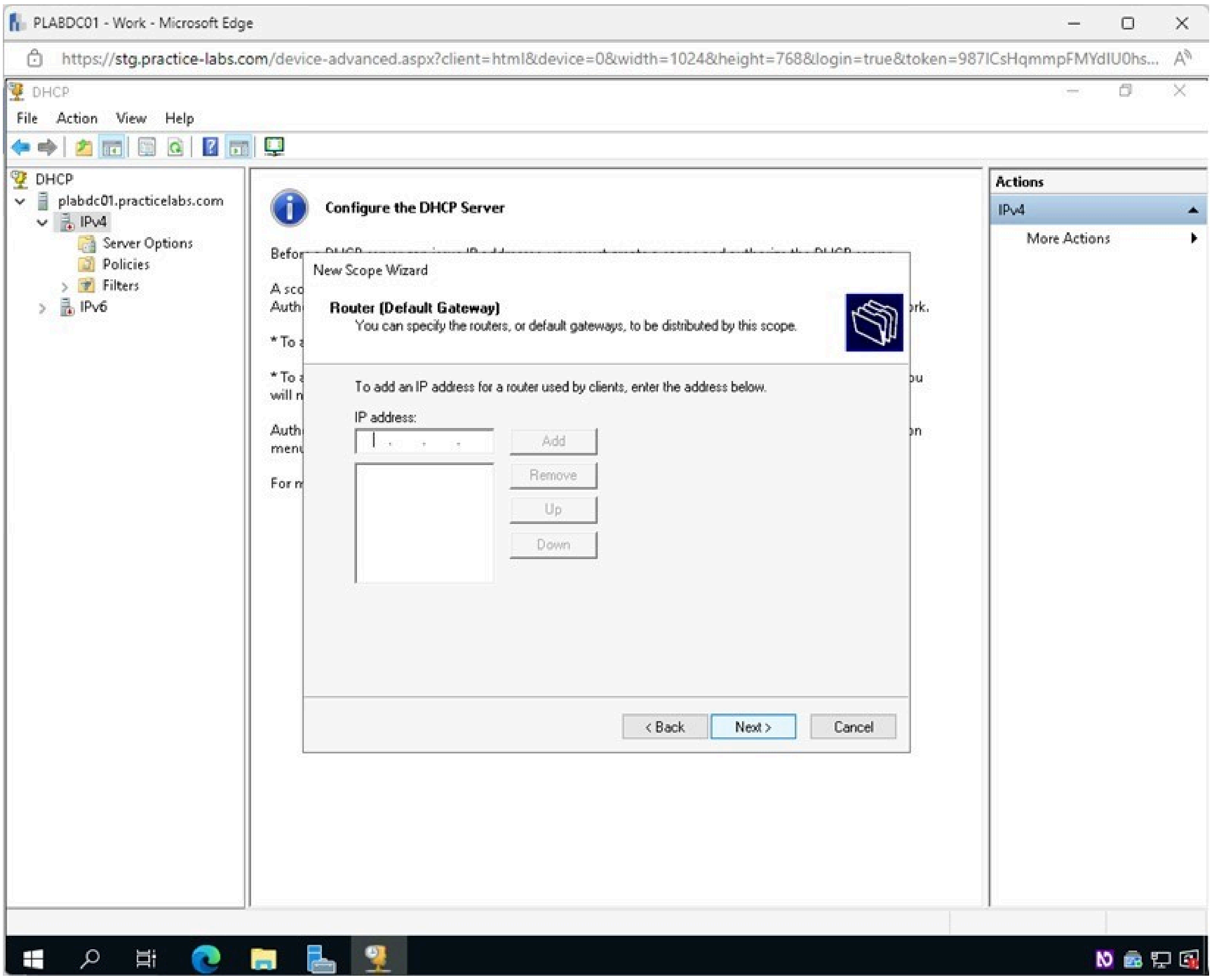

Figure 1.70 Screenshot of PLABDC01: Displaying selecting Next on the Router (Default Gateway) page.

Step 28

On the **Domain Name and DNS Servers** page, click **Next**.

> ***Note:*** *On the* ***Domain Name and DNS Servers*** *page, you can enter the network's address for the DNS Server.*

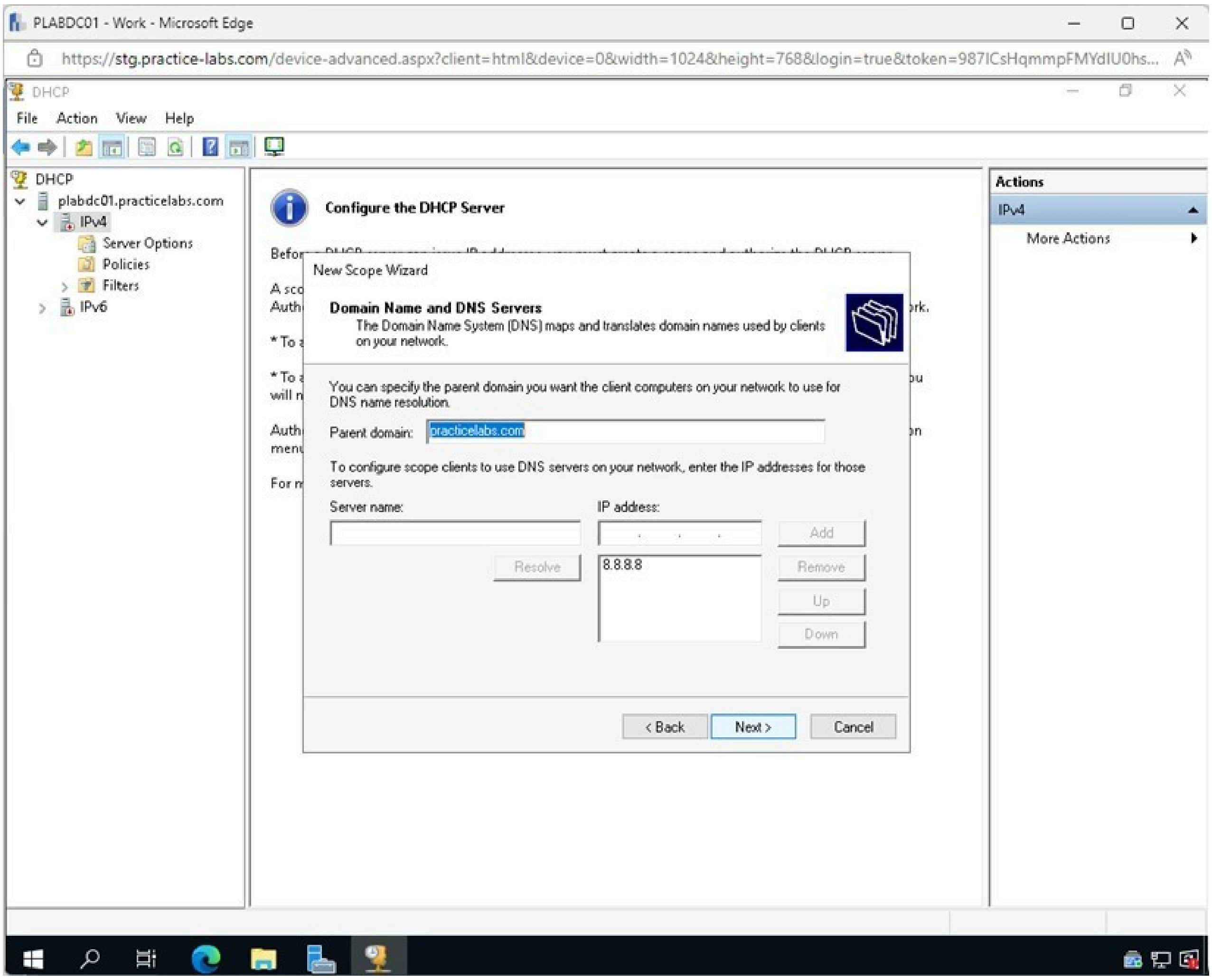

Figure 1.71 Screenshot of PLABDC01: Displaying selecting Next on the Domain Name and DNS Servers page.

Step 29

On the **WINS Servers** page, click **Next**.

> ***Note:*** *This is where you would enter the network's address for a WINS Server.*

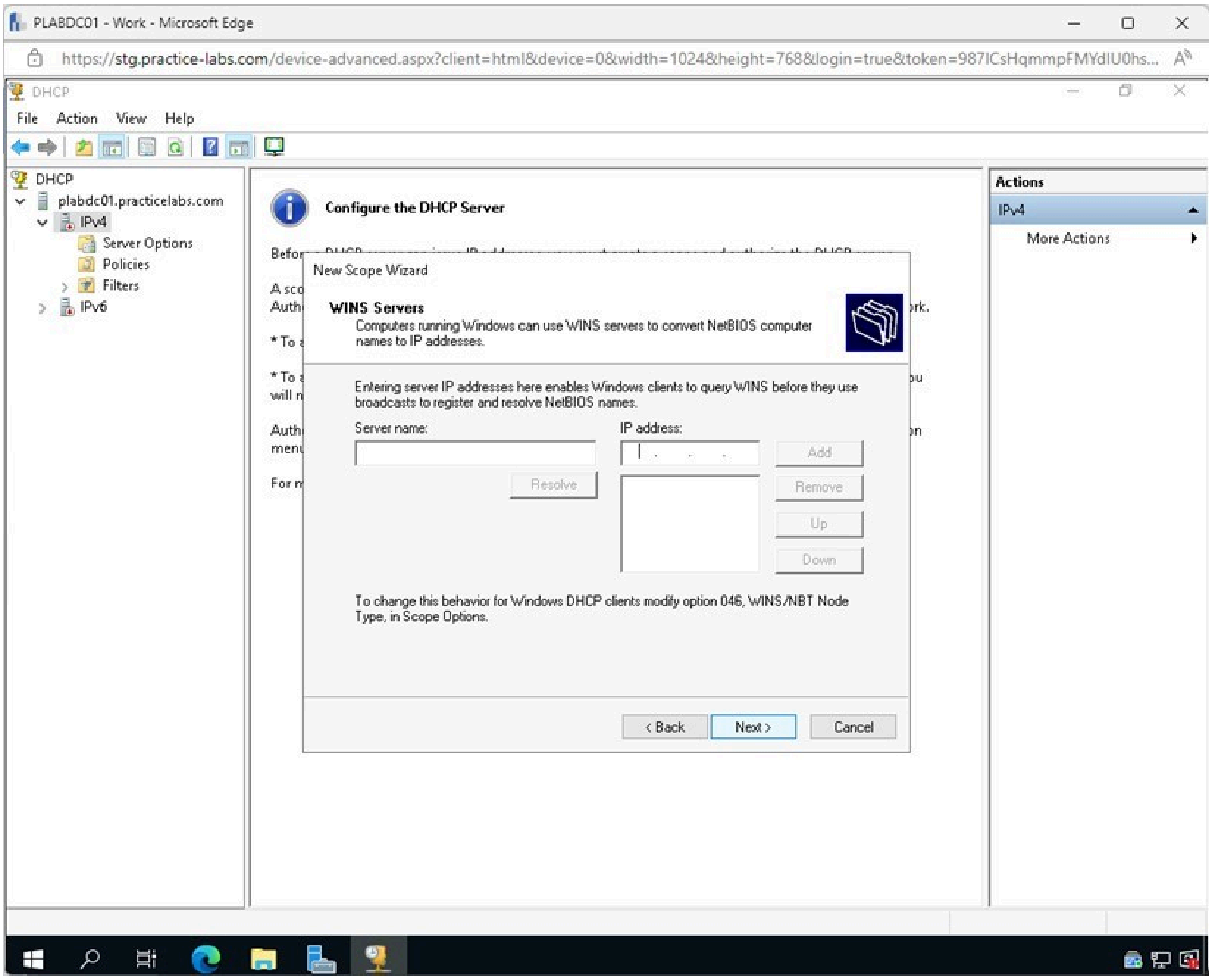

Figure 1.72 Screenshot of PLABDC01: Displaying selecting Next on the WINS Servers page.

Step 30

In the **Activate Scope** page, click **Next**.

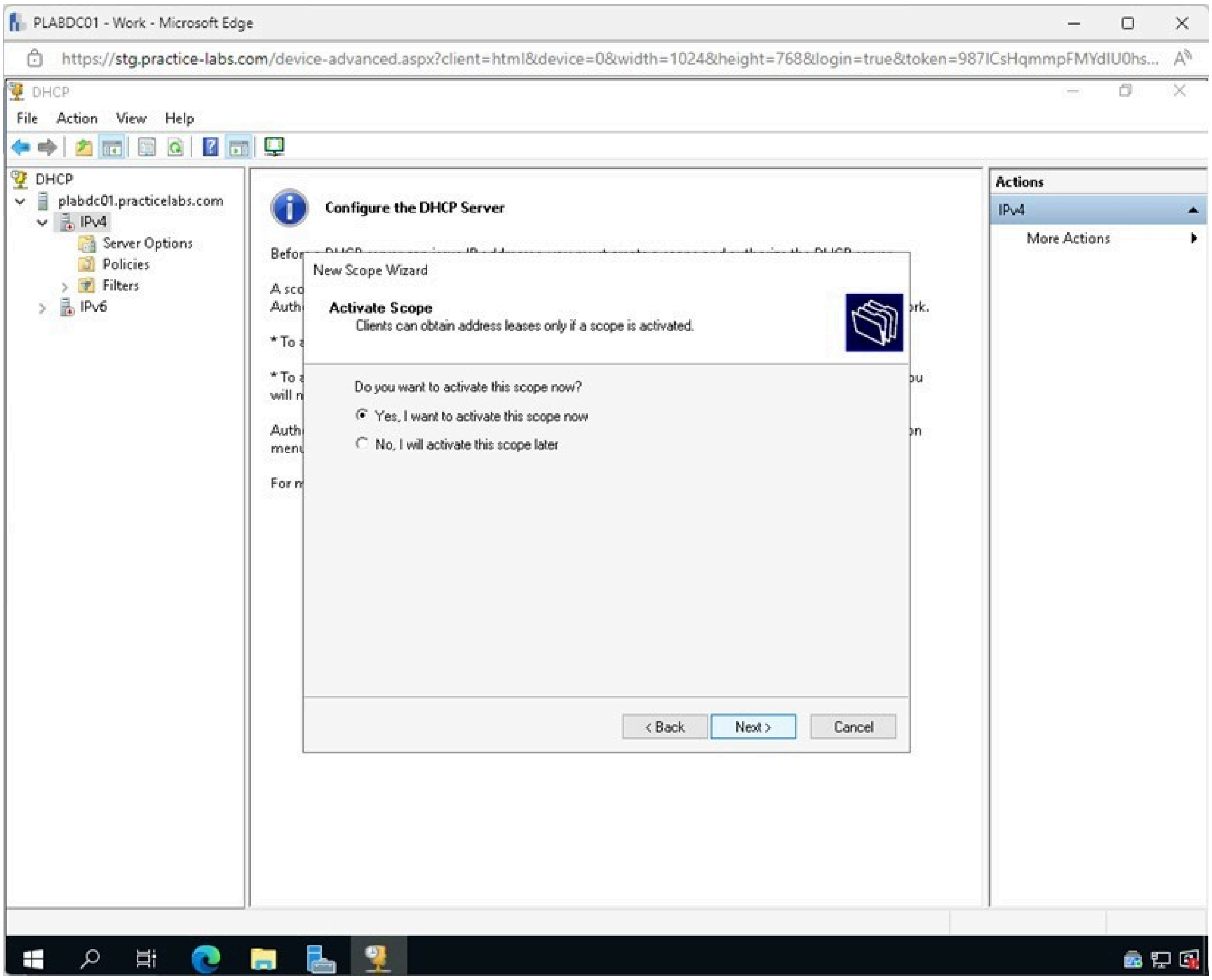

Figure 1.73 Screenshot of PLABDC01: Displaying selecting Next on the Activate Scope page.

Step 31

In the **Completing the New Scope Wizard** page, click **Finish**.

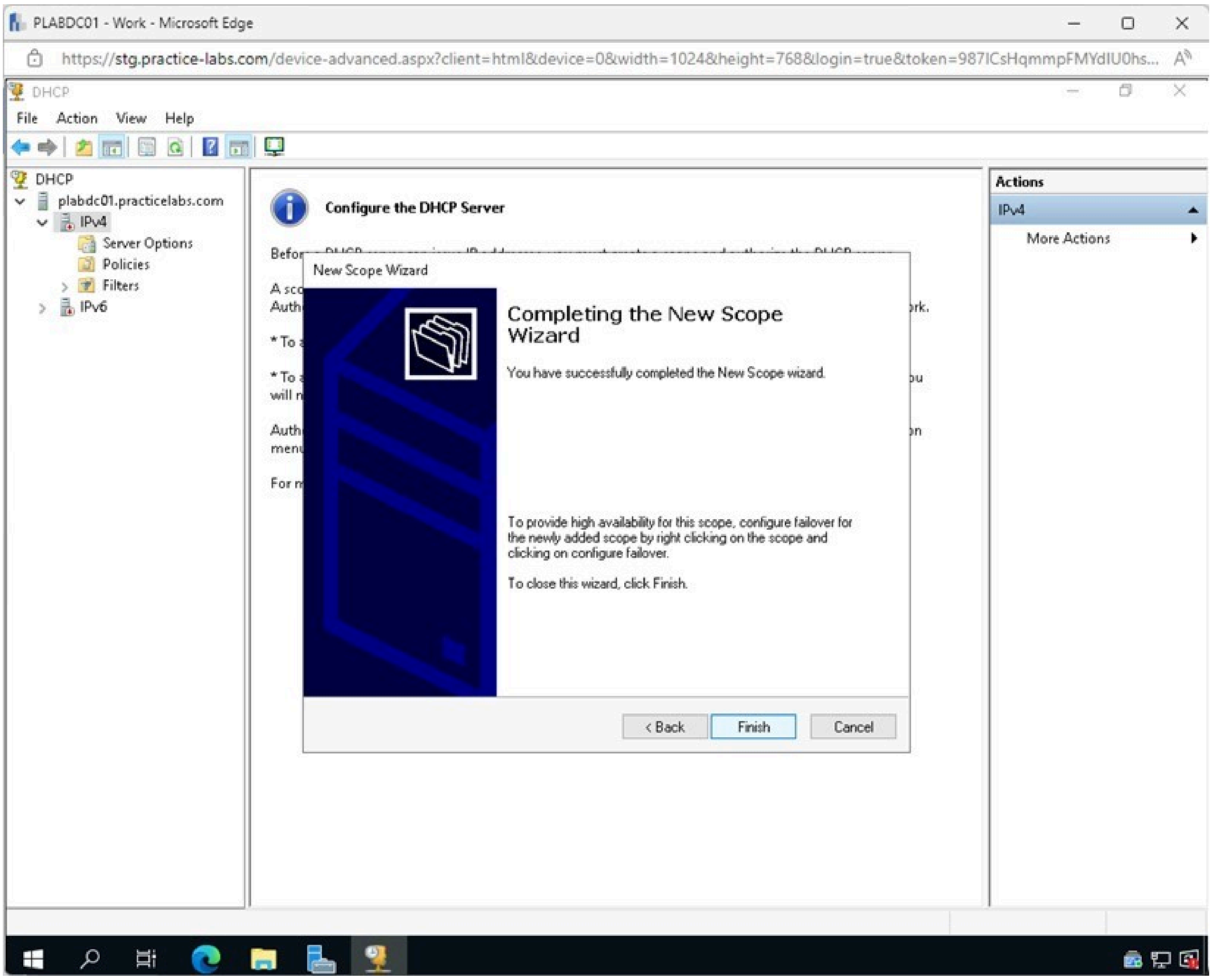

Figure 1.74 Screenshot of PLABDC01: Displaying selecting Finish on the Completing the New Scope Wizard page.

Step 32

Notice the new **Test Network** Scope now appears on the left pane.

Close the **DHCP** window.

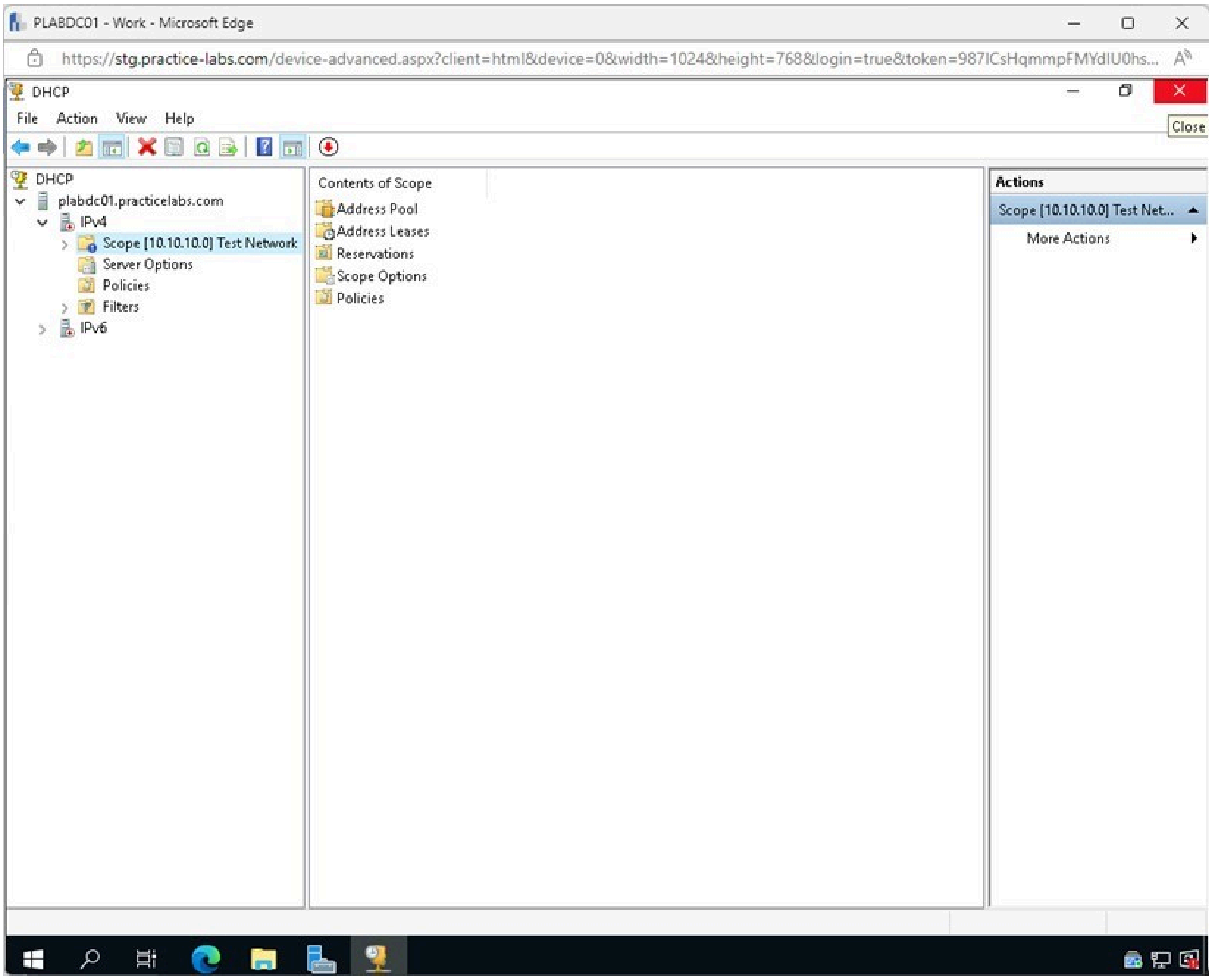

Figure 1.75 Screenshot of PLABDC01: Displaying closing the DHCP window.

Minimize the **Server Manager** window.

Exercise 2 - Command Line Tools and Techniques

The command line interface of an operating system is an extremely versatile tool for administrators. There are many built-in command line tools, and additional tools can also be added to increase the command lines' capabilities.

In this exercise, some common Windows and Linux commands will be explored.

Learning Outcomes

After completing this exercise, you should be able to:

- Explore Windows Commands Commonly used in Troubleshooting
- Explore Linux Commands

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **PLABDC01 -** (Windows Server 2019 - Domain Controller)
- **PLABUBUNTU -** (Linux Ubuntu - Stand-alone Linux Server)

Task 1 - Windows Commands Commonly used in Troubleshooting

In this task, several common command line tools that are used in troubleshooting will be explored.

- **ipconfig** - Internet Protocol Configuration can retrieve and show all information about the system's current Transmission Control Protocol/Internet Protocol (TCP/IP) configuration settings. When ipconfig is run for the first time after installing a new operating system, it will update the DNS and DHCP settings. If ipconfig is run without any arguments, it will simply list the IPv4 and IPv6 addresses, gateways, and subnet mask of the network interface cards.
- **ping -** Packet Internet or Inter-Network Groper is a simple Internet tool that allows a user to test and validate the existence of a certain destination IP address. Ping may be used from just about any OS with network support, including most embedded network administration tools.
- **tracert & traceroute -** It is a useful tool that shows how long it takes for a data packet to go from a certain IP address or domain to its destination. The output of a traceroute command is a list of the different routers known as hops that the packets interact with before arriving at the desired IP address or domain. Windows uses tracert, and Linux uses traceroute.
- **nslookup** - This command is used to discover the IP address or DNS record associated with a hostname, such as google.com. If a user has an IP address, they can also perform a reverse DNS query to discover the hostname associated with the IP address.
- **netstat** - It provides access to a wide range of network information depending on the parameters specified on the command line. It's possible to check the status of TCP and UDP connections, as well as view the contents of routing tables.
- **nbtstat -** It provides users the ability to resolve issues with naming in NETBIOS. Running the command will show active TCP/IP connections utilizing NETBIOS and other useful statistics.

Step 1

Connect to **PLABDC01**.

Click the **Start** charm and type the following:

```
cmd
```

Select **Command Prompt** from the **Best match** pop-up menu.

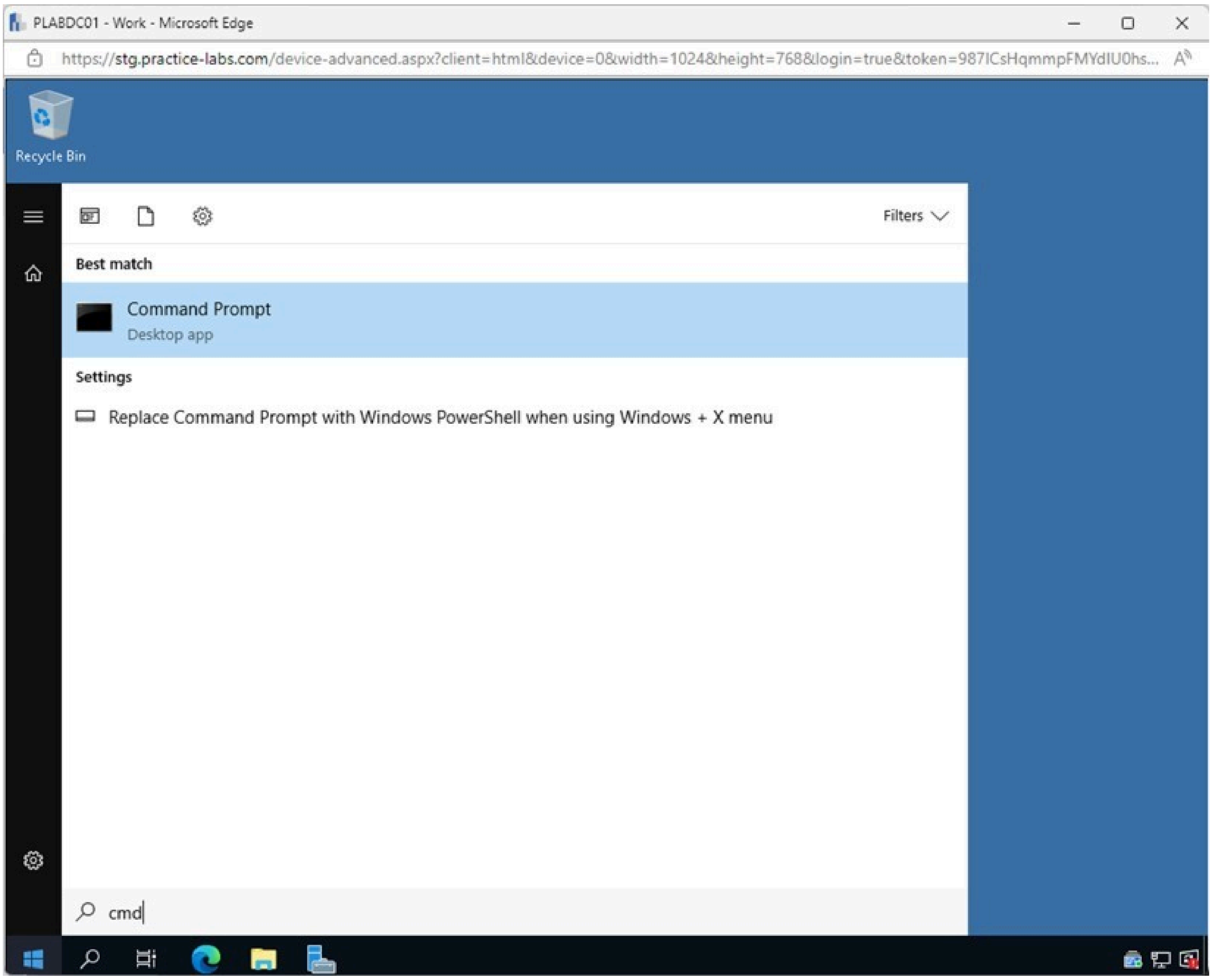

Figure 2.1 Screenshot of PLABDC01: Displaying selecting Command Prompt from the Best match pop-up menu.

Step 2

In the **Command Prompt** window, type the following command and press **Enter**:

```
ipconfig /?
```

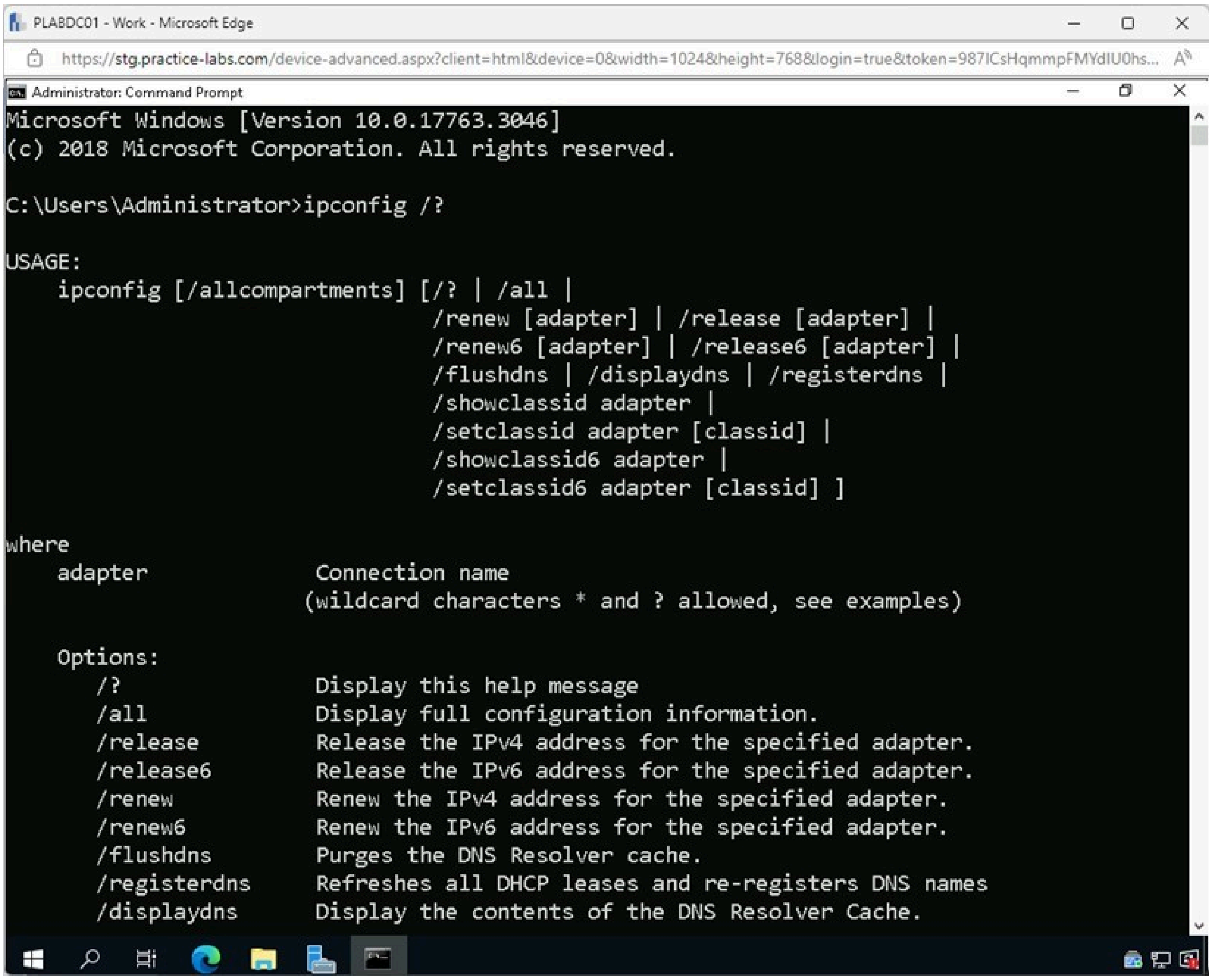

Figure 2.2 Screenshot of PLABDC01: Displaying entering and executing the required command on the Command Prompt window.

Note: *The **ipconfig** command, when used with the **/?** parameter lists all the options that are available to run with the ipconfig command.*
*In the last exercise, you configured a DHCP Server. If the server were misconfigured and fixed, a combination of the **ipconfig /release** and **ipconfig /renew** would be used to give the computer a new IP address.*
*When unable to get to an address by the web address, **ipconfig /flushdns** can be used to help flush the DNS cache and force the DNS Server to refresh its information.*

Step 3

Type the following command and press **Enter**:

```
ipconfig
```

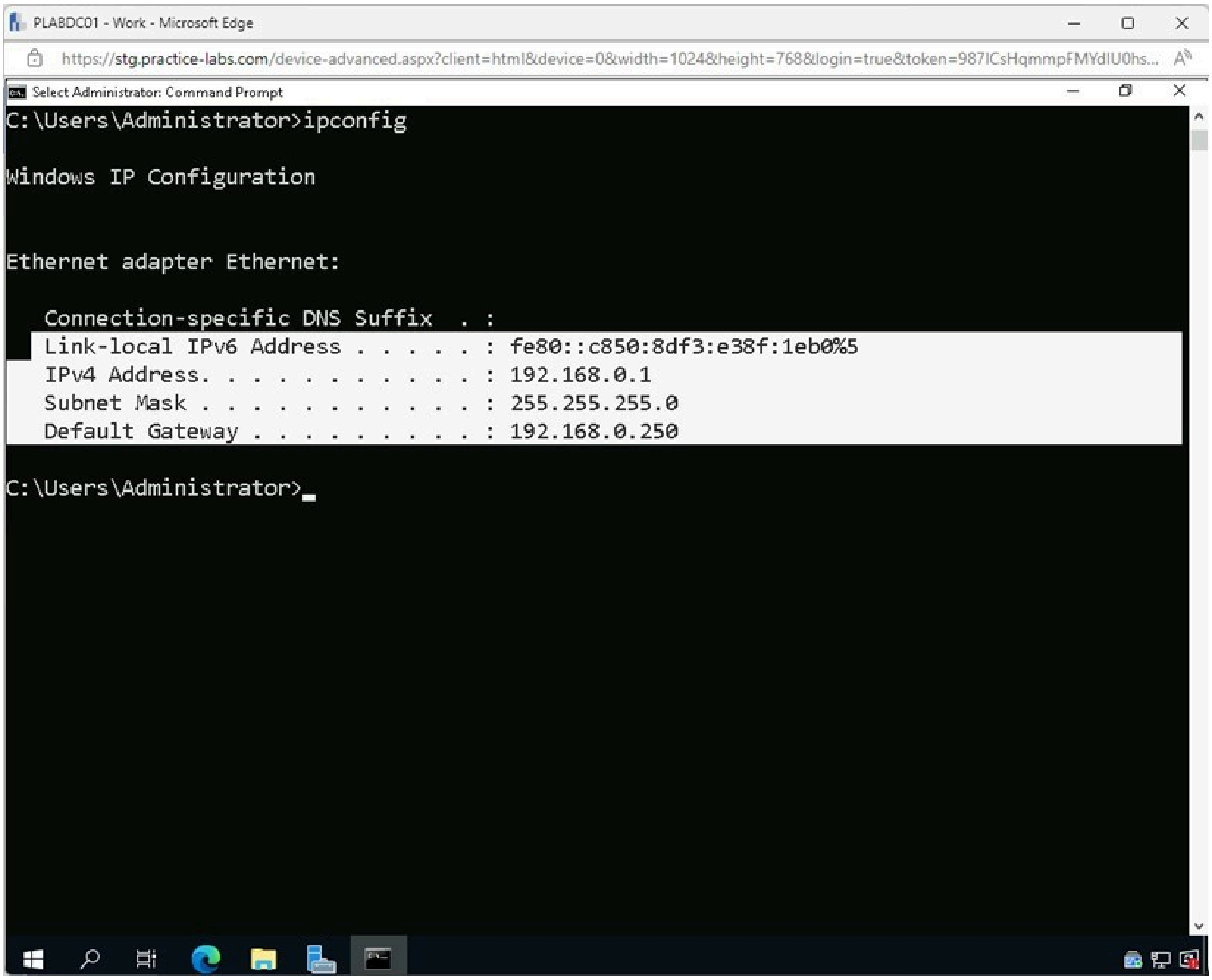

Figure 2.3 Screenshot of PLABDC01: Displaying entering and executing the required command on the Command Prompt window.

Note: *The above command will list the Link-local IPv6 Address, the IPv4 Address, Subnet Mask and Default Gateway.*

Step 4

Type the following and press **Enter** to clear the screen:

```
cls
```

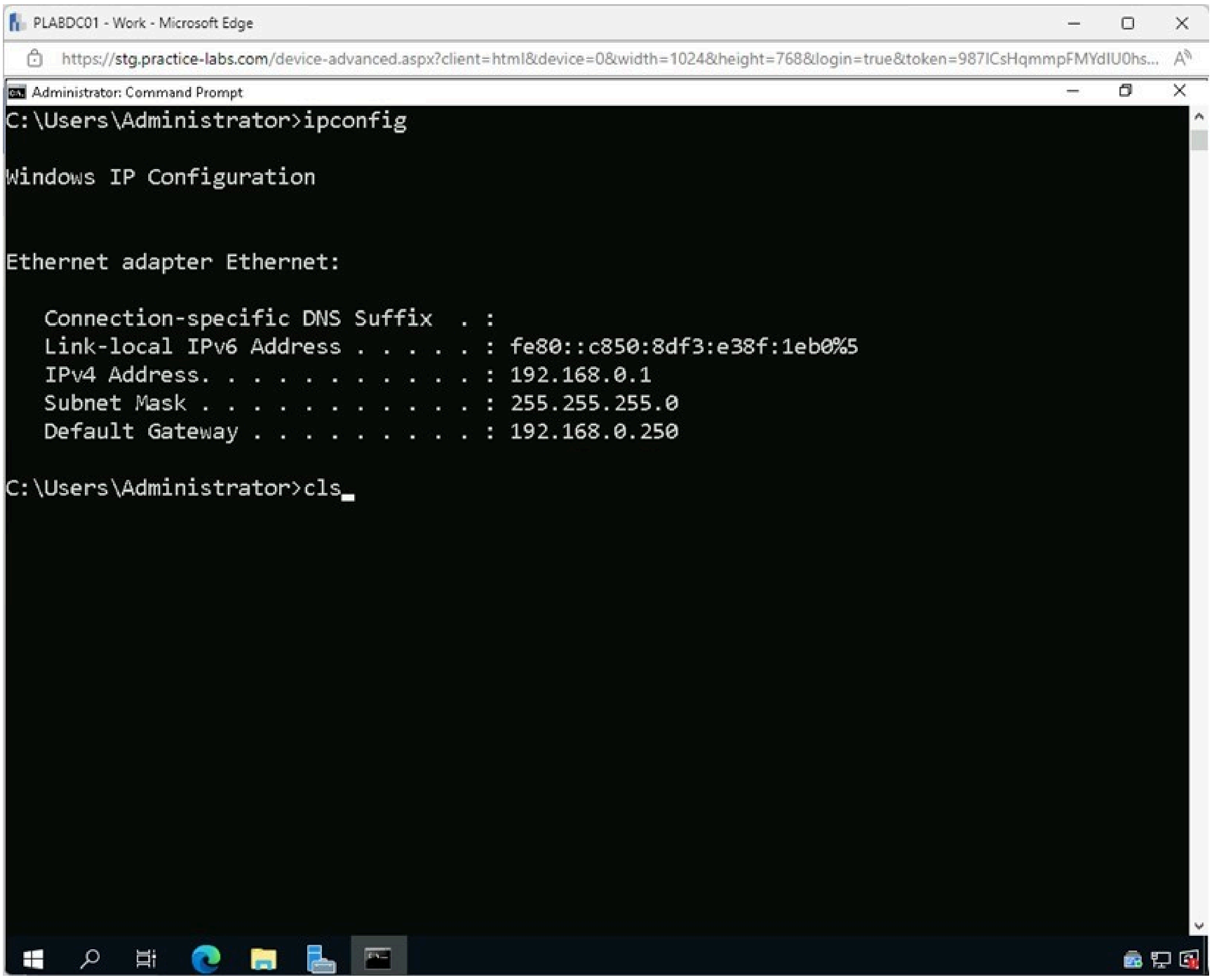

Figure 2.4 Screenshot of PLABDC01: Displaying entering the required command on the Command Prompt window.

Step 5

Type the following command and press **Enter**:

```
ping /?
```

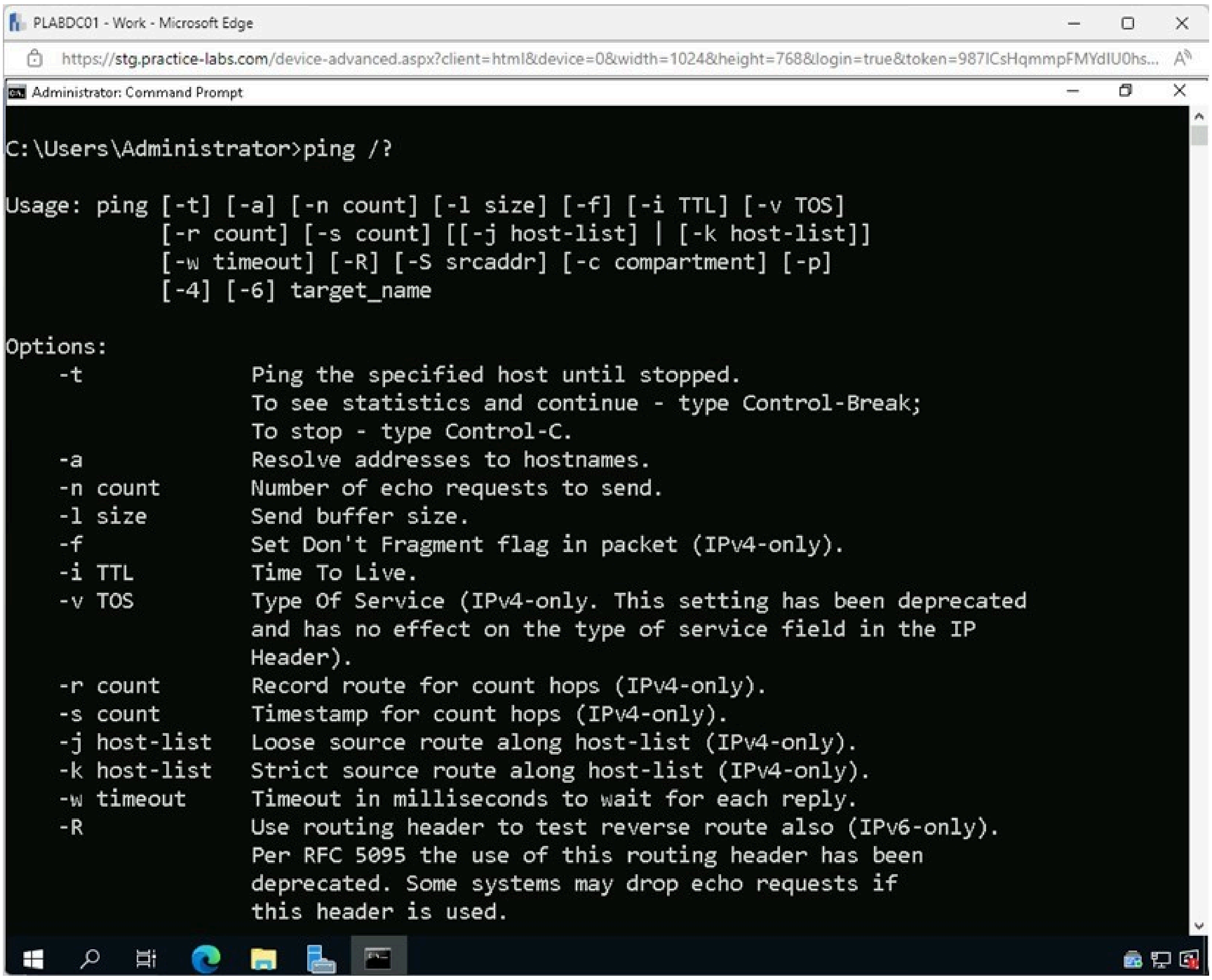

Figure 2.5 Screenshot of PLABDC01: Displaying entering and executing the required command on the Command Prompt window.

> ***Note:*** *The **ping** help page is listed.*

Step 6

Type the following command and press **Enter**:

```
ping google.com
```

> ***Note:*** *You can enter the **cls** command and press **Enter** to clear the screen before typing the other commands.*

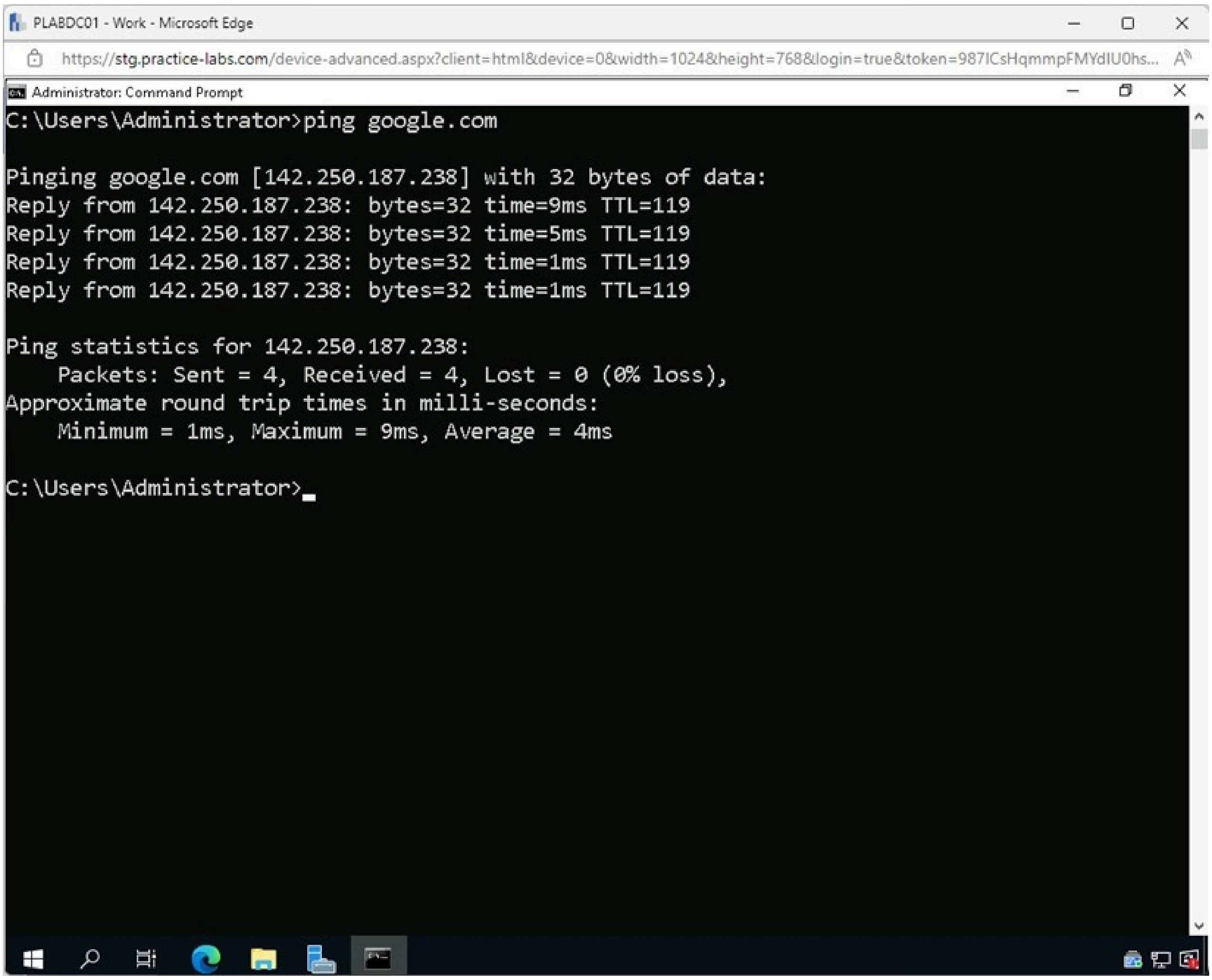

Figure 2.6 Screenshot of PLABDC01: Displaying entering and executing the required command on the Command Prompt window.

> ***Note:*** *Notice the ping sends 4 exploratory packets to see if there is connectivity with the google.com web server.*
> *Sometimes, a server may be down, or the information stored in the DNS Server may be bad. When unable to get to an address by the web address but can be reached by the IP address, the DNS will need to be flushed. Using the ipconfig /flushdns command, the DNS cache is cleared, forcing the DNS Server to refresh its information.*

Step 7

Type the following command and press **Enter**:

```
tracert /?
```

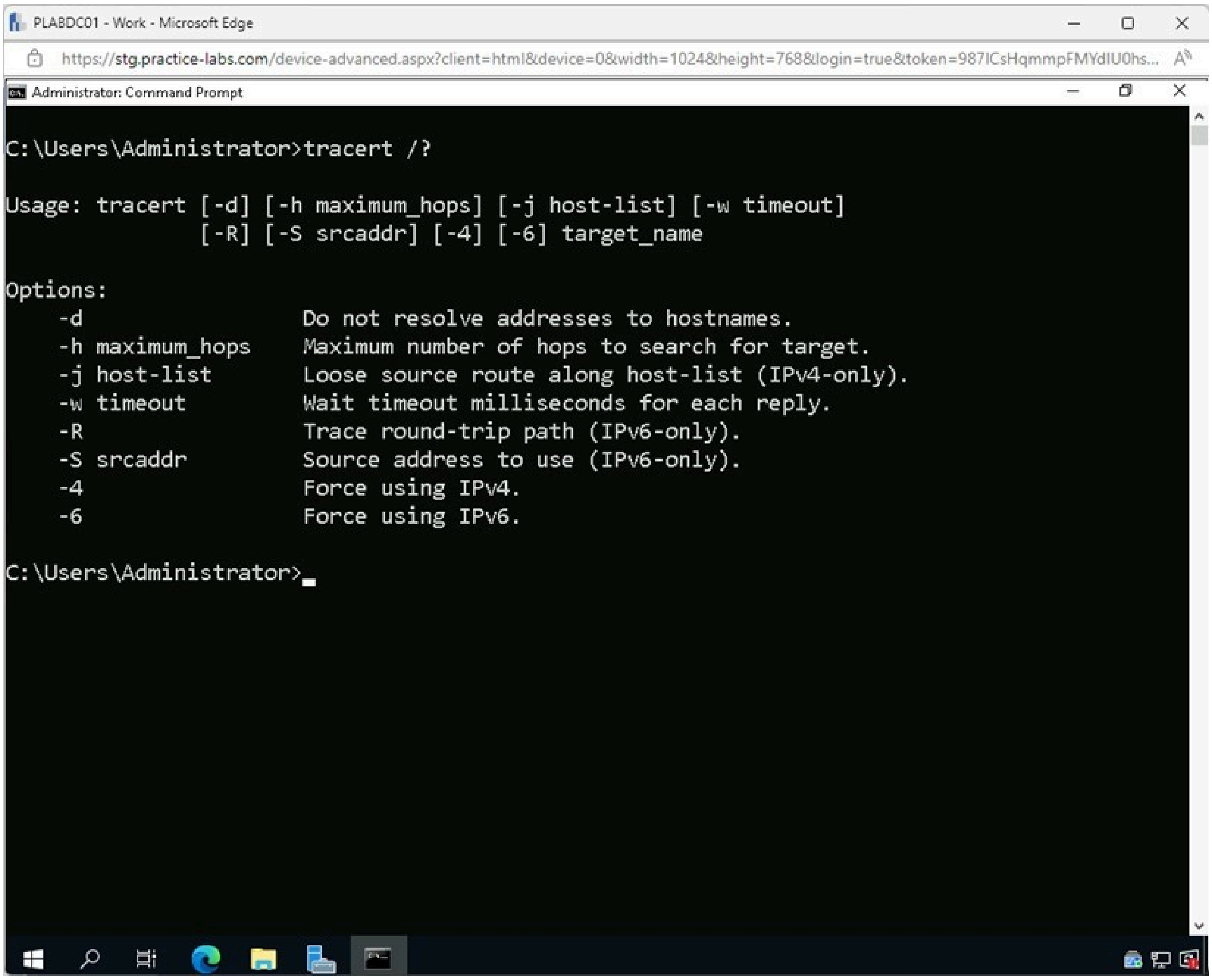

Figure 2.7 Screenshot of PLABDC01: Displaying entering and executing the required command on the Command Prompt window.

> ***Note:*** *The help page for the* ***tracert*** *command is listed.*

Step 8

Type the following command and press **Enter**:

```
nslookup /?
```

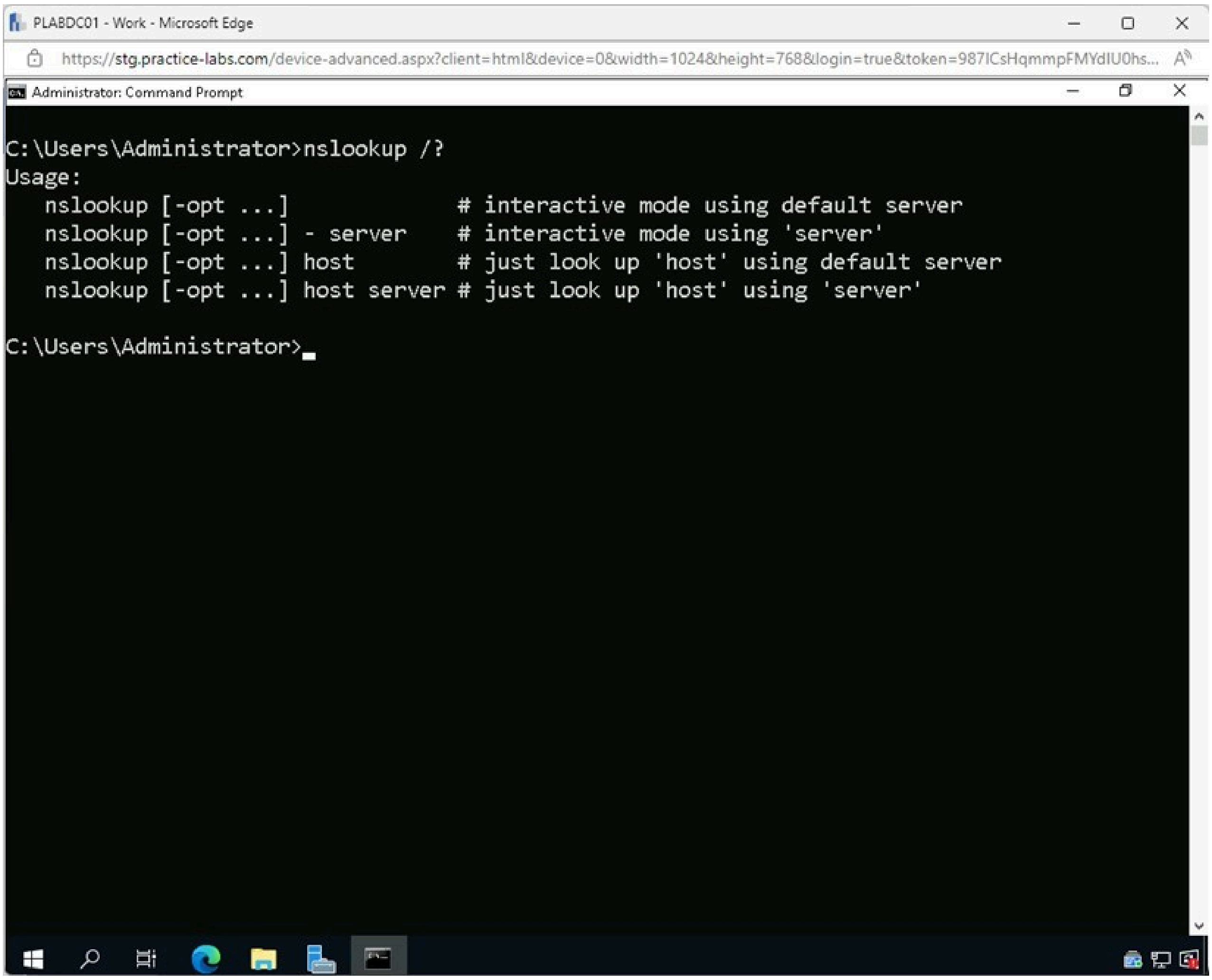

Figure 2.8 Screenshot of PLABDC01: Displaying entering and executing the required command on the Command Prompt window.

Note: *The help page for the* ***nslookup*** *command is listed.*

Step 9

Type the following command and press **Enter**:

```
nslookup google.com
```

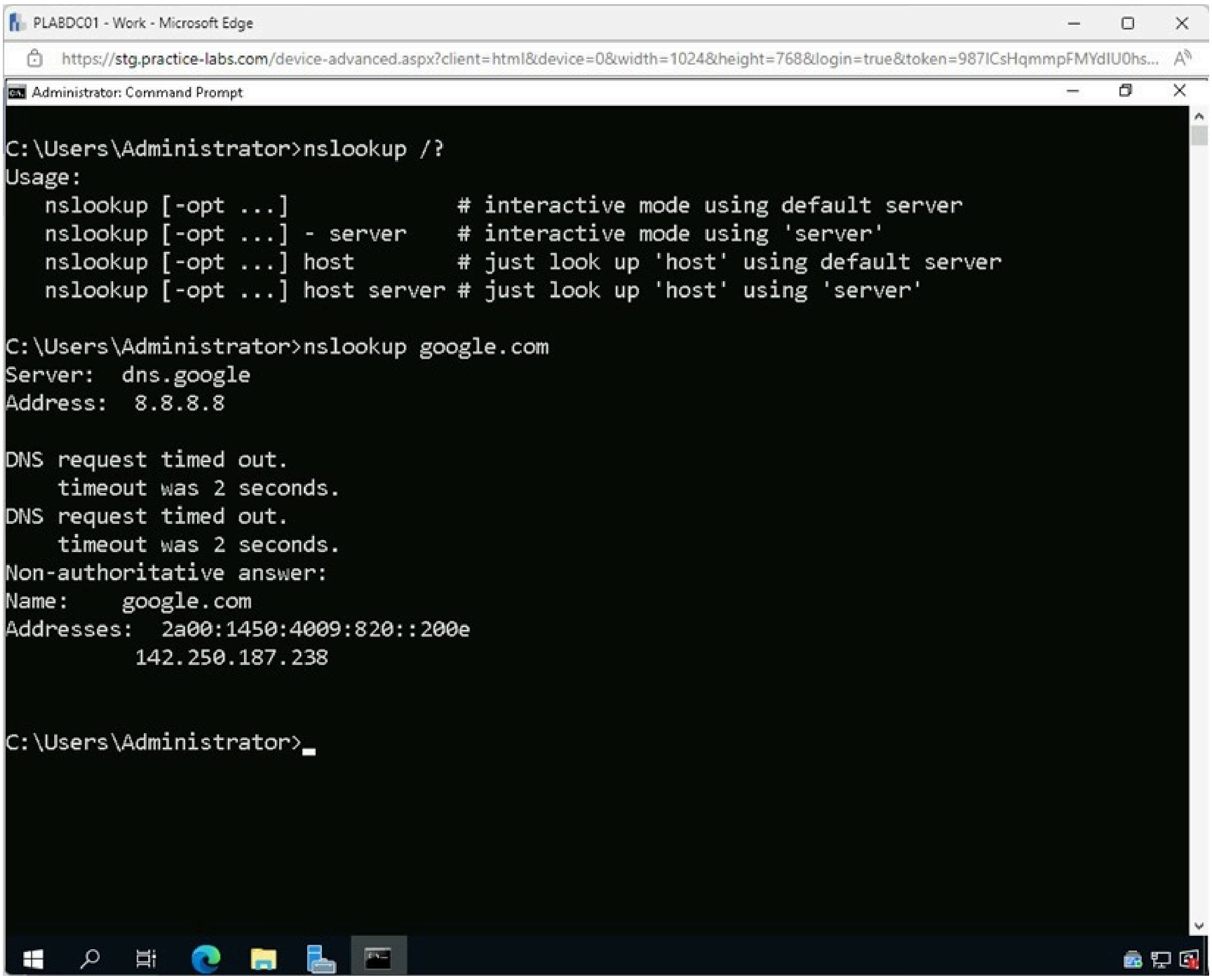

Figure 2.9 Screenshot of PLABDC01: Displaying entering and executing the required command on the Command Prompt window.

> ***Note:*** *The* ***nslookup*** *command finds the different IP addresses associated with google.com.*

Step 10

Type the following command and press **Enter**:

```
netstat /?
```

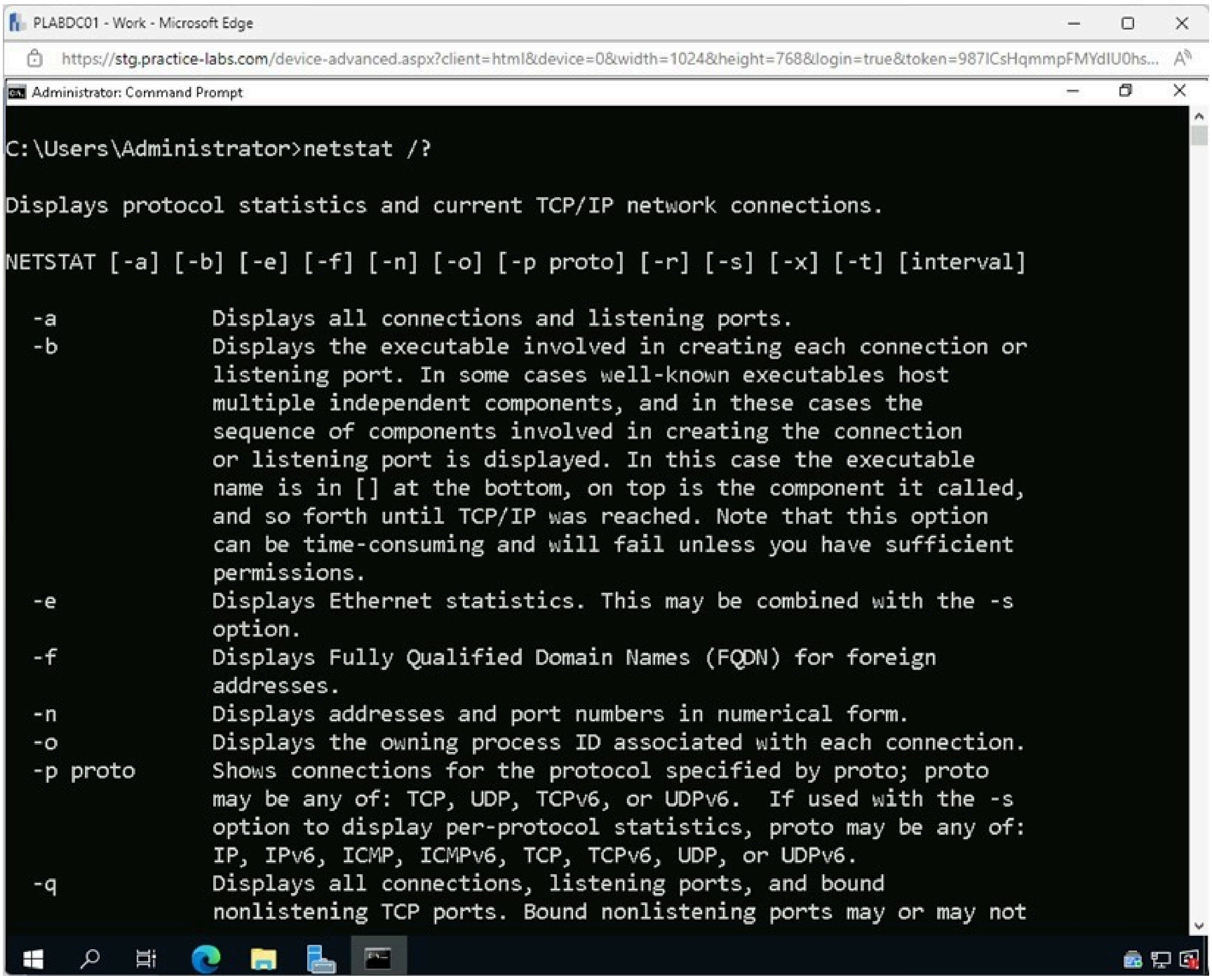

Figure 2.10 Screenshot of PLABDC01: Displaying entering and executing the required command on the Command Prompt window.

> ***Note:*** *The help page for the* ***netstat*** *command is listed.*

Step 11

Type the following command and press **Enter**:

```
netstat
```

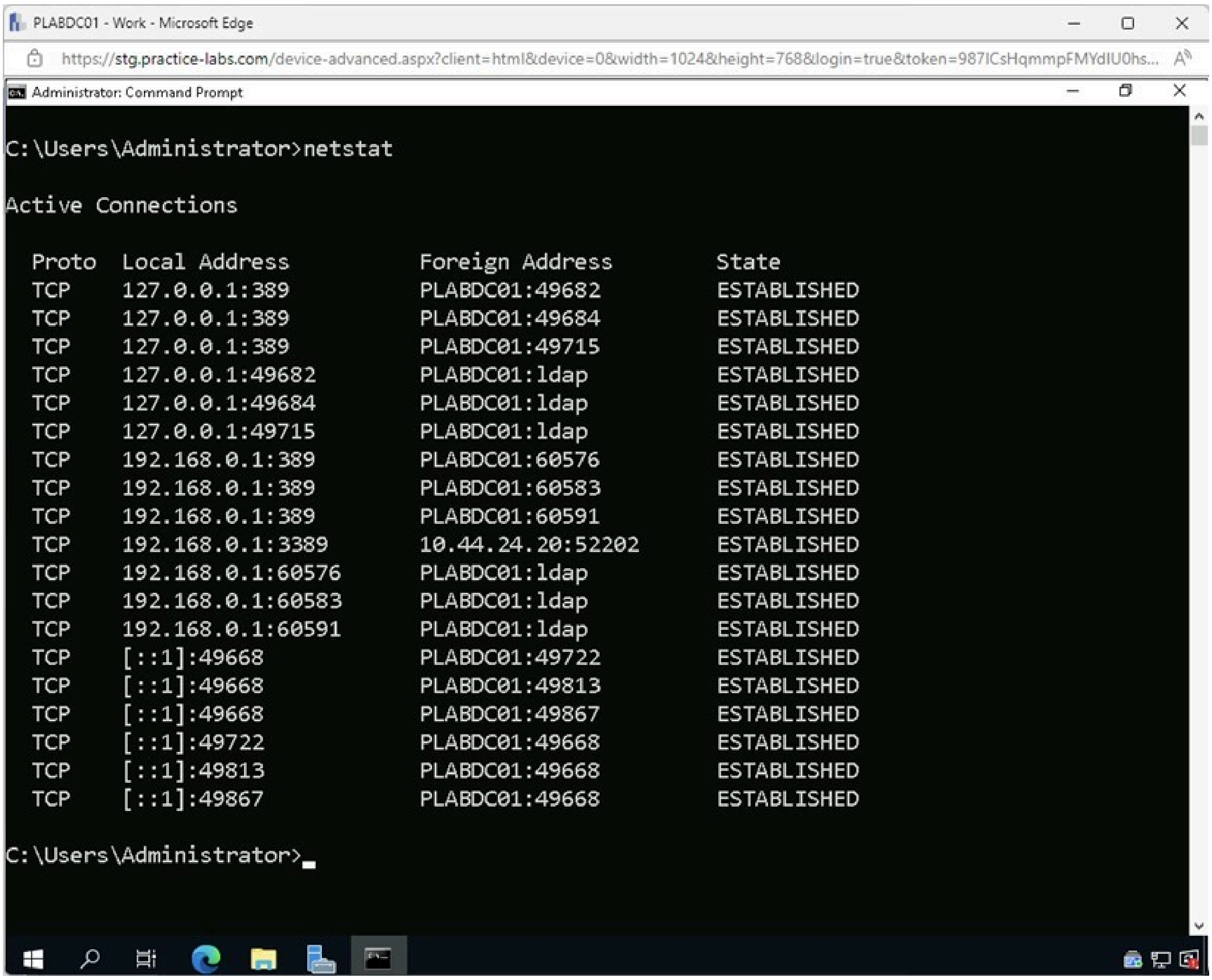

Figure 2.11 Screenshot of PLABDC01: Displaying entering and executing the required command on the Command Prompt window.

> ***Note:*** *The* **netstat** *command shows the different TCP connections that are currently active with the computer.*

Step 12

Type the following command and press **Enter**:

```
nbtstat /?
```

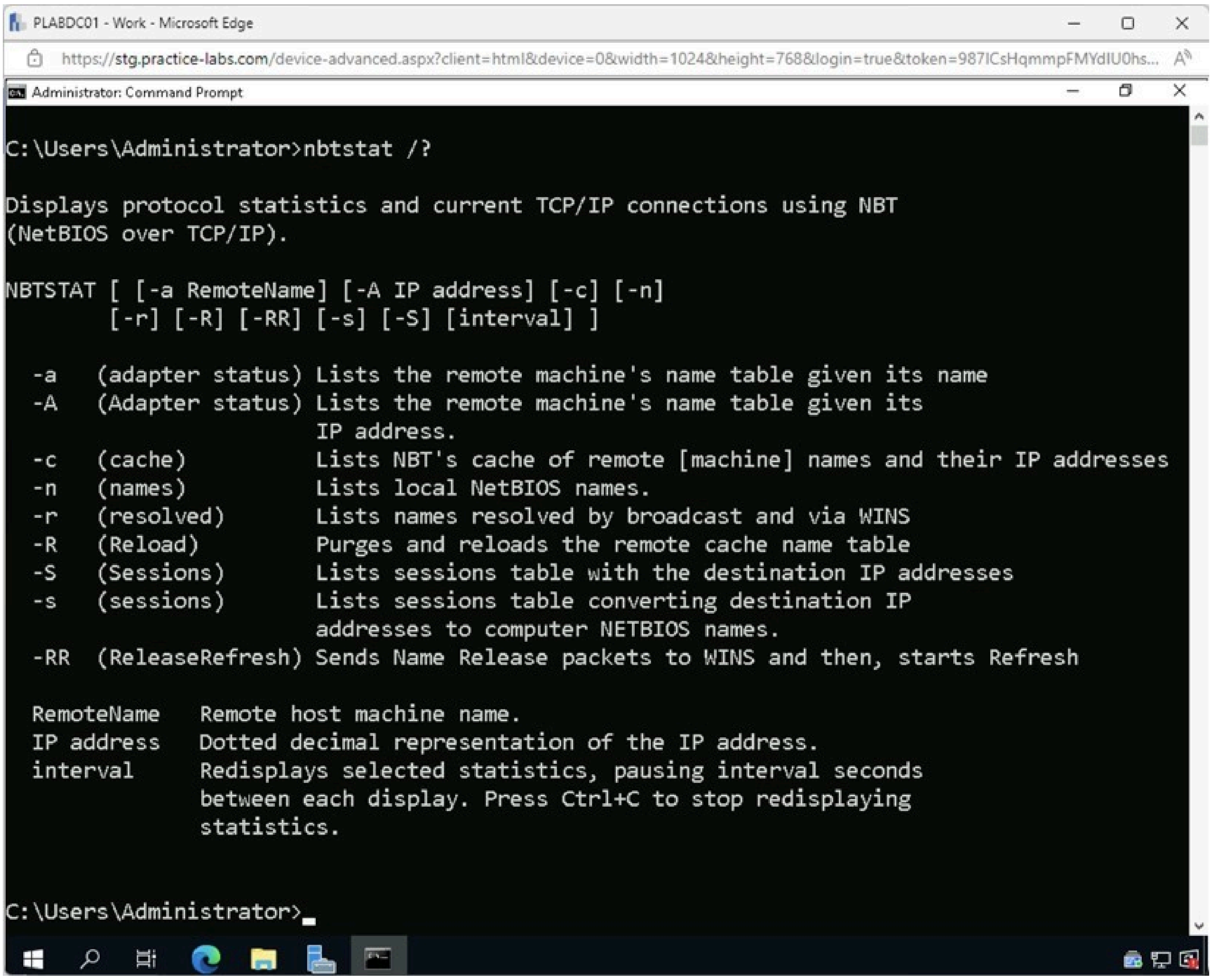

Figure 2.12 Screenshot of PLABDC01: Displaying entering and executing the required command on the Command Prompt window.

> ***Note:*** *The help page for the* ***nbtstat*** *command is listed.*

Step 13

Type the following command and press **Enter**:

```
nbtstat -n
```

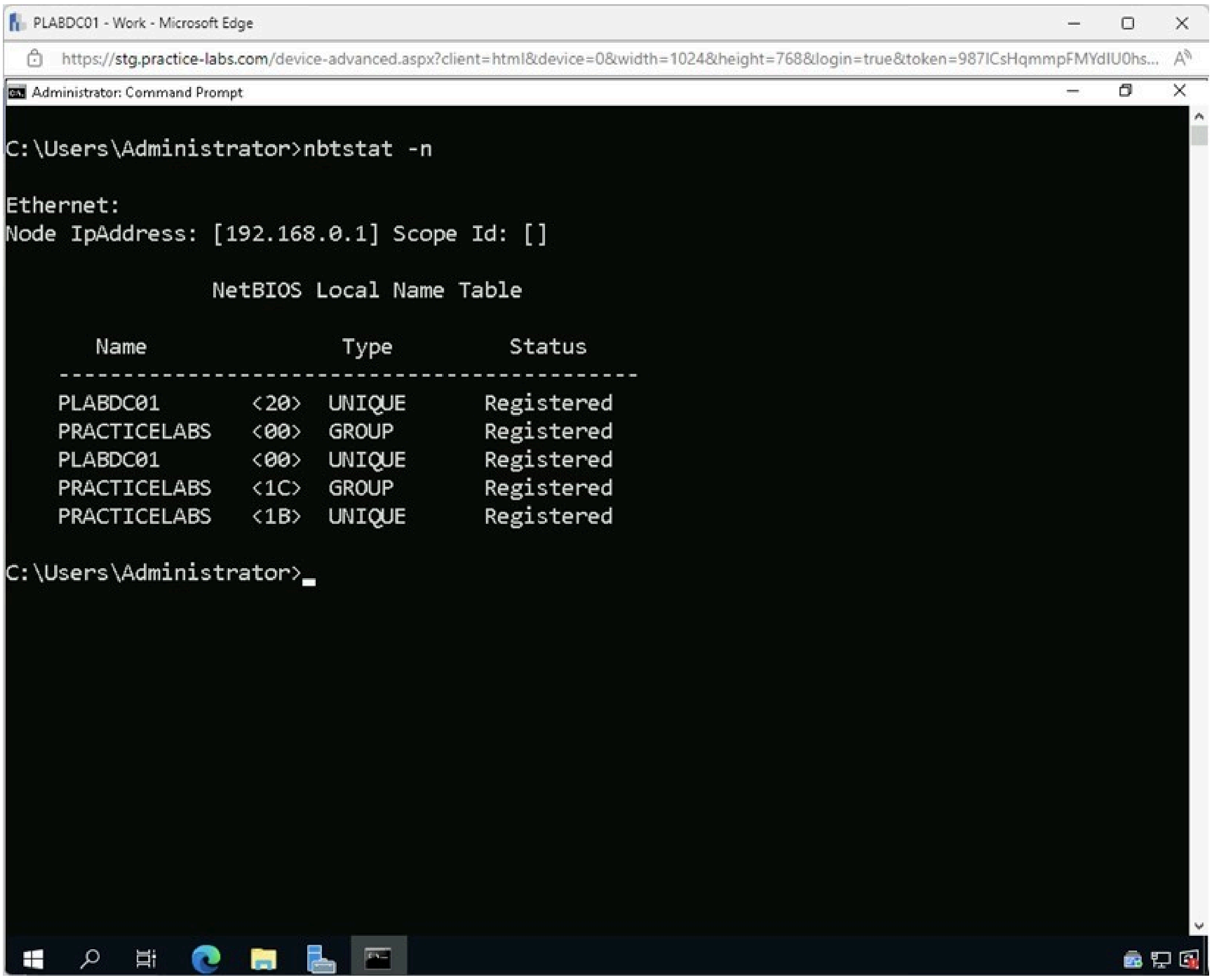

Figure 2.13 Screenshot of PLABDC01: Displaying entering and executing the required command on the Command Prompt window.

> ***Note:*** *The* ***nbtstat*** *command in the Command Prompt can be used to check and verify protocol statics and TCP/IP connections to the local host. Using the command with the* ***-n*** *parameter displays NetBIOS names of remotely connected hosts.*

Step 14

Close the **Command Prompt** window.

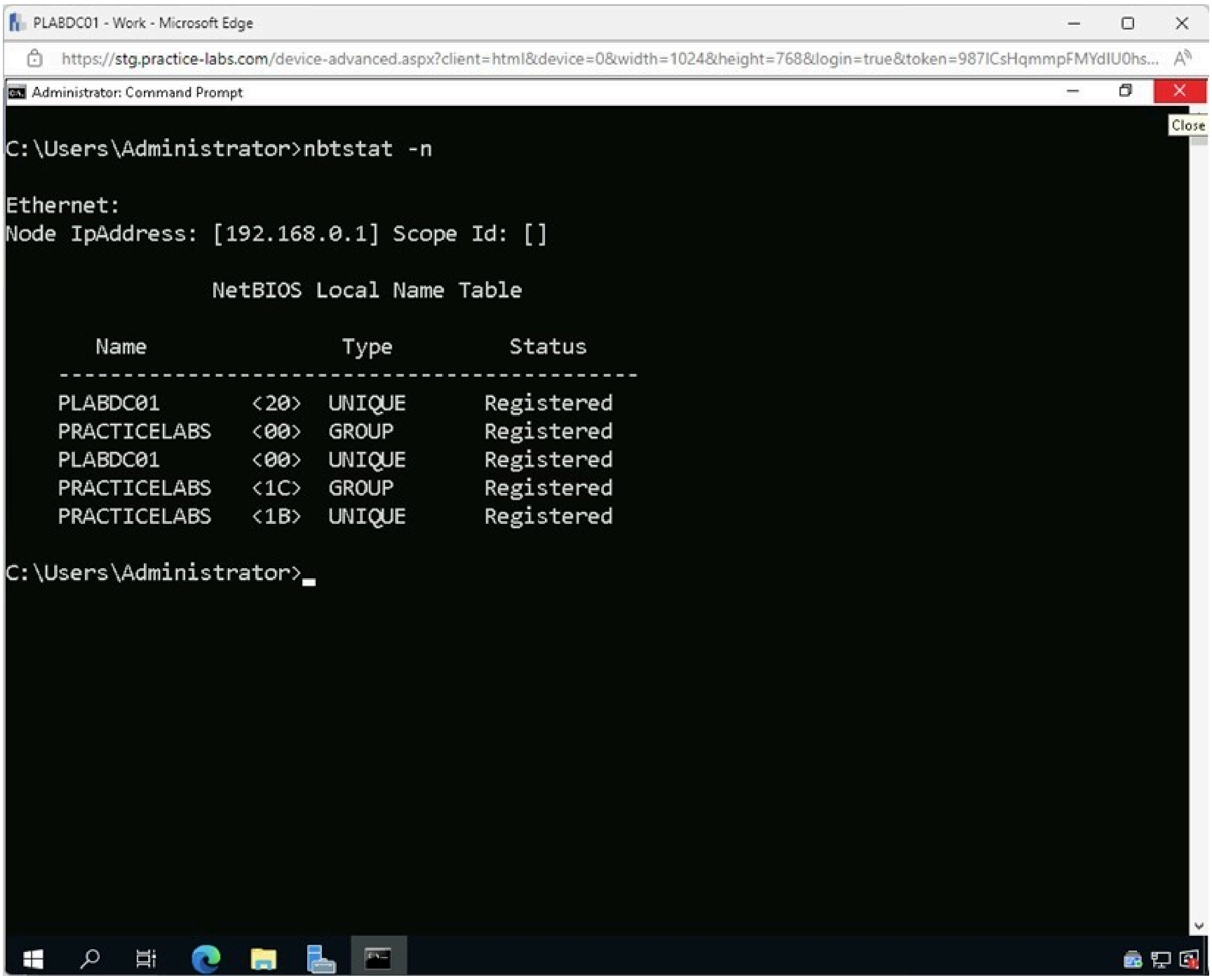

Figure 2.14 Screenshot of PLABDC01: Displaying closing the Command Prompt window.

Task 2 - Linux Commands

In this task, several common command line tools that are used in Linux for troubleshooting will be explored.

- **ip addr** - This command will give the IP address and other basic network information.
- **ifconfig** - This command will give the IP address and other basic network information for a wired network.
- **iwconfig** - This command will give the IP address and other basic network information for a wireless network.
- **dig** - This command is used to discover the IP address or DNS record associated with a hostname, such as google.com. If a user has an IP address, they can also perform a reverse DNS query to discover the hostname associated with the IP address.
- **netcat** - It is a program used for communicating across TCP/UDP networks. The command is built to be used in back-end communications that can be driven either directly by users or by other applications and scripts.

- **telnet** - Administrators often use Telnet to access a remote host's command line interface. Additionally, telnet is utilized by other protocols to set up a protocol control channel. Remember that because there is no encryption, the contents of the packet will be in plain text.
- **route** - This command is used to examine and change the IP routing table.

Step 1

Connect to **PLABUBUNTU**.

Click **Terminal** on the left pane.

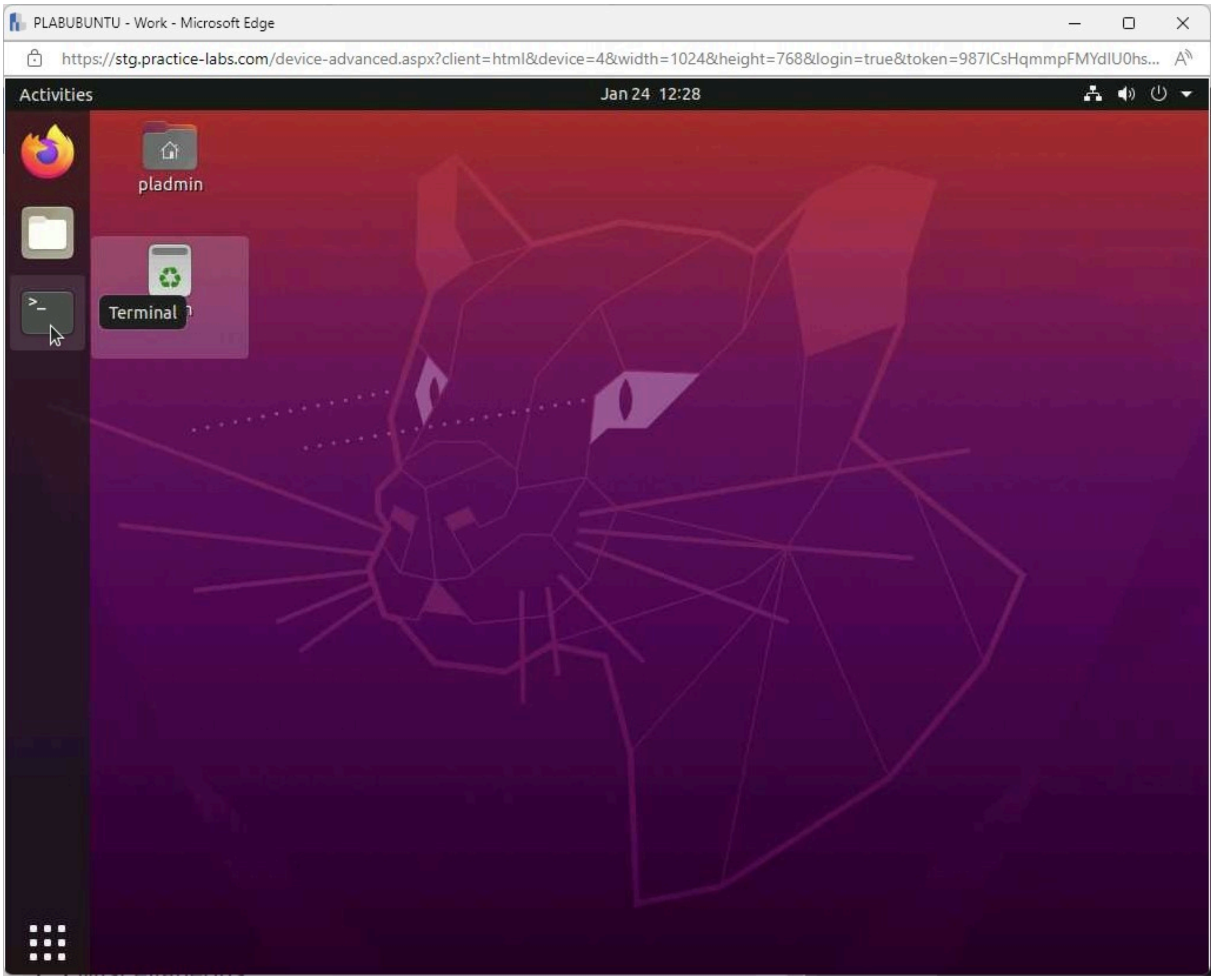

Figure 2.15 Screenshot of PLABUBUNTU: Displaying selecting Terminal on the left pane.

Step 2

In the **Terminal** window, type the following command and press **Enter**:

```
man ip addr
```

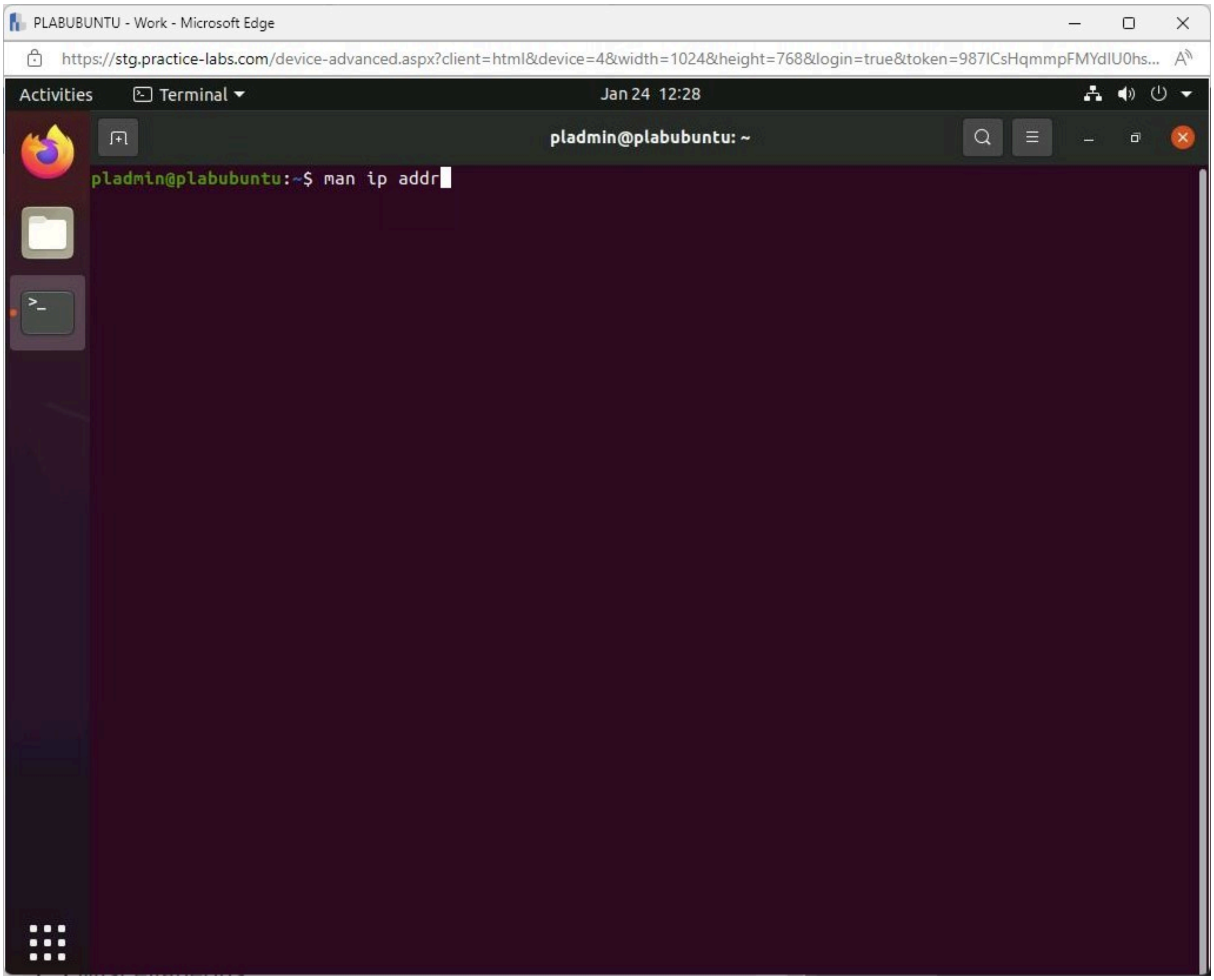

Figure 2.16 Screenshot of PLABUBUNTU: Displaying entering the required command in the Terminal window.

Step 3

The **man page** for the **ip addr** command is shown. This is a comprehensive view of what the command can do and how to use it.

Press **page down** on the keyboard to review the man page and press **q** when ready to quit.

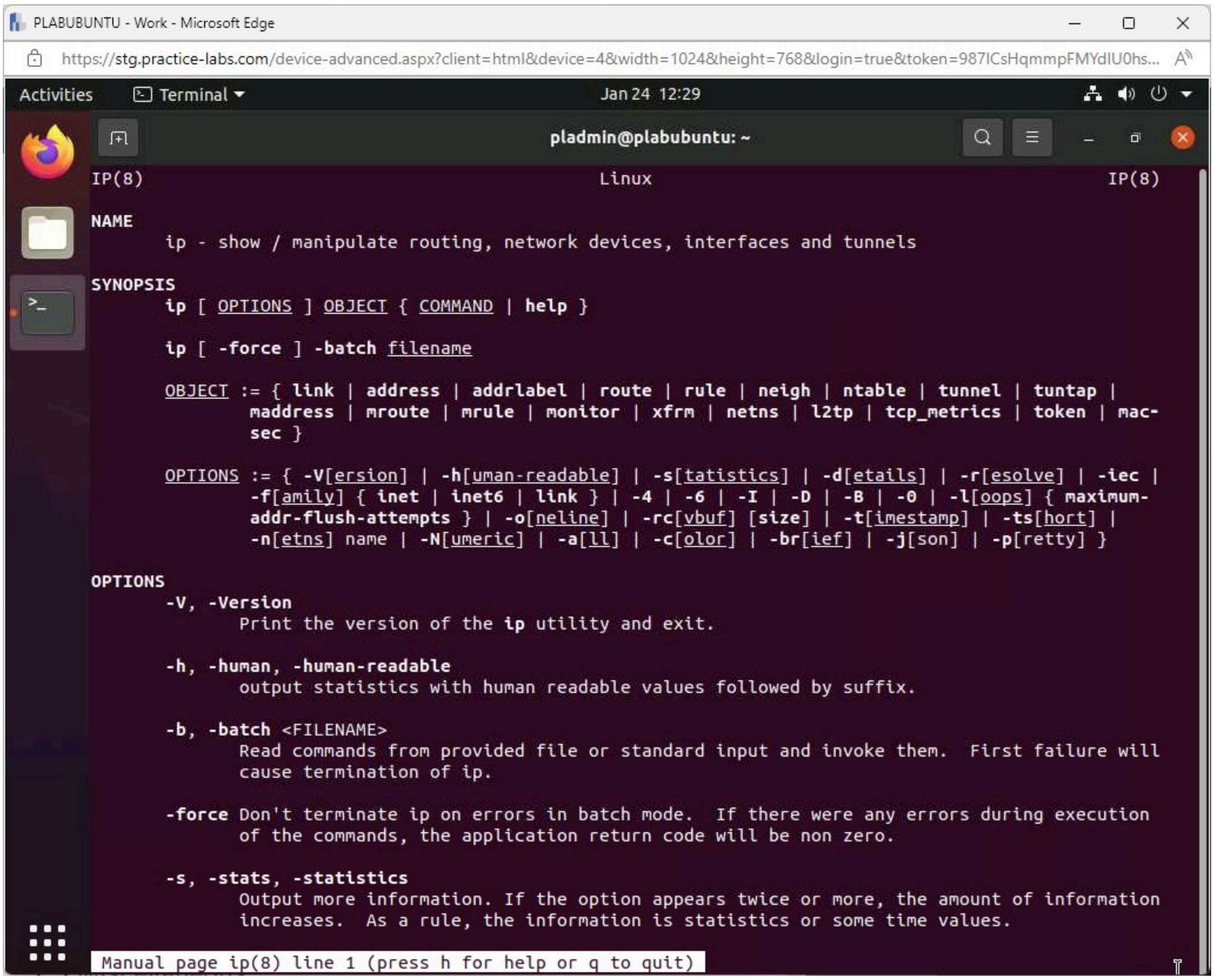

Figure 2.17 Screenshot of PLABUBUNTU: Displaying the output of the command in the Terminal window.

Step 4

Type the following command and press **Enter**:

```
ip addr
```

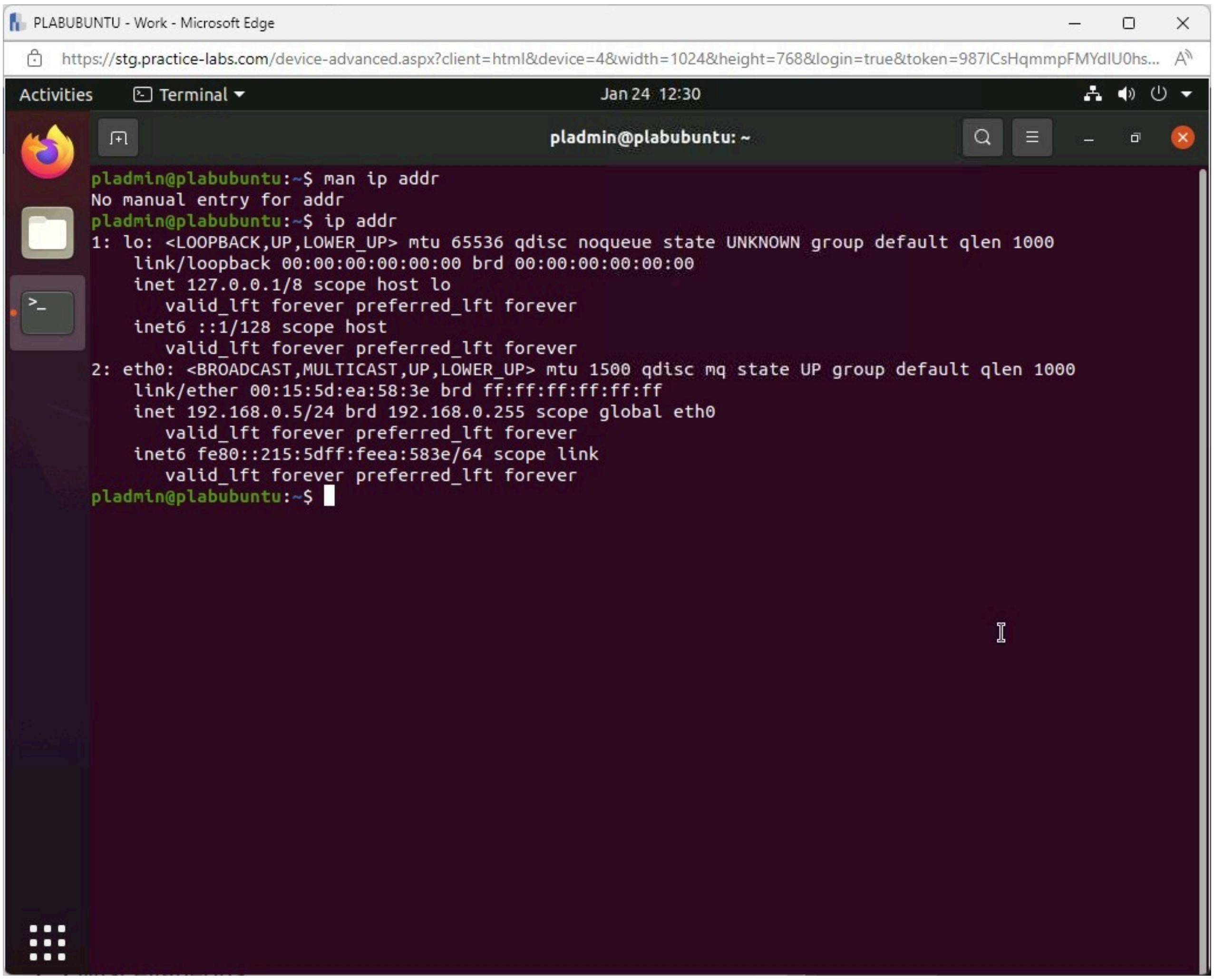

Figure 2.18 Screenshot of PLABUBUNTU: Displaying entering and executing the required command on the Terminal window.

> ***Note:*** *Notice the IP address of the* ***PLABUBUNTU*** *device.*

Step 5

Type the following command and press **Enter** to clear the screen:

```
clear
```

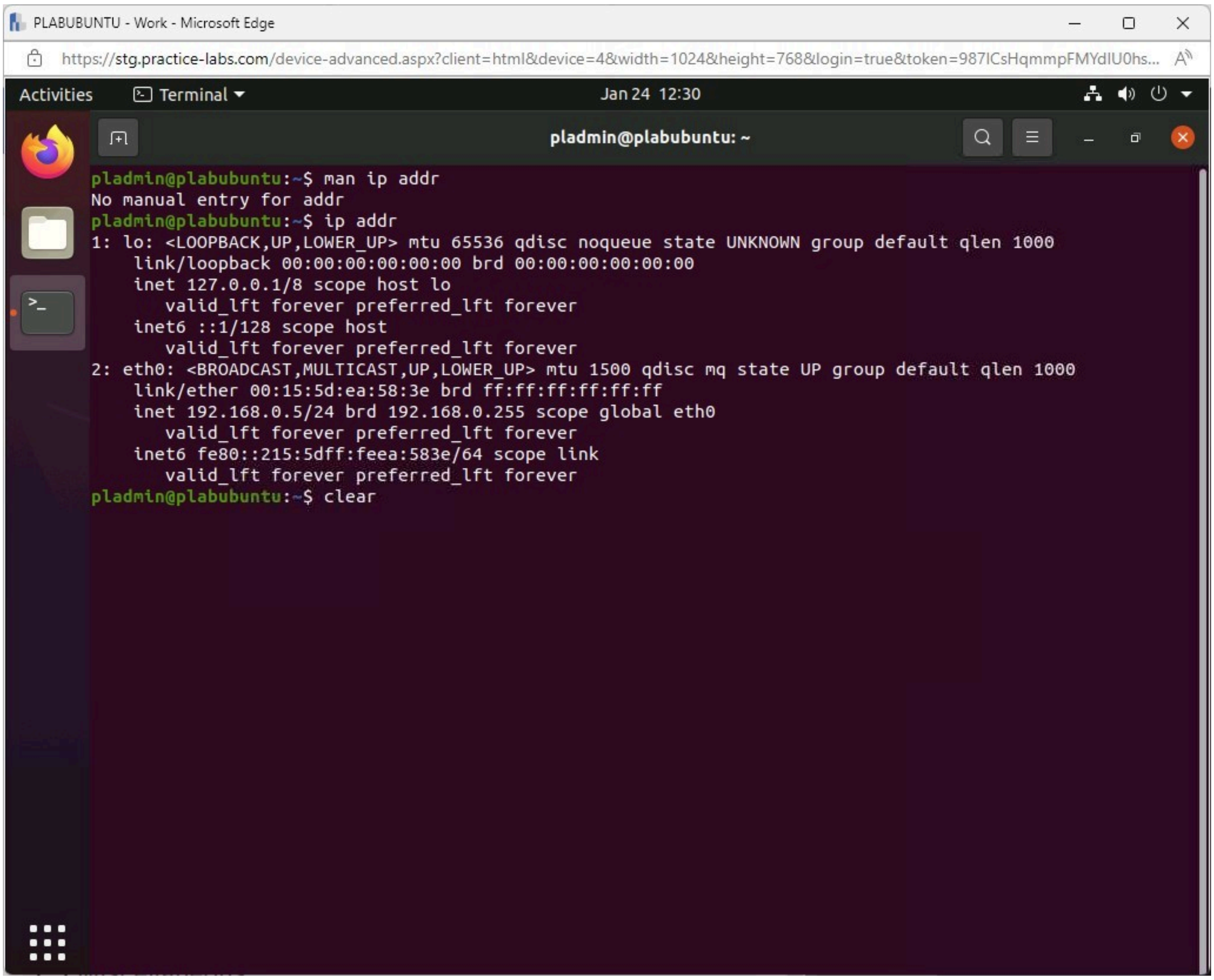

Figure 2.19 Screenshot of PLABUBUNTU: Displaying entering the required command on the Terminal window.

Step 6

Type the following command and press **Enter**:

```
ifconfig
```

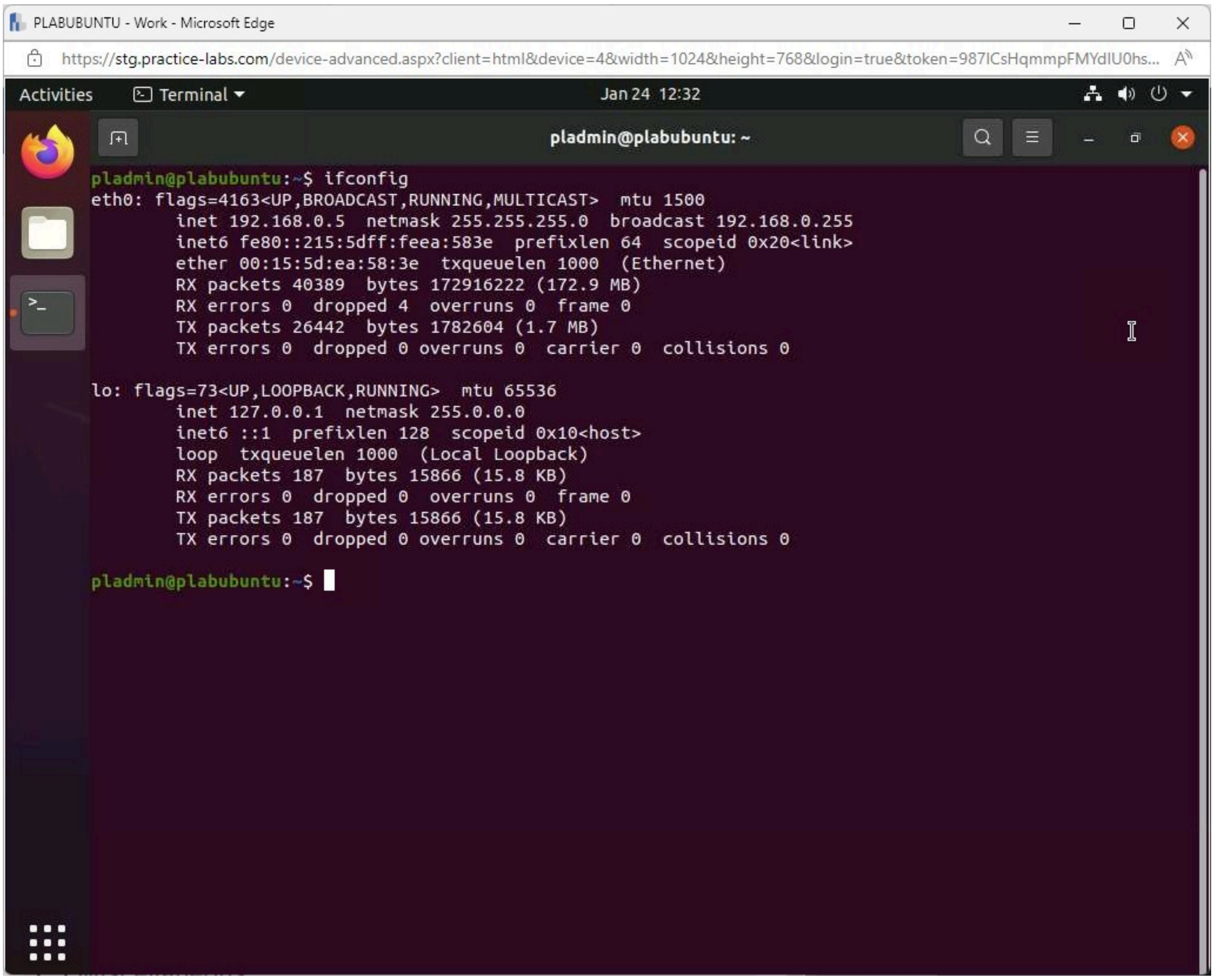

Figure 2.20 Screenshot of PLABUBUNTU: Displaying entering and executing the required command on the Terminal window.

> ***Note:** Notice the **IP address** of the **DNS Server** for this device. It is a loopback address, so we did not get the results as when we altered the **PLABWIN10** hosts file.*

Step 7

Type the following command and press **Enter**:

```
iwconfig
```

> ***Note:** You can enter the **clear** command and press **Enter** to clear the screen before typing the other commands.*

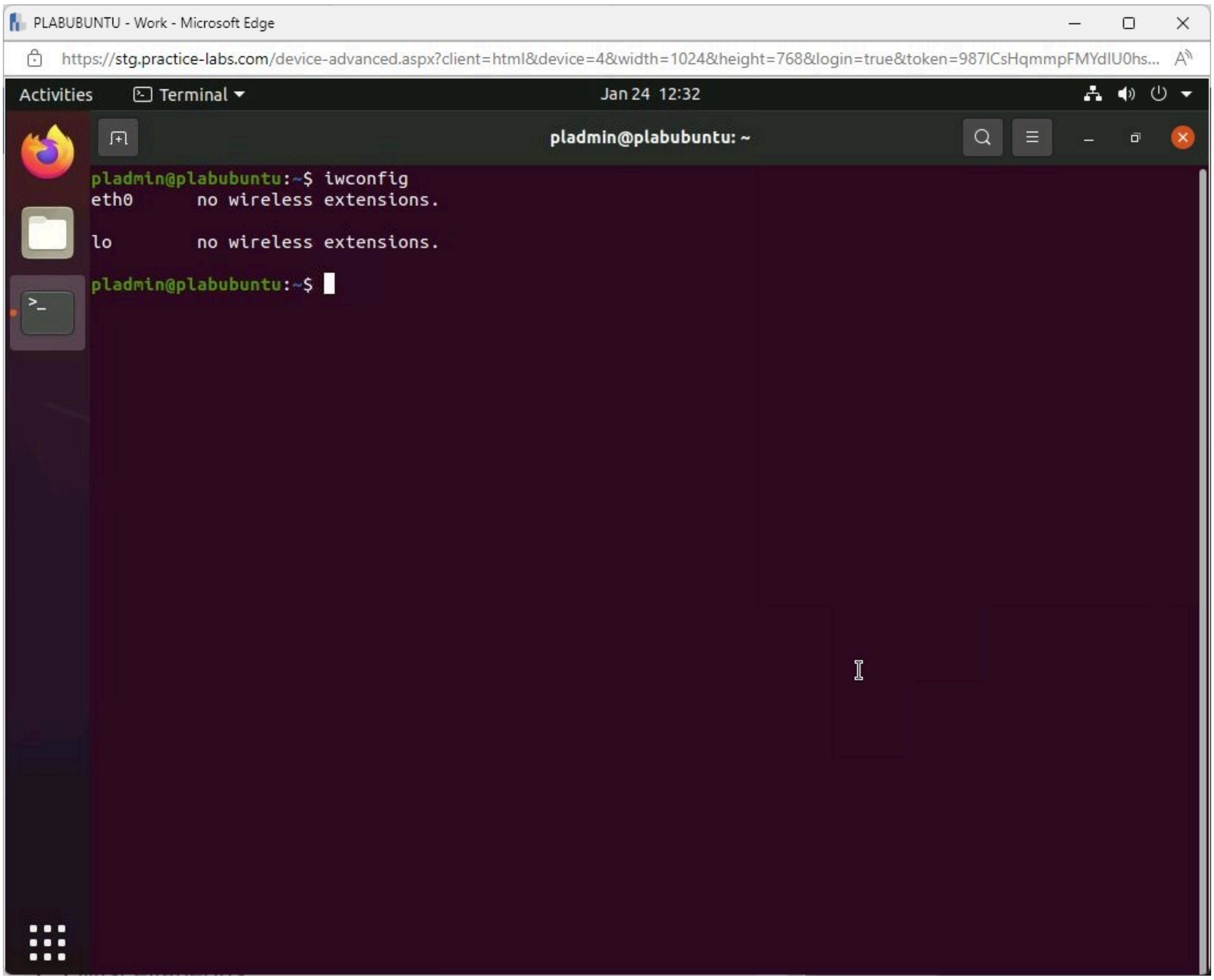

Figure 2.21 Screenshot of PLABUBUNTU: Displaying entering and executing the required command on the Terminal window.

Note: *Notice there are no wireless adapters.*

Step 8

Type the following command and press **Enter**:

```
dig www.google.com
```

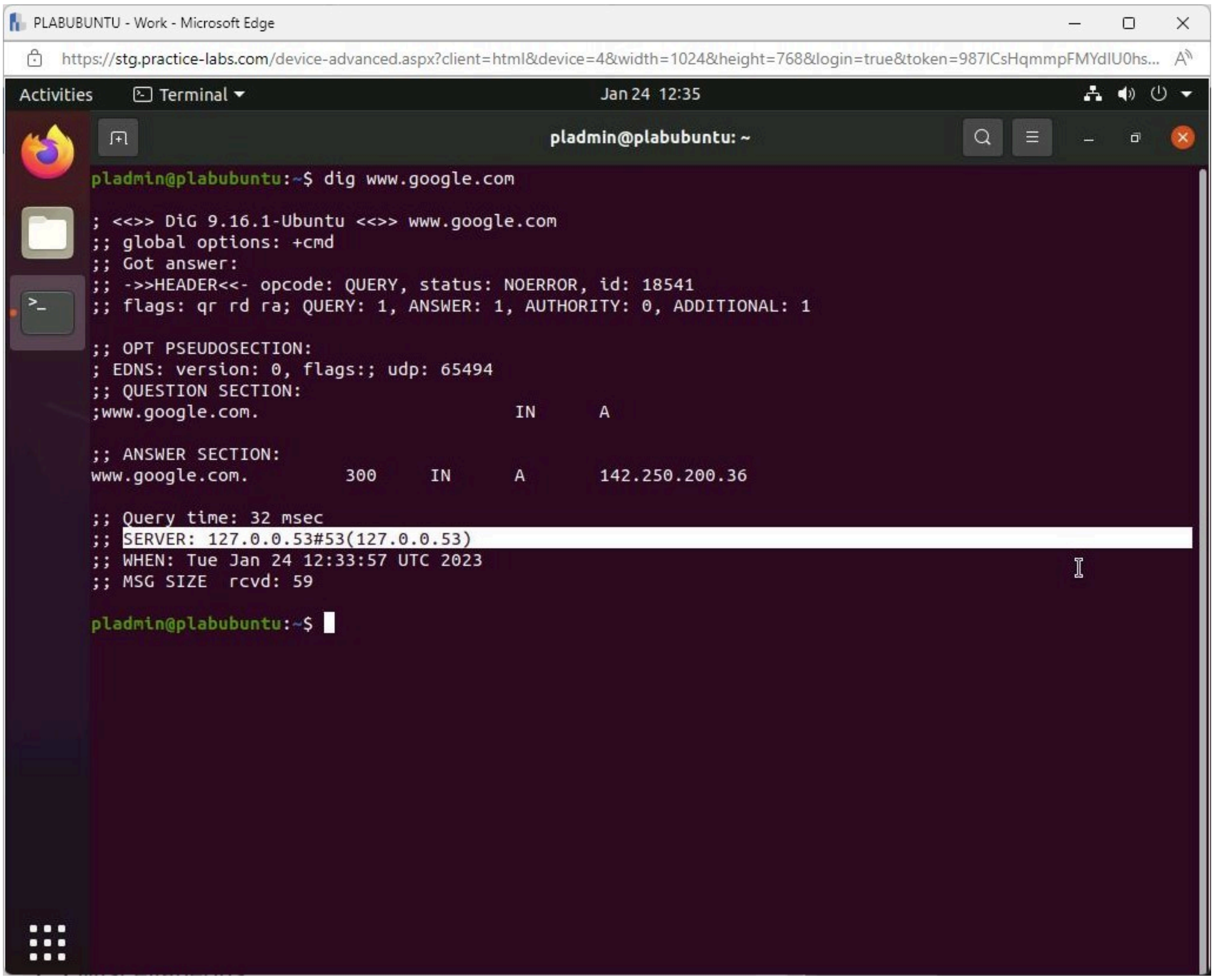

Figure 2.22 Screenshot of PLABUBUNTU: Displaying entering and executing the required command on the Terminal window.

> ***Note:*** *Notice the **IP address** of the **DNS Server** for the device. It is a loopback address, so we did not get the results as when we altered the **PLABWIN10** hosts file.*

Step 9

Type the following command and press **Enter**:

```
man nc
```

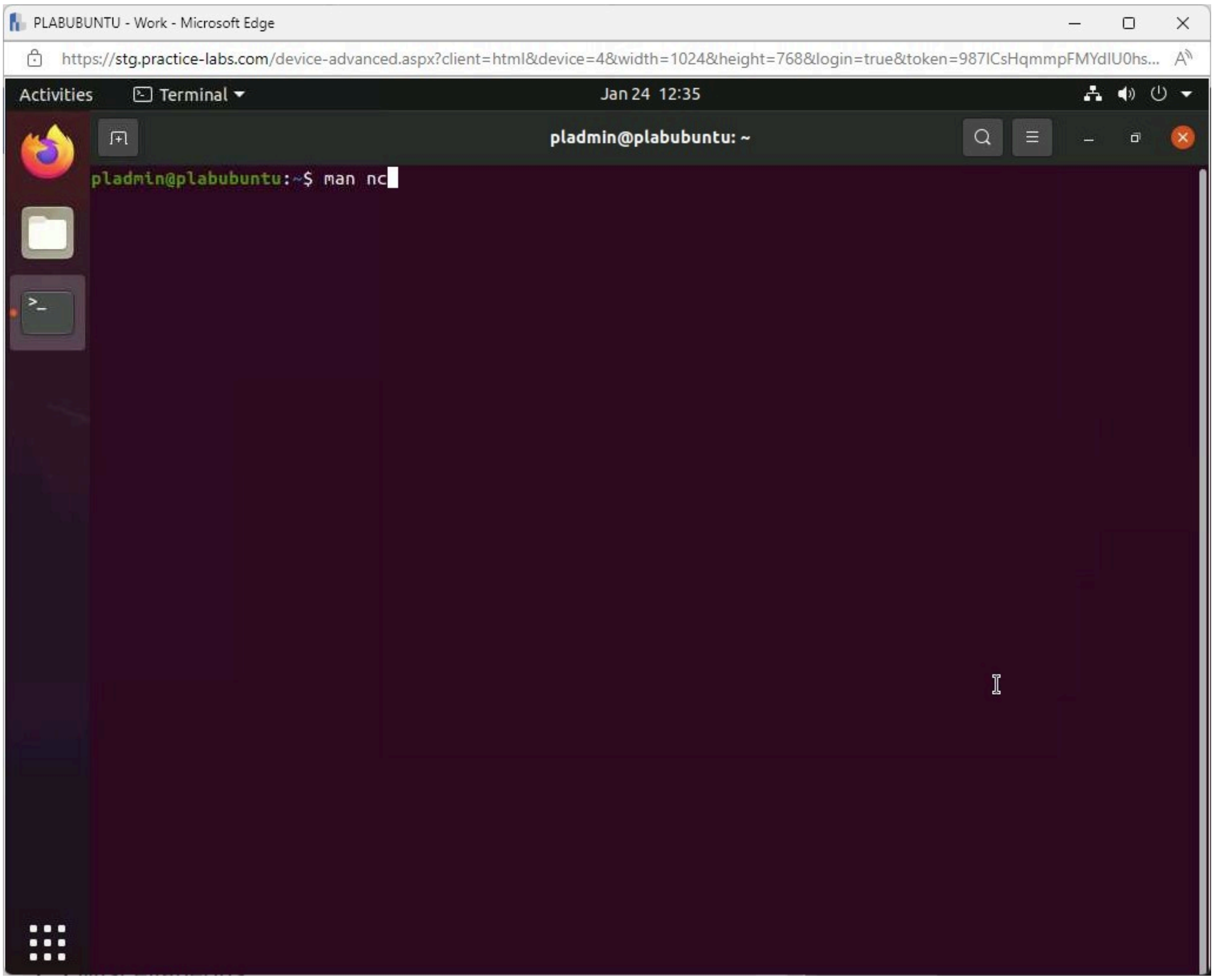

Figure 2.23 Screenshot of PLABUBUNTU: Displaying entering the required command on the Terminal window.

Step 10

The **man page** for the **netcat** command is shown. This is a comprehensive view of what the command can do and how to use it.

Press **page down** on the keyboard to review the man page and press **q** when ready to quit.

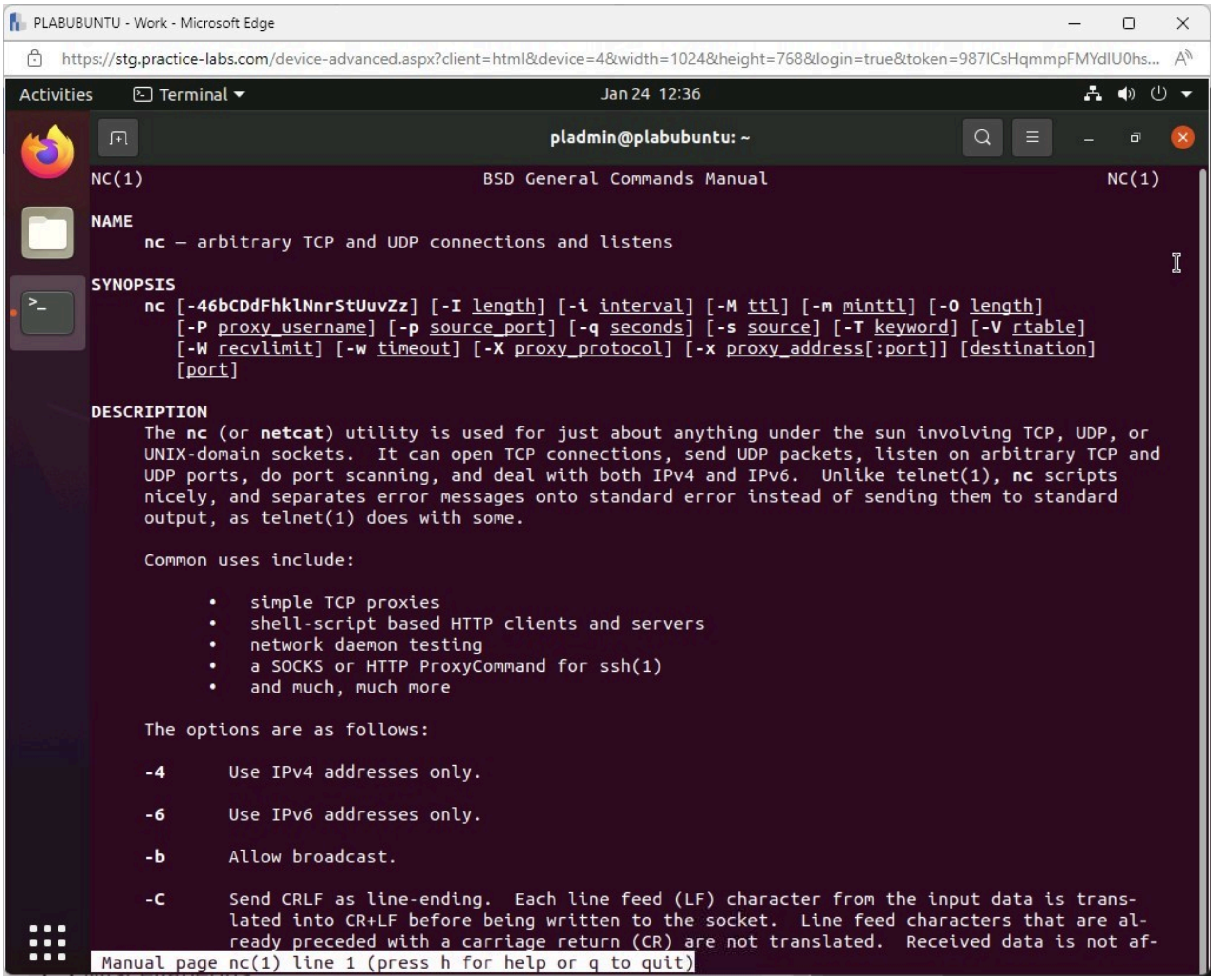

Figure 2.24 Screenshot of PLABUBUNTU: Displaying the output of the command in the Terminal window.

Step 11

Type the following command and press **Enter**:

```
man telnet
```

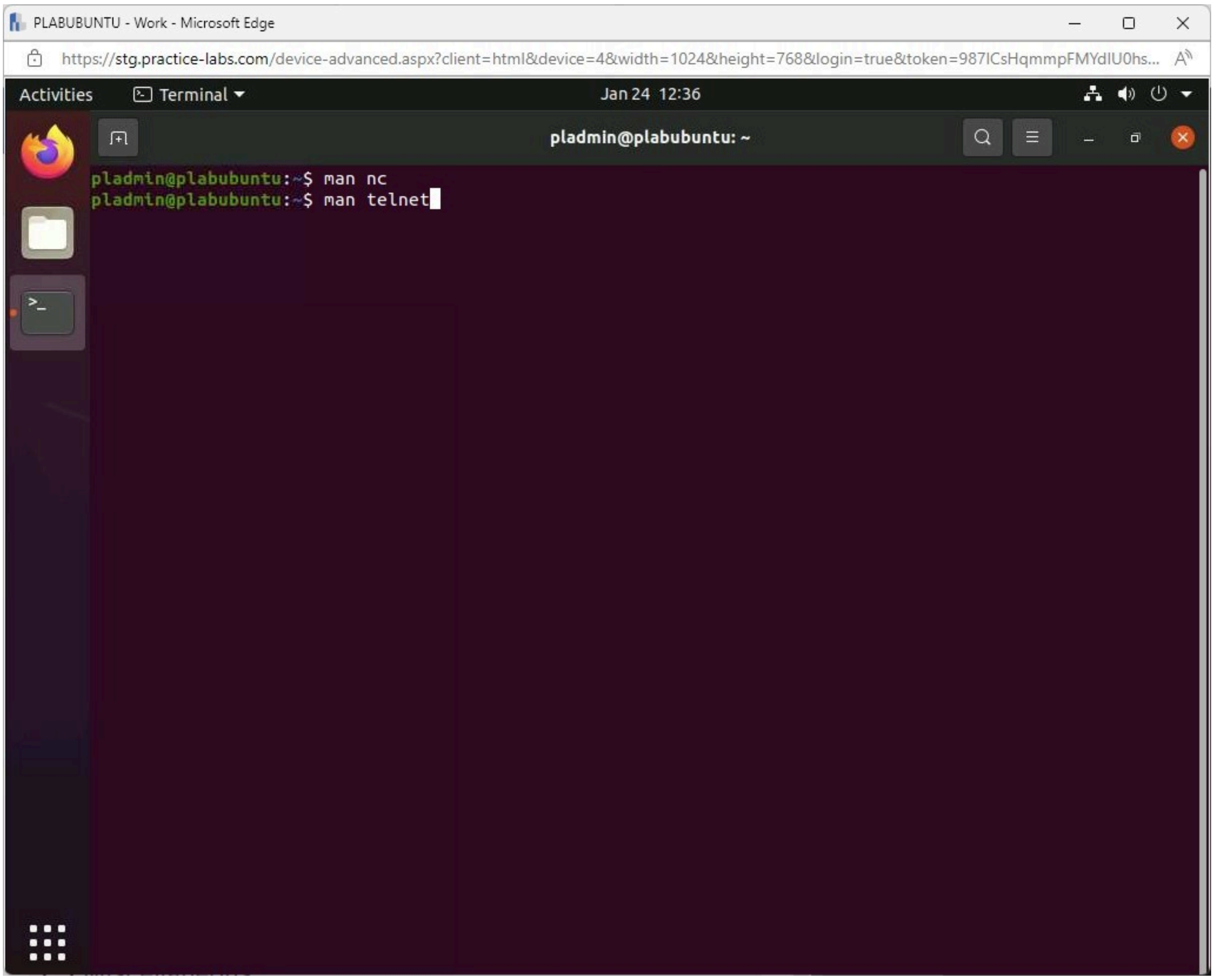

Figure 2.25 Screenshot of PLABUBUNTU: Displaying entering the required command on the Terminal window.

Step 12

The **man page** for the **telnet** command is shown. This is a comprehensive view of what the command can do and how to use it.

Press **page down** on the keyboard to review the man page and press **q** when ready to quit.

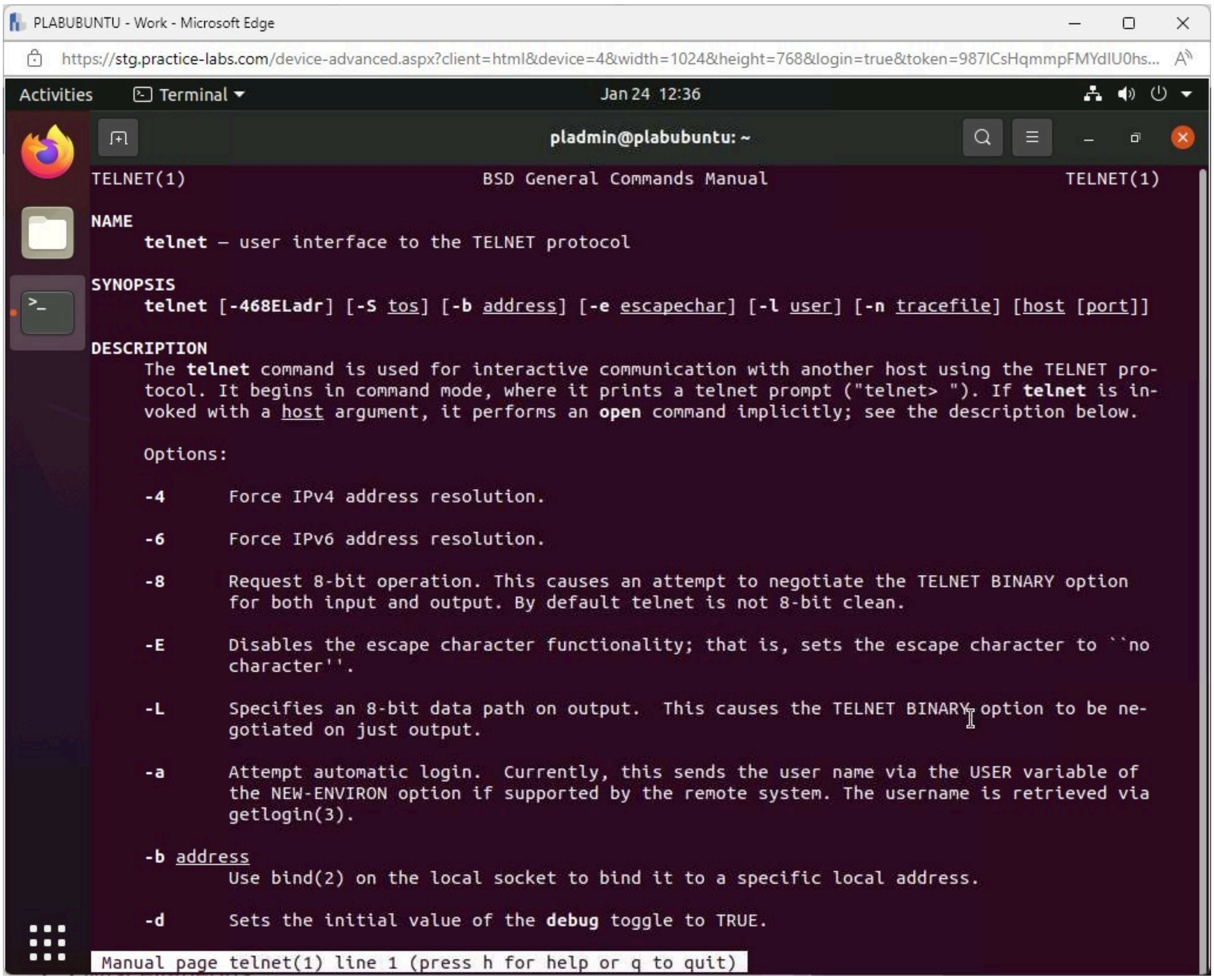

Figure 2.26 Screenshot of PLABUBUNTU: Displaying the output of the command in the Terminal window.

Step 13

Type the following command and press **Enter**:

```
man route
```

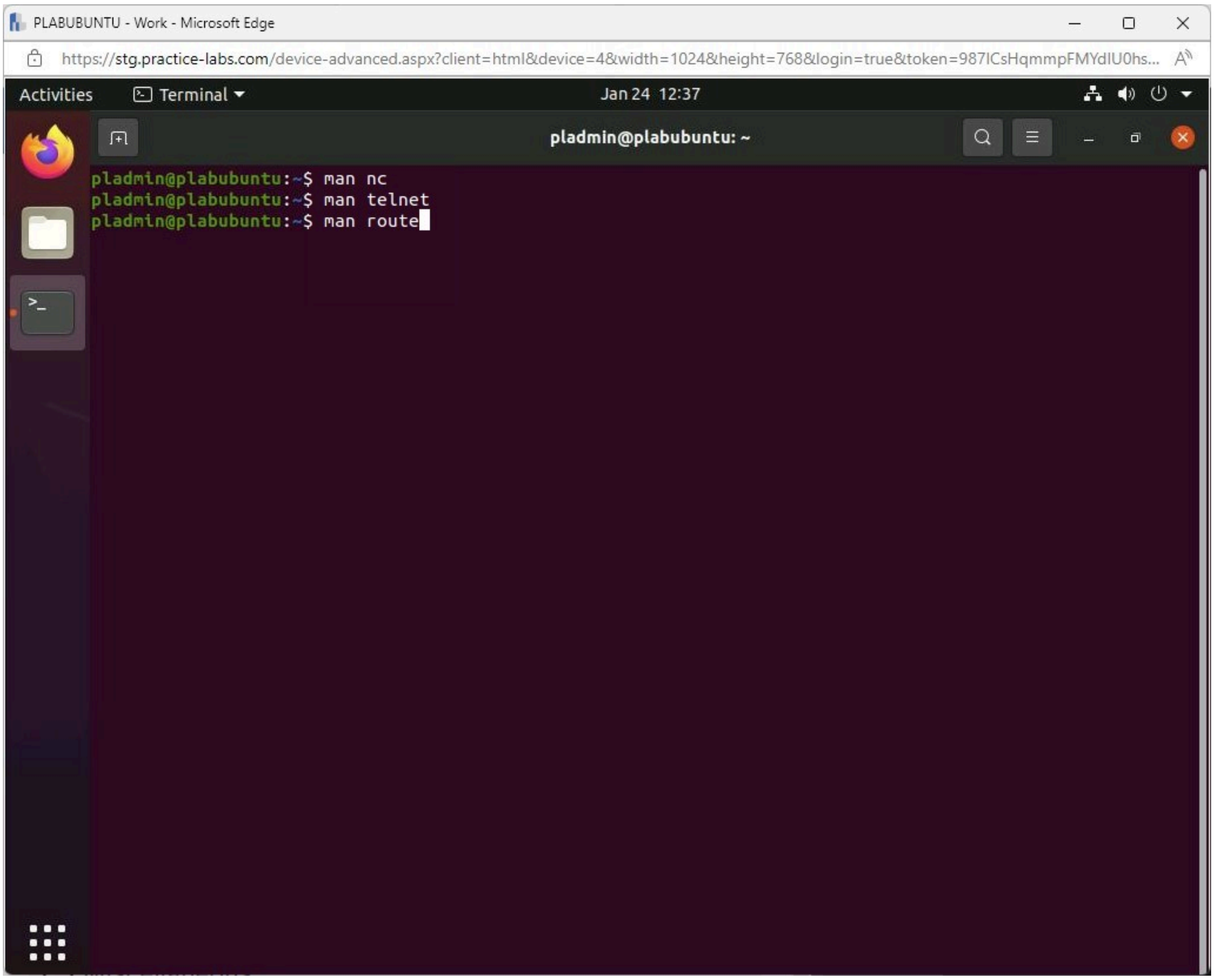

Figure 2.27 Screenshot of PLABUBUNTU: Displaying entering the required command on the Terminal window.

Step 14

The **man page** for the **route** command is shown. This is a comprehensive view of what the command can do and how to use it.

Press **page down** on the keyboard to review the man page and press **q** when ready to quit.

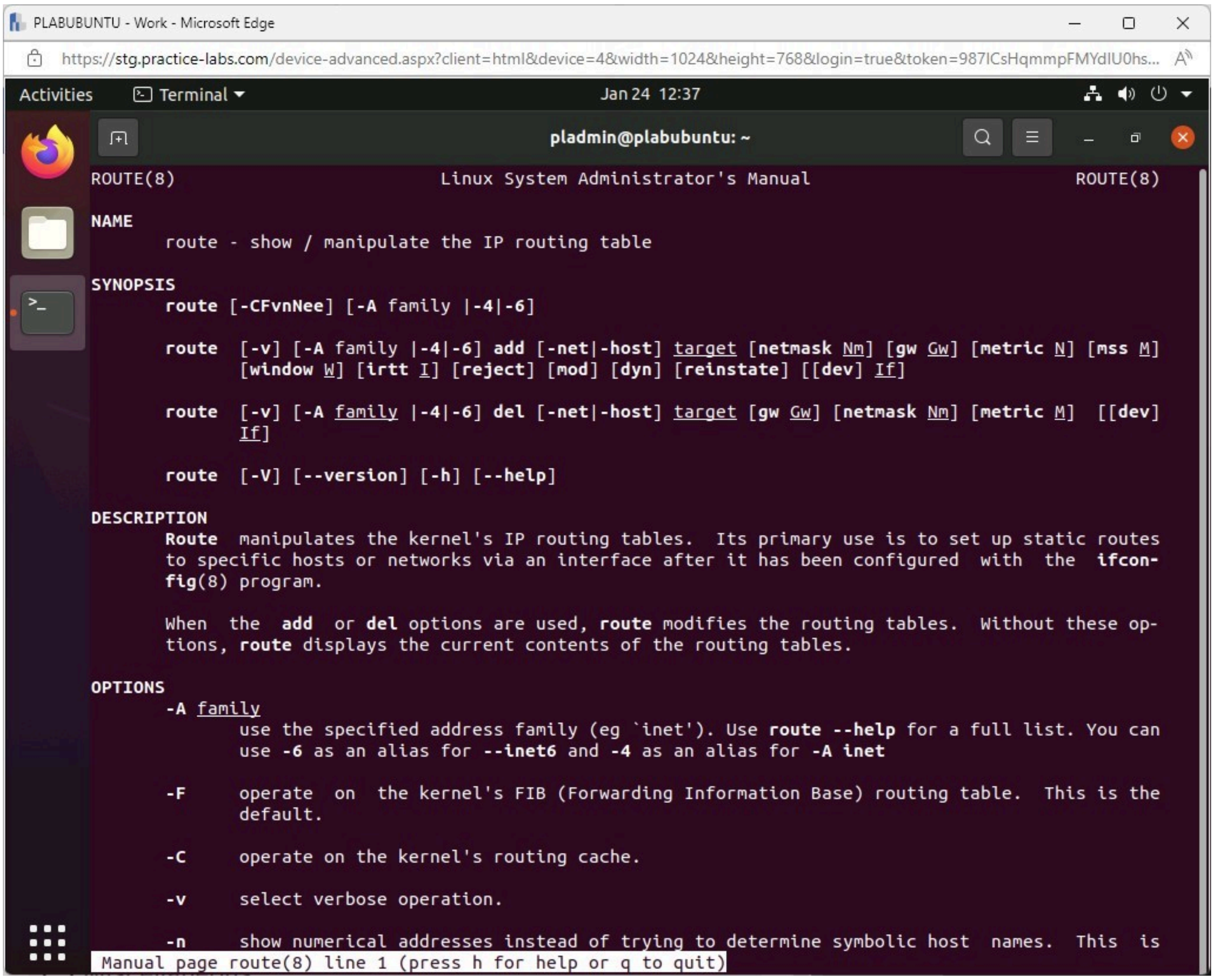

Figure 2.28 Screenshot of PLABUBUNTU: Displaying the output of the command in the Terminal window.

Step 15

Close the **Terminal** window.

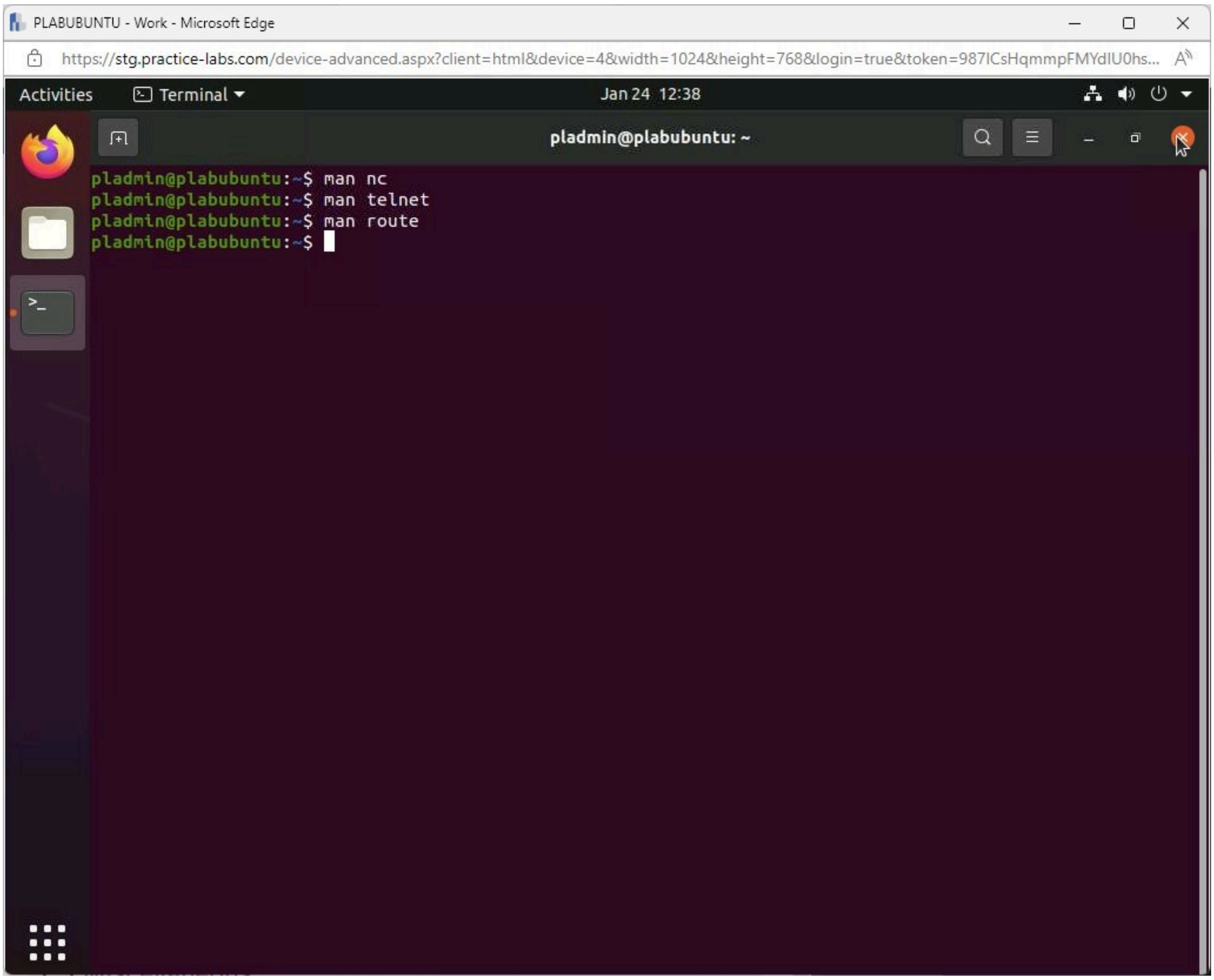

Figure 2.29 Screenshot of PLABUBUNTU: Displaying closing the Terminal window.

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Troubleshooting Network Connectivity Issues** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Common Problems
- Exercise 2 - Command Line Tools and Techniques

You should now be able to:

- Explore Host File Misconfiguration
- Work with Network Troubleshooter
- Explore Internet Options
- Configure Network Adapters
- Configure Network Adapters IPv4 and IPv6 Properties
- Install and Configure DHCP Server
- Explore Windows Commands Commonly used in Troubleshooting
- Explore Linux Commands

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.